35. If a and b are the degree and order of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + 4\frac{dy}{dx} + \frac{d^3y}{dx^3} = 14$ then the value of $2a - 3b$ is

(a) 7 (b) -7

(c) 3 (d) -3

Ans. (b)

36. The order of differential equation is always

(a) negative integer

(b) rational number

(c) positive integer

(d) whole number

Ans. (c)

****

UNIT-IV. Differential Calculus

4.1 CURVATURE

♦ 4.1.(a) Curvature of Curve

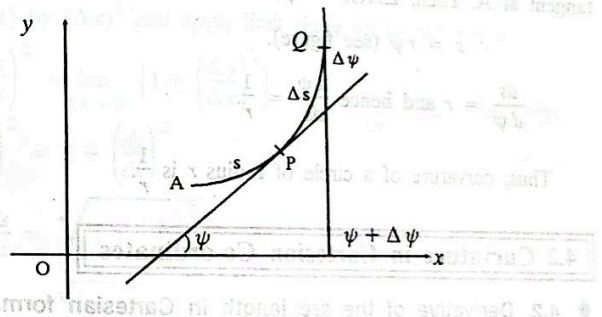
The rate of bending of a curve in any interval is called the curvature of the curve in that interval.

Note : The curvature of a circle at any point on it is the same and is equal to the reciprocal of its radius.

Note : The curvature of a straight line is zero.

♦ 4.1.(b) Definition of Curvature

Let P and Q be any two close points on a plane curve. Let the actual distances of P and Q measured from a fixed point A on the given curve s and $s + \Delta s$, so that arc PQ (the arcual length of PQ) is Δs.



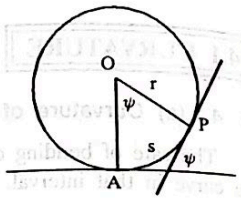
Let the tangents at P and Q to the curve make angles ψ and $\psi + \Delta\psi$ with a fixed line in the plane of the curve, say, the x axis. Then, the angle between the tangents at P and Q is equal to $\Delta\psi$.

Thus, for a change of Δs in the arcual length of the curve, the direction of the tangent to the curve changes by $\Delta\psi$.

Hence, $\frac{\Delta\psi}{\Delta s}$ is the average rate of bending of the curve (or average rate of change of direction of the tangent to the curve in the arcual interval $\hat{PQ}$) or average curvature of the arc PQ.

$\lim_{\Delta s \to 0} \left(\frac{\Delta \psi}{\Delta s}\right) = \frac{d\psi}{ds}$ is the rate of bending of the curve with respect to arcual distance P or the curvature of the curve at the point P. The curvature is denoted by κ (kappa).

(or)



Definition : Consider a curve given by the equation $y = f(x)$. Suppose, the curve has a definite tangent at each point. Let A be a fixed point on the curve and P be an arbitrary point on the curve. Let s denote the arc length AP. Let ψ be the angle made by the tangent with the x axis. Then $\left(\frac{d\psi}{ds}\right)$ is called the **curvature** of the curve at P. Thus, the curvature is the rate of turning of the tangent w.r.to the arc length.

Theorem : The curvature of a circle of radius r at any point is $\frac{1}{r}$.

Proof : Let A be a fixed point on the circle and P be any point on the circle. Let arc AP $= s$. Let the tangent at P make on angle ψ with the tangent at A. Then, $\angle AOP = \psi$.

$\therefore s = r\psi$ (see figure).

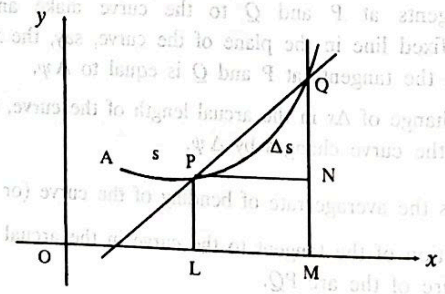
$\therefore \frac{ds}{d\psi} = r$ and hence $\frac{d\psi}{ds} = \frac{1}{r}$.

Thus, curvature of a circle of radius r is $\frac{1}{r}$.

4.2 Curvature in Cartesian Co-ordinates

♦ 4.2. Derivative of the arc length in Cartesian form.

Let $y = f(x)$ be the equation of the Cartesian curve.



Let A be a fixed point on the curve and $P(x,y)$, $Q\left((x+\Delta x), (y+\Delta y)\right)$ be two neighbouring points on the curve.

Let the lengths of the arcs AP and AQ be s and $s + \Delta s$ respectively.

(i.e.,) $\hat{AP} = s$ and $\hat{AQ} = s + \Delta s$

$\therefore \hat{PQ} = \Delta s$

Draw PL and QM perpendiculars onto the X axis and PN perpendicular to QM.

From the figure, we have

$$PN = LM = OM - OL = (x + \Delta x) - x = \Delta x$$

$$QN = QM - MN = QM - PL = (y + \Delta y) - y = \Delta y$$

Also $PQ^2 = PN^2 + QN^2$; $PQ^2 = (\Delta x)^2 + (\Delta y)^2$

If the point Q is close enough to P then, arc PQ = chord $PQ = \Delta s$

Thus, $(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$ ... (1)

As $Q \longrightarrow P$ along the curve, $\Delta x \longrightarrow 0$ and $\Delta y \longrightarrow 0$

Let us divide (1) by $(\Delta x)^2$ and apply limit there on as $\Delta x \longrightarrow 0$

(i.e.,) $\lim_{\Delta x \to 0} \left(\frac{\Delta s}{\Delta x}\right)^2 = \lim_{\Delta x \to 0} \left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right]$

$$\left(\frac{ds}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$$

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= \sqrt{1 + y_1^2} \quad \text{... (2)}$$

Similarly, dividing (1) by $(\Delta y)^2$ and taking limit there on as $\Delta y \to 0$ we obtain

$$\frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \quad \text{... (3)}$$

Coralaray 1.

Suppose, we have a parametric curve $x = x(t)$ and $y = y(t)$ then

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$$

Also $\frac{ds}{dt} = \frac{ds}{dx}\frac{dx}{dt}$

$$\frac{ds}{dt} = \sqrt{1 + \left(\frac{dy}{dt} / \frac{dx}{dt}\right)^2}\ \frac{dx}{dt} \quad \text{by (2)}$$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\ \frac{1}{\left(\frac{dx}{dt}\right)}\left(\frac{dx}{dt}\right)$$

Thus, $$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \quad \ldots (4)$$

Coralaray 2.

It may be observed from the figure that as $Q \to P$, the chord PQ becomes the tangent at the point P.

If ψ is the angle made by the tangent at P, with the X-axis then we know from the geometrical meaning of the derivative that

$$\tan\psi = \frac{dy}{dx} \text{ or } \cot\psi = \frac{dx}{dy}$$

$\therefore$ (2) $\Rightarrow \frac{ds}{dx} = \sqrt{1 + \tan^2\psi} = sec\,\psi$ ... (5)

(3) $\Rightarrow \frac{ds}{dy} = \sqrt{1 + \cot^2\psi} = cosec\,\psi$ ... (6)

(5) $\Rightarrow \cos\psi = \frac{dx}{ds}$ ... (7)

(6) $\Rightarrow \sin\psi = \frac{dy}{ds}$ ... (8)

4.3 CENTRE AND RADIUS OF CURVATURE

♦ 4.3.(a) Circle of curvature, radius of curvature and centre of curvature

The curvature at any point P of a curve is equal to the curvature of the circle which passes through P and two close points on the curve on either side of P. Such a circle exists for each point of the curve. It is called the "Circle of curvature" of the curve at the point. The radius of this circle is called the radius of curvature of the curve at that point.

♦ 4.3. (b) Formula for the radius of curvature

The reciprocal of the curvature of a curve at any point is called the radius of curvature at the point and is denoted by ρ.

Hence, we have $\rho = \frac{ds}{d\psi}$

1. Cartesian form

We know that $\frac{dy}{dx}$ represents the slope of the tangent to the curve $y = f(x)$ at (x, y). Hence, $\frac{dy}{dx} = \tan\psi$.

Differentiating w.r.t. s, we get

$$\left(\frac{d^2y}{dx^2}\right)\left(\frac{dx}{ds}\right) = \sec^2\psi\left(\frac{d\psi}{ds}\right).$$

$$\therefore \quad y_2 \cos\psi = \sec^2\psi\left(\frac{d\psi}{ds}\right).$$

$$\therefore \frac{ds}{d\psi} = \frac{\sec^3\psi}{y_2} = \frac{(1+\tan^2\psi)^{3/2}}{y_2} = \frac{(1+y_1^2)^{3/2}}{y_2}$$

$$\therefore \rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

Note 1 : If we start with the equation $\cot\psi = \frac{dx}{dy}$ and proceed as above, we arrive at the following alternative formula for the radius of curvature.

$$\rho = \frac{\left(1 + \left(\frac{dx}{dy}\right)^2\right)^{3/2}}{\left(\frac{d^2x}{dy^2}\right)}$$

This alternative formula can be used to find ρ when $\frac{dy}{dx}$ becomes ∞ at the given point.

2. Parametric form

Let $x = f(t)$ and $y = g(t)$ be the parametric equations of the given curve.

$$\therefore \frac{dy}{dx} = \left(\frac{dy}{dt}\right)\left(\frac{dt}{dx}\right) = \frac{g'(t)}{f'(t)} \quad \ldots (1)$$

where $f'(t) = \frac{df}{dt}$ and $g'(t) = \frac{dg}{dt}$.

$$\therefore \frac{d^2y}{dx^2} = \frac{g''f' - g'f''}{(f')^3} \text{ (verify).} \quad \ldots (2)$$

Substituting (1) and (2) in the cartesian form and simplifying, we get

$$\rho = \frac{(f'^2 + g'^2)^{3/2}}{f'g'' - f''g'}$$

3. Implicit form

Let $f(x,y) = 0$ be the implicit form of the given curve.

Differentiating $f(x,y) = 0$, we get $f_x + f_y y' = 0$, where f_x and f_y denote partial derivatives w.r.to x and y respectively.

$$\therefore y' = -\frac{f_x}{f_y} \quad \ldots (1)$$

$$\text{Now, } y'' = -\frac{1}{(f_y)^2}[f_y(f_{xx} + f_{yx}y') - f_x(f_{xy} + f_{yy}y')]$$

$$\text{By (1), } y'' = -\frac{1}{(f_y)^3}[f_{xx}(f_y)^2 - 2f_{xy}f_xf_y + f_{yy}(f_x)^2] \quad \ldots (2)$$

Substituting (1) & (2) in the cartesian form and simplifying, we get

$$\rho = \frac{(f_x^{\,2} + f_y^{\,2})^{3/2}}{f_{xx}f_y^{\,2} - 2f_{xy}f_xf_y + f_{yy}f_x^{\,2}} \text{ (taking +ve value)}$$

4. Polar form

Let $r = f(\theta)$ be the given curve in polar coordinates.

$\therefore x = r\cos\theta$ and $y = r\sin\theta$ may be regarded as the parametric equations of the given curve, the parameter being θ.

$$\therefore \frac{dx}{d\theta} = \cos\theta\frac{dr}{d\theta} - r\sin\theta$$

$$\text{and } \frac{dy}{d\theta} = \sin\theta\frac{dr}{d\theta} + r\cos\theta$$

$$\therefore \frac{d^2x}{d\theta^2} = \cos\theta\frac{d^2r}{d\theta^2} - 2\sin\theta\frac{dr}{d\theta} - r\cos\theta$$

$$\text{and } \frac{d^2y}{d\theta^2} = \sin\theta\frac{d^2r}{d\theta^2} + 2\cos\theta\frac{dr}{d\theta} - r\sin\theta.$$

Substituting these values in the formula for ρ in parametric form and simplifying, we get $\rho = \dfrac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$ where $r_1 = \dfrac{dr}{d\theta}$ and $r_2 = \dfrac{d^2r}{d\theta^2}$.

Note :

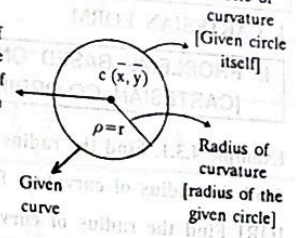
1. **Given curve is a circle**

$(x-h)^2 + (y-k)^2 = r^2$

(or) $x^2 + y^2 = a^2$

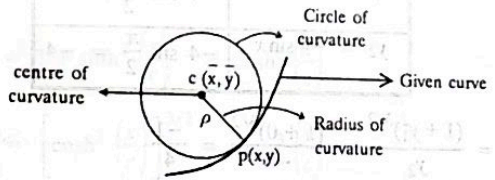
(or) $x^2 + y^2 + 2gx + 2fy + c = 0$

(or) $a(x^2 + y^2) + 2gx + 2fy + c = 0$



Circle	At any point		
	Radius of curvature	Centre of curvature	Circle of curvature
1. $(x-h)^2+(y-k)^2=r^2$	$\rho = r$	$\bar{x} = h, \bar{y} = k$	$(x-h)^2+(y-k)^2=r^2$
2. $x^2 + y^2 = a^2$	$\rho = a$	$\bar{x} = 0, \bar{y} = 0$	$x^2 + y^2 = a^2$
3. $x^2 + y^2 + 2gx + 2fy + c = 0$	$\rho = \sqrt{g^2+f^2-c}$	$\bar{x} = -g, \bar{y} = -f$	$x^2 + y^2 + 2gx + 2fy + c = 0$
4. $a(x^2+y^2)+2gx + 2fy + c = 0$	$\rho = \sqrt{\frac{g^2}{a^2}+\frac{f^2}{a^2}-\frac{c}{a}}$	$\bar{x} = -\frac{g}{a}, \bar{y} = -\frac{f}{a}$	$a(x^2 + y^2) + 2gx + 2fy + c = 0$

2. **Given curve is not a circle :**



The following results depend on the given point C (x,y) on the given curve.

radius of curvature	Centre of curvature c $(\bar{x}, \bar{y})$	Circle of curvature
$\rho = \dfrac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \dfrac{[1 + y_1^2]^{3/2}}{y_2}$	$\bar{x} = x - \dfrac{y_1}{y_2}(1 + y_1^2)$ $\bar{y} = y + \dfrac{1}{y_2}(1 + y_1^2)$	$(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$

Given curve is not a circle

I. CARTESIAN FORM

I. PROBLEMS BASED ON RADIUS OF CURVATURE [CARTESIAN CO-ORDINATES]

Example 4.3.1. Find the radius of curvature of the curve $y = e^x$ at (0, 1).[OR] Find the radius of curvature for $y = e^x$ at the point where it cuts the y-axis. [OR] Find the radius of curvature for $y=e^x$ at x=0 [A.U N/D 2014, R-13]

Solution : [A.U. CBT J/F 2010, A.U Jan. 2010, 2012, M/J 2013]

$y = e^x$	at (0, 1)
$y_1 = e^x$	$e^0 = 1$
$y_2 = e^x$	$e^0 = 1$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+1)^{3/2}}{1} = 2\sqrt{2}.$$

Example 4.3.2. Find the radius of curvature at $x=\frac{\pi}{2}$ on the curve $y=4\sin x$.

Solution :

$y = 4\sin x$	at $x = \frac{\pi}{2}$
$y_1 = 4\cos x$	$4\cos\frac{\pi}{2} = 0$
$y_2 = -4\sin x$	$-4\sin\frac{\pi}{2} = -4$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+0)^{3/2}}{-4} = \frac{-1}{4}$$

$$|\rho| = \frac{1}{4}$$

Example 4.3.3. Find the radius of curvature at the point $\left(\frac{1}{4}, \frac{1}{4}\right)$ on the curve $\sqrt{x} + \sqrt{y} = 1$. [A.U Nov. 2001, N/D 2006, A.U. Jan. 2008]

Solution : Given $\sqrt{x} + \sqrt{y} = 1$

$$\Rightarrow \quad \sqrt{y} = 1 - \sqrt{x}$$

squaring on both sides, we get

$$y = (1-\sqrt{x})^2$$

$$= 1 + x - 2\sqrt{x}$$

$y = 1 + x - 2\sqrt{x}$	at $\left(\frac{1}{4}, \frac{1}{4}\right)$
$y_1 = 1 - 2\left(\frac{1}{2\sqrt{x}}\right) = 1 - \frac{1}{\sqrt{x}}$	$= 1 - \frac{1}{\sqrt{1/4}} = 1 - 2 = -1$
$y_2 = \frac{1}{2}x^{-3/2}$	$= \frac{1}{2}\left(\frac{1}{4}\right)^{-3/2} = \frac{8}{2} = 4$

$$\therefore \rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+1)^{3/2}}{4} = \frac{1}{\sqrt{2}}.$$

Example 4.3.4. Show that the radius of curvature at any point of the catenary $y = c\cosh\left(\frac{x}{c}\right)$ is y^2/c. Find curvature, also find ρ at (0, c).

[A.U. M/J 2009, 2012] [A.U Tvli. Jan 2010] [A.U N/D 2014 R-08] [A.U D15 / J16 R-13, R-8]

Solution : Given : $y = c\cosh\left(\frac{x}{c}\right)$ ... (1)

$y_1 = c\sinh\left(\frac{x}{c}\right)\left(\frac{1}{c}\right)$ $= \sinh\left(\frac{x}{c}\right)$	$y_2 = \frac{1}{c}\cosh\left(\frac{x}{c}\right)$

$$y_1^2 = \sinh^2\left(\frac{x}{c}\right)$$

$$1 + y_1^2 = 1 + \sinh^2\left(\frac{x}{c}\right) = \cosh^2\left(\frac{x}{c}\right)$$

$$\left(1+y_1^2\right)^{3/2} = \left[\cosh^2\left(\frac{x}{c}\right)\right]^{3/2} = \cosh^3\left(\frac{x}{c}\right)$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\cosh^3\left(\frac{x}{c}\right)}{\frac{1}{c}\cosh\left(\frac{x}{c}\right)}$$

$$= c\cosh^2\left(\frac{x}{c}\right) = c\left(\frac{y}{c}\right)^2 \text{ by (1)}$$

i.e., $\rho = \frac{y^2}{c}$, $\rho_{(0,c)} = \frac{c^2}{c} = c$

Curvature $= \frac{1}{\rho} = \frac{1}{\left(\frac{y^2}{c}\right)} = \frac{c}{y^2}$

Example 4.3.5. Prove that at the point $x = \frac{\pi}{2}$ of the curve

$$y = 4 \sin x - \sin 2x, \ \rho = \frac{5\sqrt{5}}{4}.$$

Solution :

$y = 4 \sin x - \sin 2x$	at $x = \frac{\pi}{2}$
$y_1 = 4 \cos x - 2 \cos 2x$	$= 4 \cos \frac{\pi}{2} - 2 \cos \pi = 4(0) - 2(-1) = 2$
$y_2 = -4 \sin x + 4 \sin 2x$	$= -4 \sin \frac{\pi}{2} + 4 \sin \pi = -4(1) + 4(0) = -4$

$$\therefore \rho = \frac{[1 + y_1^2]^{3/2}}{y_2} = \frac{[1 + 2^2]^{3/2}}{(-4)} = \frac{(1+4)^{3/2}}{-4}$$

$$= \frac{5^{3/2}}{-4} = \frac{-5\sqrt{5}}{4}, \quad |\rho| = \frac{5\sqrt{5}}{4}$$

Note : 1. $\sin n\pi = 0, \ n = 1, 2, \ldots$

2. $\cos n\pi = (-1)^n, \ n = 1, 2, 3, \ldots$

3. $\sin \frac{\pi}{2} = 1$

4. $\cos \pi = -1$

Example 4.3.6. Find the radius of curvature at any point (x, y) on

$$y = c \log \sec \frac{x}{c}.$$

[A.U N/D 2006]

Solution : $y = c \log \sec \frac{x}{c}$.

$y_1 = c \frac{1}{\sec \frac{x}{c}} \left(\tan \frac{x}{c}\right) \cdot \left(\sec \frac{x}{c}\right) \left(\frac{1}{c}\right) = \tan \frac{x}{c}$	$y_2 = \frac{1}{c} \sec^2 \frac{x}{c}$

$$y_1^2 = \tan^2 \left(\frac{x}{c}\right)$$

$$1 + y_1^2 = 1 + \tan^2 \left(\frac{x}{c}\right) = \sec^2 \left(\frac{x}{c}\right)$$

$$(1 + y_1^2)^{3/2} = \left[\sec^2 \left(\frac{x}{c}\right)\right]^{3/2} = \sec^3 \left(\frac{x}{c}\right)$$

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{\sec^3 \left(\frac{x}{c}\right)}{\frac{1}{c} \sec^2 \left(\frac{x}{c}\right)} = c \sec \left(\frac{x}{c}\right)$$

Example 4.3.7. Find the radius of curvature at $x = 1$ on $y = \frac{\log x}{x}$.

Solution :

$y = \left(\frac{1}{x}\right) \log x$	at $x = 1$
$y_1 = \left(\frac{1}{x}\right)\left[\frac{1}{x}\right] + \log x \left[\frac{-1}{x^2}\right]$ $= \frac{1}{x^2} + \log x \left[-\frac{1}{x^2}\right]$ $= \frac{1}{x^2}[1 - \log x]$	$y_1 = \frac{1}{1^2}[1 - \log 1]$ $= 1 \quad [\because \log 1 = 0]$
$y_2 = \left(\frac{1}{x^2}\right)\left[0 - \frac{1}{x}\right] + (1 - \log x)\left[\frac{-2}{x^3}\right]$ $= -\frac{1}{x^3} - \frac{2}{x^3}(1 - \log x)$	$y_2 = \frac{-1}{1^3} - \frac{2}{1^3}[1 - \log 1]$ $= -1 - 2 = -3$

$$\therefore \text{Radius of curvature } \rho = \frac{[1 + y_1^2]^{3/2}}{y_2}$$

$$= \frac{[1+1]^{3/2}}{-3} = \frac{2^{3/2}}{-3}$$

$$= -\frac{2^{3/2}}{3} = \frac{-2\sqrt{2}}{3}$$

$$|\rho| = \frac{2\sqrt{2}}{3}$$

Example 4.3.8. Find the radius of curvature at the point (c, c) on the curve $xy = c^2$. **[AU, Nov. 2002] [A.U Jan. 2014, R008] [A.U M/J 2014 R-08]**

Solution : $xy = c^2$ **[A.U Trichy Jan 2009, CBT A/M 2011]**

$y = \frac{c^2}{x}$	at (c, c)
$y_1 = \frac{dy}{dx} = -\frac{c^2}{x^2}$	$y_1 = \frac{-c^2}{c^2} = -1$
$y_2 = \frac{d^2y}{dx^2} = \frac{2c^2}{x^3}$	$y_2 = \frac{2c^2}{c^3} = \frac{2}{c}$

$$\rho = \frac{\left(1+y_1^2\right)^{3/2}}{y_2}$$

$$= \frac{[1+(-1)^2]^{3/2}}{\left(\frac{2}{c}\right)}$$

$$= \frac{c\,(2)^{3/2}}{2} = \frac{c\,2\sqrt{2}}{2} = c\sqrt{2}.$$

We know that

$y = \frac{1}{x}$

$y' = \frac{-1}{x^2}$

$y'' = \frac{2}{x^3}$

Example 4.3.9. Show that the measure of curvature of the curve $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$ at any point (x, y) on it is $\frac{ab}{2(ax+by)^{3/2}}$.

Solution : The equation of the curve is **[A.U N/D 2003]**

$$\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$$

$$\frac{\sqrt{x}}{\sqrt{a}} + \frac{\sqrt{y}}{\sqrt{b}} = 1 \qquad \ldots (1)$$

diff. (1) w.r.to x, we get

$$\frac{1}{\sqrt{a}}\frac{1}{2\sqrt{x}} + \frac{1}{\sqrt{b}}\frac{1}{2\sqrt{y}}y_1 = 0$$

$$\frac{1}{\sqrt{a}}\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{b}}\frac{1}{\sqrt{y}}y_1 = 0$$

$$\frac{1}{\sqrt{b}}\frac{1}{\sqrt{y}}y_1 = -\frac{1}{\sqrt{a}}\frac{1}{\sqrt{x}}$$

$$y_1 = -\frac{\sqrt{b}}{\sqrt{a}}\frac{\sqrt{y}}{\sqrt{x}} \qquad \ldots (2)$$

$$y_1 = -\frac{\sqrt{b}}{\sqrt{a}}\frac{\sqrt{b}}{\sqrt{x}}\left(\frac{\sqrt{y}}{\sqrt{b}}\right)$$

$$= -\frac{b}{\sqrt{a}}\frac{1}{\sqrt{x}}\left[1 - \frac{\sqrt{x}}{\sqrt{a}}\right] \qquad \text{by (1)}$$

$$= -\frac{b}{\sqrt{a}}\frac{1}{\sqrt{x}} + \frac{b}{a}$$

$$y_2 = \frac{-b}{\sqrt{a}}\frac{-1/2}{x^{3/2}} = \frac{b}{2\sqrt{a}}\frac{1}{x^{3/2}}$$

$$1 + y_1^2 = 1 + \frac{b}{a}\frac{y}{x} = \frac{ax+by}{ax} \qquad \text{by (2)}$$

$$(1+y_1^2)^{3/2} = \frac{(ax+by)^{3/2}}{a^{3/2}-x^{3/2}}$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(ax+by)^{3/2}}{a^{3/2}-x^{3/2}}\left[\frac{2a^{1/2}x^{3/2}}{b}\right] = \frac{2(ax+by)^{3/2}}{ab}$$

$$\text{Curvature} = \frac{1}{\rho} = \frac{ab}{2(ax+by)^{3/2}}$$

Example 4.3.10. Find the radius of curvature at any point (x, y) on the curve $y = \frac{1}{2}a\left(e^{x/a} + e^{-x/a}\right)$.

[AU, Nov. 2001]

Solution : Radius of curvature $\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$

$$y = \frac{a}{2}\left[e^{x/a} + e^{-x/a}\right] = a\left[\frac{e^{x/a}+e^{-x/a}}{2}\right]$$

$$y = a\cosh\left[\frac{x}{a}\right] \qquad \left[\because \cosh x = \frac{e^x + e^{-x}}{2}\right]$$

$$y_1 = a\sinh\left[\frac{x}{a}\right]\frac{1}{a} = \sinh\frac{x}{a}, \quad y_2 = \left[\cosh\frac{x}{a}\right]\left[\frac{1}{a}\right]$$

$$y_1^2 = \sinh^2\frac{x}{a}$$

$$1 + y_1^2 = 1 + \sinh^2\frac{x}{a} \qquad \left[\because \sin^2\theta + \cos^2\theta = 1,\ \cosh^2\theta - \sinh^2\theta = 1\right]$$

$$= \cosh^2\frac{x}{a}$$

$$\left(1+y_1^2\right)^{3/2} = \left(\cosh^2\frac{x}{a}\right)^{3/2} = \cosh^3\frac{x}{a}$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

i.e., $$\rho = \frac{\cosh^3\frac{x}{a}}{\frac{1}{a}\cosh\frac{x}{a}} = a\cosh^2\frac{x}{a}$$

Example 4.3.11. If ρ is the radius of curvature of any point (x, y) on the curve $y = \frac{ax}{a+x}$ show that $\left(\frac{2\rho}{a}\right)^{2/3} = \left(\frac{x}{y}\right)^2 + \left(\frac{y}{x}\right)^2$

[A.U. A/M 2005, N/D 2006] [A.U Jan. 2012] [A.U Jan. 2014, R 2008]

Solution : Given : $y = \frac{ax}{a+x}$, $\Rightarrow \frac{y}{x} = \frac{a}{a+x}$ ... (1)

$$y_1 = \frac{(a+x)(a) - (ax)(1)}{(a+x)^2} = \frac{a^2 + ax - ax}{(a+x)^2} = \frac{a^2}{(a+x)^2}$$

$$\Rightarrow \boxed{y_1 = \left[\frac{a}{a+x}\right]^2 = \left(\frac{y}{x}\right)^2} \text{ by (1)}$$

$$y_2 = \frac{(a+x)^2(0) - a^2 2(a+x)}{(a+x)^4} = \frac{-2a^2}{(a+x)^3}$$

$$\Rightarrow \boxed{y_2 = -\frac{2}{a}\frac{a^3}{(a+x)^3} = -\frac{2}{a}\left(\frac{y}{x}\right)^3 \text{ by (1)}}$$

$$\boxed{y_2^{2/3} = \left(\frac{-2}{a}\right)^{2/3}\left(\frac{y}{x}\right)^2 = \left(\frac{2}{a}\right)^{2/3}\left(\frac{y}{x}\right)^2} \quad \therefore \rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

$$\Rightarrow \rho^{2/3} = \frac{(1+y_1^2)}{(y_2)^{2/3}} = \frac{1 + \left(\frac{y}{x}\right)^4}{\left(\frac{2}{a}\right)^{2/3}\left(\frac{y}{x}\right)^2}$$

$$\left(\frac{2\rho}{a}\right)^{2/3} = \frac{1}{(y/x)^2} + \left(\frac{y}{x}\right)^2 \Rightarrow \left(\frac{2\rho}{a}\right)^{2/3} = \left(\frac{x}{y}\right)^2 + \left(\frac{y}{x}\right)^2$$

II. PARAMETRIC FORM

II. PROBLEMS BASED ON RADIUS OF CURVATURE [PARAMETRIC FORM]

Example 4.3.12. Find ρ at any point $P(at^2, 2at)$ on the parabola $y^2 = 4ax$.

Solution : $x = at^2$, $y = 2at$ (Given)

$$x' = \frac{dx}{dt} = 2at, \; y' = 2a \; ; \qquad x'' = \frac{d^2x}{dt^2} = 2a, \; y'' = 0$$

$$\rho = \frac{[(x')^2 + (y')^2]^{3/2}}{x'y'' - y'x''} = \frac{(4a^2t^2 + 4a^2)^{3/2}}{0 - (2a)(2a)} = -2a(t^2+1)^{3/2}$$

$$|\rho| = 2a(1+t^2)^{3/2}$$

Example 4.3.13. Find the radius of curvature at any point P (a cos θ, b sin θ) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

[A.U. M/J 2006] [A.U. CBT J/F 2010]

Solution : $x = a\cos\theta$, $\quad y = b\sin\theta$ (Given)

$$x' = \frac{dx}{d\theta} = -a\sin\theta, \qquad y' = \frac{dy}{d\theta} = b\cos\theta$$

$$x'' = \frac{d^2x}{d\theta^2} = -a\cos\theta, \qquad y'' = \frac{d^2y}{d\theta^2} = -b\sin\theta$$

$$\rho = \frac{[(x')^2 + (y')^2]^{3/2}}{x'y'' - y'x''}$$

$$= \frac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{(-a\sin\theta)(-b\sin\theta) - (b\cos\theta)(-a\cos\theta)}$$

$$= \frac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{ab\sin^2\theta + ab\cos^2\theta} = \frac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{ab}$$

Example 4.3.14. Find the radius of curvature at any point $x = a\cos^3\theta$, $y = a\sin^3\theta$ of the curve $x^{2/3} + y^{2/3} = a^{2/3}$. Also show that $\rho^3 = 27axy$.

Solution : We have, [A.U. Tvli. A/M 2009] [A.U M/J 2009] [A.U Trichy Jan. 2009, A.U CBT Sep. 2009, Tvli. Jan. 2011] [A.U A/M 2015 R-13]

$$x = a\cos^3\theta \; ... (1) \qquad y = a\sin^3\theta \; ... (2)$$

$$\frac{dx}{d\theta} = 3a\cos^2\theta(-\sin\theta) = -3a\cos^2\theta\sin\theta \qquad \frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta} = -\tan\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{d}{dx}[-\tan\theta] = -\sec^2\theta\frac{d\theta}{dx}$$

$$= \frac{-\sec^2\theta}{\left(\frac{dx}{d\theta}\right)} = \frac{-\sec^2\theta}{-3a\cos^2\theta\sin\theta} = \frac{1}{3a\cos^4\theta\sin\theta}$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+\tan^2\theta)^{3/2}}{\left[\frac{1}{3a\cos^4\theta\sin\theta}\right]}$$

$$= (\sec^2\theta)^{3/2}\, 3a\cos^4\theta\sin\theta = \sec^3\theta\, 3a\cos^4\theta\sin\theta = 3a\sin\theta\cos\theta$$

$$\rho^3 = (3a\sin\theta\cos\theta)^3 = 27a^3\sin^3\theta\cos^3\theta = 27a^3\left(\frac{y}{a}\right)\left(\frac{x}{a}\right) \text{ by (1) \& (2)}$$

$$\rho^3 = 27axy$$

Example 4.3.15. Prove that the radius of curvature at any point of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1-\cos\theta)$ is $4a\cos\frac{\theta}{2}$. [A.U Trichy Jan. 2010]

Solution : Given : [A.U M/J 2010, 2012] [A.U Jan. 2013, Jan. 2014]

$x = a(\theta + \sin\theta)$ | $y = a(1-\cos\theta)$

$x' = \frac{dx}{d\theta} = a(1+\cos\theta)$ | $y' = \frac{dy}{d\theta} = a[0+\sin\theta] = a\sin\theta$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a\sin\theta}{a(1+\cos\theta)} = \frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}$$

$$\frac{dy}{dx} = \tan\frac{\theta}{2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx} = \frac{\frac{d}{d\theta}\left(\tan\frac{\theta}{2}\right)}{\left(\frac{dx}{d\theta}\right)}$$

$$= \frac{\frac{1}{2}\sec^2\frac{\theta}{2}}{a(1+\cos\theta)} = \frac{\frac{1}{2}\sec^2\frac{\theta}{2}}{a2\cos^2\frac{\theta}{2}} = \frac{1}{4a\cos^4\frac{\theta}{2}}$$

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\left(1+\tan^2\frac{\theta}{2}\right)^{3/2}}{\left[\frac{1}{4a\cos^4\frac{\theta}{2}}\right]} = \left[\sec^2\frac{\theta}{2}\right]^{3/2} \times 4a\cos^4\frac{\theta}{2}$$

$$\rho = 4a\cos\frac{\theta}{2}$$

Example 4.3.16. Show that the radius of curvature at any point of the curve $x = ae^\theta(\sin\theta - \cos\theta)$, $y = ae^\theta(\sin\theta + \cos\theta)$ is twice the perpendicular distance of the tangent at the point from the origin. [A.U Tvli. Jan. 2010]

Solution : $x = ae^\theta(\sin\theta - \cos\theta)$ (Given)

$$\frac{dx}{d\theta} = ae^\theta[\cos\theta + \sin\theta] + a(\sin\theta - \cos\theta)e^\theta = 2ae^\theta\sin\theta$$

$y = ae^\theta(\sin\theta + \cos\theta)$ (Given)

$$\frac{dy}{d\theta} = ae^\theta[\cos\theta - \sin\theta] + a(\sin\theta + \cos\theta)e^\theta = 2ae^\theta\cos\theta$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{2ae^\theta\cos\theta}{2ae^\theta\sin\theta} = \cot\theta$$

$$\boxed{\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}}$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}(\cot\theta)\left[\frac{1}{\left(\frac{dx}{d\theta}\right)}\right] = -\operatorname{cosec}^2\theta\left[\frac{1}{2ae^\theta\sin\theta}\right] = \frac{-\operatorname{cosec}^3\theta}{2ae^\theta}$$

$$\rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left(\frac{d^2y}{dx^2}\right)} = \frac{(1+\cot^2\theta)^{3/2}}{\left[\frac{-\operatorname{cosec}^3\theta}{2ae^\theta}\right]} = -2ae^\theta$$

$$|\rho| = 2ae^\theta \quad \ldots (1)$$

Equation of tangent is

$$y - ae^\theta(\sin\theta + \cos\theta) = \frac{\cos\theta}{\sin\theta}[x - ae^\theta(\sin\theta - \cos\theta)]$$

$$y\sin\theta - ae^\theta\sin^2\theta - ae^\theta\sin\theta\cos\theta = x\cos\theta - ae^\theta\sin\theta\cos\theta + ae^\theta\cos^2\theta$$

$$y\sin\theta - x\cos\theta = ae^\theta$$

$\therefore$ The length of the perpendicular from the origin to the tangent is

$$p = \frac{ax_1 + by_1 + c}{\sqrt{a^2+b^2}} = \frac{0\sin\theta - 0\cos\theta - ae^\theta}{\sqrt{\sin^2\theta + \cos^2\theta}} = -ae^\theta$$

$|p| = ae^{\theta}$ ... (2)

From (1) and (2), we get $\rho = 2p$

Radius of Curvature in Polar Co-ordinates

Example 4.3.17. Find the radius of curvature ρ for the cardioid $r = a(1 - \cos\theta)$. Prove $\frac{\rho^2}{r}$ is a constant.

Solution : Given : $r = a(1 - \cos\theta)$

$$\therefore \frac{dr}{d\theta} = a\sin\theta \; ; \quad \frac{d^2r}{d\theta^2} = a\cos\theta$$

$$r^2 + r_1^2 = a^2(1-\cos\theta)^2 + a^2\sin^2\theta = 2a^2(1-\cos\theta) = 4a^2\sin^2\theta/2$$

$$r^2 + 2r_1^2 - r_2 = a^2(1-\cos\theta)^2 + 2a^2\sin^2\theta - a^2\cos\theta(1-\cos\theta)$$

$$= 3a^2(1-\cos\theta) = 6a^2\sin^2(\theta/2)$$

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r_2}$$

$$= \frac{\left(4a^2\sin^2\frac{\theta}{2}\right)^{3/2}}{6a^2\sin^2\theta/2} = \frac{4a}{3}\sin\frac{\theta}{2} = \frac{2}{3}\sqrt{2ar} \quad \text{i.e., } \rho = \frac{2}{3}\sqrt{2ar}$$

$$\therefore \frac{\rho^2}{r} = \text{constant.}$$

Example 4.3.18. For the curve $r^n = a^n\cos n\theta$, prove $\rho = \frac{a^n r^{1-n}}{n+1}$

Solution : Given : $r^n = a^n\cos n\theta$. Taking logarithm and differentiating with respect to θ, we get

$$\frac{n}{r}\cdot\frac{dr}{d\theta} = -n\frac{\sin n\theta}{\cos n\theta}$$

$$r_1 = \frac{dr}{\partial\theta} = -r\tan n\theta$$

$$\frac{d^2r}{d\theta^2} = -nr\sec^2\theta n\theta - \frac{dr}{d\theta}\tan n\theta = r\tan^2 n\theta - nr\sec^2 n\theta$$

$$\rho = \frac{(r^2 + r^2\tan^2 n\theta)^{3/2}}{r^2 + 2r^2\tan^2 n\theta - r^2\tan^2 n\theta + nr^2\sec^2 n\theta}$$

$$= \frac{r^3\sec^3 n\theta}{(n+1)r^2\sec^2 n\theta} = \frac{r}{(n+1)\cos n\theta}$$

$$= \frac{r}{n+1}\cdot\frac{a^n}{r^n} = \frac{a^n}{n+1}r^{1-n}. \quad \text{i.e., } \rho = \frac{a^n}{n+1}r^{1-n}$$

Note : In the above problem put $n = 1$, we get the circle $r = a\cos\theta$ whose ρ is $\frac{a}{2}$; put $n = 2$, we get lemniscate of Bernoulli, $r^2 = a^2\cos 2\theta$, whose ρ is $\frac{a^2}{3r}$

When $n = \frac{1}{2}$, we get cardioid and $\rho = \frac{2}{3}\sqrt{ar}$.

Example 4.3.19. Find the radius of curvature at the pole for the curve $r = a\sin n\theta$.

Solution : Given : $r = a\sin n\theta$

$$r_1 = \frac{dr}{d\theta} = na\cos n\theta \; ; \; r_2 = \frac{d^2r}{d\theta^2} = -n^2 a\sin n\theta$$

$$\rho = \frac{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{3/2}}{r^2 + 2\left(\frac{dr}{d\theta}\right)^2 - r\frac{d^2r}{d\theta^2}}$$

$r_{\text{at }\theta=0}$ is $\frac{n^3a^3}{2n^2a^2} = \frac{na}{2}$

Example 4.3.20. Find the curvature for $r = ae^{\theta\cos\alpha}$.

Solution : Given : $r = ae^{\theta\cos\alpha}$

$$r_1 = a\cot\alpha\,.e^{\theta\cot\alpha} = r\cot\alpha$$

$$r_2 = r\cot^2\alpha$$

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2} = \frac{(r^2 + r^2\cot^2\alpha)^{3/2}}{r^2 + 2r^2\cot^2\alpha - r^2\cot^2\alpha}$$

$$\rho = r\,\text{cosec}\,\alpha.$$

III. IMPLICIT FORM

Example 4.3.21. Find the radius of curvature of the curve given by $x^3 - 2x^2y + 3xy^2 - 4y^3 + 5x^2 - 6xy + 7y^2 - 8y = 0$ at (0, 0).

Solution :

Let $f(x,y) = x^3 - 2x^2y + 3xy^2 - 4y^3 + 5x^2 - 6xy + 7y^2 - 8y$	At (0, 0)
$f_x = 3x^2 - 4xy + 3y^2 + 10x - 6y$	$f_x = 0$
$f_y = -2x^2 + 6xy - 12y^2 - 6x + 14y - 8$	$f_y = -8$
$f_{xx} = 6x - 4y + 10$	$f_{xx} = 10$
$f_{yy} = 6x - 24y + 14$	$f_{yy} = 14$
$f_{xy} = -4x + 6y - 6$	$f_{xy} = -6$

$$\rho = \frac{(f_x^2 + f_y^2)^{3/2}}{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}$$

$$\rho = \frac{[0 + (-8)^2]^{3/2}}{(10)(-8)^2 - 2(-6)(0)(-8) + (14)(0)}$$

$$= \frac{(8^2)^{3/2}}{(10)(8^2)} = \frac{8^3}{(10)(8^2)} = \frac{8}{10} = \frac{4}{5}$$

Example 4.3.22. Find the curvature of the curve $f(x, y) = x^3 + 3xy^2 + 5x^2 + 7y^2 - 6x = 0$ at (0, 0).

Solution :

$f(x,y) = x^3 + 3xy^2 + 5x^2 + 7y^2 - 6x$	At (0, 0)
$f_x = 3x^2 + 3y^2 + 10x - 6$	$f_x = -6$
$f_y = 6xy + 14y$	$f_y = 0$
$f_{xy} = 6y$	$f_{xy} = 0$
$f_{xx} = 6x + 10$	$f_{xx} = 10$
$f_{yy} = 6x + 14$	$f_{yy} = 14$

$$\rho = \frac{\left(f_x^2 + f_y^2\right)^{3/2}}{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2} = \frac{[(-6)^2 + (0)^2]^{3/2}}{(10)(0^2) - 2(0)(-6)(0) + (14)(-6)^2}$$

$$= \frac{[6^2]^{3/2}}{(14)(6^2)} = \frac{6^3}{14(6^2)} = \frac{6}{14} = \frac{3}{7}$$

$$\text{Curvature} = \frac{1}{\text{radius of curvature}} = \frac{1}{\rho} = \frac{1}{(3/7)} = \frac{7}{3}$$

Example 4.3.23 : Find the radius of curvature at (a, 0) of the curve $xy^2 = a^3 - x^3$ **[A.U Jan. 2010] [A.U N/D 2010] [A.U Jan. 2014, R 2008] [A.U N/D 2014 R-13]**

Solution :

Let $f(x,y) = xy^2 - a^3 + x^3$	At $(a, 0)$
$f_x = y^2 + 3x^2$	$f_x = 0^2 + 3a^2 = 3a^2$
$f_y = 2xy$	$f_y = 2(a)(0) = 0$
$f_{xx} = 6x$	$f_{xx} = 6(a) = 6a$
$f_{yy} = 2x$	$f_{yy} = 2(a) = 2a$
$f_{xy} = 2y$	$f_{xy} = 2(0) = 0$

$$\text{Radius of curvature } \rho = \frac{[f_x^2 + f_y^2]^{3/2}}{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}$$

$$= \frac{[(3a^2)^2 + 0^2]^{3/2}}{(6a)(0) - 2(0)(3a^2)(0) + (2a)(3a^2)^2}$$

$$= \frac{[(3a^2)^2]^{3/2}}{(2a)(3a^2)^2} = \frac{(3a^2)^3}{(2a)(3a^2)^2} = \frac{3a^2}{2a} = \frac{3}{2}a$$

Example 4.3.24 : Show that the numerical value of the radius of curvature of the curve $x^2y = a(x^2 + y^2)$ at $(-2a, 2a)$ is $2a$

Solution :

Let $f(x,y) = x^2y - a(x^2 + y^2)$	At $(-2a, 2a)$
$f_x = 2xy - 2ax$	$f_x = 2(-2a)(2a) - 2a(-2a)$ $= -8a^2 + 4a^2 = -4a^2$
$f_y = x^2 - 2ay$	$f_y = (-2a)^2 - 2a(2a)$ $= 4a^2 - 4a^2 = 0$
$f_{xx} = 2y - 2a$	$f_{xx} = 2(2a) - 2a$ $= 4a - 2a = 2a$
$f_{yy} = -2a$	$f_{yy} = -2a$
$f_{xy} = 2x$	$f_{xy} = 2(-2a) = -4a$

$$\text{Radius of curvature } \rho = \frac{[f_x^2 + f_y^2]^{3/2}}{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}$$

$$= \frac{[(-4a^2)^2 + 0^2]^{3/2}}{(2a)(0^2) - 2(-4a)(-4a^2)(0) + (-2a)(-4a^2)^2}$$

$$= \frac{[(4a^2)^2]^{3/2}}{(-2a)[4a^2]^2} = \frac{[4a^2]^3}{(-2a)(4a^2)^2} = \frac{4a^2}{-2a} = -2a$$

$$|\rho| = 2a$$

Hence, the proof.

Example 4.3.25. Find the radius of curvature at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on the curve $x^3 + y^3 = 3axy$ **[AU CBT J/F 2009]**

Solution : **[A.U. CBT J/F 2010] [A.U Madurai Jan. 2011]**

Let $f(x,y) = x^3 + y^3 - 3axy$	At $\left(\frac{3a}{2}, \frac{3a}{2}\right)$
$f_x = 3x^2 - 3ay$	$f_x = 3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right) = \frac{27}{4}a^2 - \frac{9}{2}a^2 = \frac{9}{4}a^2$
$f_y = 3y^2 - 3ax$	$f_y = 3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right) = \frac{9}{4}a^2$
$f_{xx} = 6x$	$f_{xx} = 6\left(\frac{3a}{2}\right) = 9a$
$f_{yy} = 6y$	$f_{yy} = 6\left(\frac{3a}{2}\right) = 9a$
$f_{xy} = -3a$	$f_{xy} = -3a$

$$\text{Radius of curvature } \rho = \frac{[f_x^2 + f_y^2]^{3/2}}{f_{xx}f_y^2 - 2f_xf_xf_y + f_{yy}f_x^2}$$

$$= \frac{\left[\left(\frac{9}{4}a^2\right)^2 + \left(\frac{9}{4}a^2\right)^2\right]^{3/2}}{(9a)\left(\frac{9}{4}a^2\right)^2 - 2(-3a)\left(\frac{9}{4}a^2\right)\left(\frac{9}{4}a^2\right) + (9a)\left(\frac{9}{4}a^2\right)^2}$$

$$= \frac{\left[2\left(\frac{9}{4}a^2\right)^2\right]^{3/2}}{(9a + 6a + 9a)\left(\frac{9}{4}a^2\right)^2}$$

$$= \frac{(2)^{3/2}\left[\left(\frac{9}{4}a^2\right)^2\right]^{3/2}}{(24a)\left(\frac{9}{4}a^2\right)^2} = \frac{2\sqrt{2}\left(\frac{9}{4}a^2\right)^3}{(24a)\left(\frac{9}{4}a^2\right)^2}$$

$$= \frac{2\sqrt{2}\left(\frac{9}{4}a^2\right)}{24a} = \frac{2\sqrt{2}\,9a}{4 \times 24}$$

$$= \frac{3\sqrt{2}\,a}{16} = \frac{3a}{8\sqrt{2}}$$

Example 4.3.26 : Find ρ at (1, 1) on $x^3 + y^3 = 2$

Solution :

Let $f(x,y) = x^3 + y^3 - 2$	at (1, 1)
$f_x = 3x^2$	$f_x = 3$
$f_y = 3y^2$	$f_y = 3$
$f_{xx} = 6x$	$f_{xx} = 6$
$f_{yy} = 6y$	$f_{yy} = 6$
$f_{xy} = 0$	$f_{xy} = 0$

$$\text{Radius of curvature } \rho = \frac{[f_x^2 + f_y^2]^{3/2}}{f_{xx}f_y^2 - 2f_{xy}f_xf_y + f_{yy}f_x^2}$$

$$= \frac{[3^2 + 3^2]^{3/2}}{6(3)^2 - 2(0)(3)(3) + 6(3)^2}$$

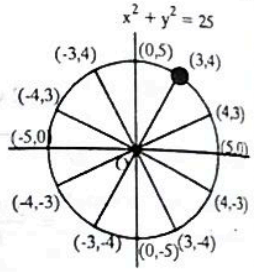
$$= \frac{[9 + 9]^{3/2}}{(6)(9) + (6)(9)} = \frac{18^{3/2}}{108}$$

$$= \frac{(18)\sqrt{18}}{108} = \frac{1}{6}(3)\sqrt{2}$$

$$= \frac{1}{2}(\sqrt{2}) = \frac{1}{\sqrt{2}}$$

Given Curve is a circle :

Example 4.3.27. What is the radius of curvature at (3, 4) on the curve $x^2 + y^2 = 25$? **[A.U CBT Jan. 2011]**



Solution : Given : $x^2 + y^2 = 25$

$$\Rightarrow (x-0)^2 + (y-0)^2 = 5^2$$

which is a circle, whose centre (0, 0) and radius is 5.

$\therefore$ Radius of curvature $\rho = 5$ for any point

$\Rightarrow$ Radius of curvature $\rho = 5$ for (3, 4)

Example 4.3.28. Find the curvature and radius of curvature of

(i) $2x^2 + 2y^2 + 2x - 5y + 1 = 0$ at any point.

(ii) $2x^2 + 2y^2 + 5x - 2y + 1 = 0$ [A.U M/J 2005] [A.U. Jan. 2010]

(iii) $x^2 + y^2 - 4x - 6y + 10 = 0$ [A.U. N/D 2006] [A.U N/D 2014 R-08]

Solution : (i) Given : $2x^2 + 2y^2 + 2x - 5y + 1 = 0$

$$\Rightarrow x^2 + y^2 + x - \frac{5}{2}y + \frac{1}{2} = 0$$

This equation is of the form

$x^2 + y^2 + 2gx + 2fy + c = 0$, it represents a circle.

$$r = \sqrt{g^2 + f^2 - c}$$

Here, $g = \frac{1}{2}$, $f = -\frac{5}{4}$, $c = \frac{1}{2}$

$$\therefore r = \sqrt{\frac{1}{4} + \frac{25}{16} - \frac{1}{2}}$$

$$= \sqrt{\frac{4 + 25 - 8}{16}} = \sqrt{\frac{21}{16}} = \frac{\sqrt{21}}{4}$$

$\Rightarrow$ Radius of curvature $\rho = \frac{\sqrt{21}}{4}$

$\Rightarrow$ Curvature $= \frac{1}{\rho} = \frac{4}{\sqrt{21}}$

(ii) Given : $2x^2 + 2y^2 + 5x - 2y + 1 = 0$

$$\Rightarrow x^2 + y^2 + \frac{5}{2}x - y + \frac{1}{2} = 0$$

This equation is of the form

$x^2 + y^2 + 2gx + 2fy + c = 0$, it represents a circle.

$$r = \sqrt{g^2 + f^2 - c}$$

Here, $g = \frac{5}{4}$, $f = -\frac{1}{2}$, $c = \frac{1}{2}$

$$\therefore r = \sqrt{\frac{25}{16} + \frac{1}{4} - \frac{1}{2}}$$

$$= \sqrt{\frac{25 + 4 - 8}{16}}$$

$$= \frac{\sqrt{21}}{4}$$

$\therefore$ Radius of curvature $\rho = \frac{\sqrt{21}}{4}$

$\Rightarrow$ Curvature $= \frac{1}{\rho} = \frac{4}{\sqrt{21}}$

(iii) Given : $x^2 + y^2 - 4x - 6y + 10 = 0$

This equation is of the form

$x^2 + y^2 + 2gx + 2fy + c = 0$, it represents a circle.

$$r = \sqrt{g^2 + f^2 - c}$$

Here, $g = -2$, $f = -3$, $c = 10$

$$\therefore r = \sqrt{4 + 9 - 10}$$

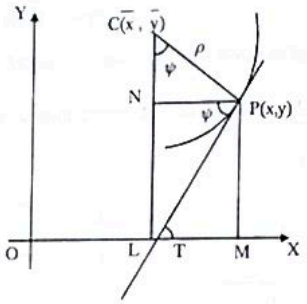
$$= \sqrt{3}$$

$\therefore$ Radius of curvature $\rho = \sqrt{3}$

$\Rightarrow$ Curvature $= \frac{1}{\rho} = \frac{1}{\sqrt{3}}$

◆ 4.3 (c) Centre of Curvature - Coordinates

Let P (x, y) be any point on the curve $y = f(x)$, having radius at curvature ρ. Draw the tangent and the normal at P. Let C $(\bar{x}, \bar{y})$ be the point on the inward normal such that CP $= \rho$.

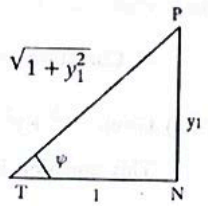


Draw perpendiculars CL and PM on the x axis and also PN is drawn perpendicular to CL.

Let ψ be the angle made by the tangent at P with the x-axis it can be easily seen from the figure that $\angle NCP = \psi$.

Now, from the Δ CNP, $\sin\psi = \frac{NP}{\rho}$ and $\cos\psi = \frac{CN}{\rho}$

(i.e.,) $NP = \rho \sin\psi$; $CN = \rho\cos\psi$



Also, we have

$\bar{x} = OL = OM - LM$

$= x - NP = x - \rho\sin\psi$ ... (1)

$\bar{y} = CL = NL + CN$

$= PM + CN = y + \rho\cos\psi$ ... (2)

Also, we have $\tan\psi = \frac{dy}{dx} = y_1 = \frac{y_1}{1}$

Observe the Δ PMT, we have

$$\sin\psi = \frac{y_1}{\sqrt{1+y_1^2}} \; ; \quad \cos\psi = \frac{1}{\sqrt{1+y_1^2}}$$

$$(1) \Rightarrow \bar{x} = x - \frac{(1+y_1^2)^{3/2}}{y_2}\cdot\frac{y_1}{\sqrt{1+y_1^2}} = x - \frac{y_1}{y_2}(1+y_1^2)$$

$$\bar{y} = y + \frac{(1+y_1^2)^{3/2}}{y_2}\cdot\frac{1}{\sqrt{1+y_1^2}} = y + \frac{1+y_1^2}{y_2}$$

Thus, $(\bar{x}, \bar{y}) = \left[x - \frac{y_1}{y_2}(1+y_1^2),\ y + \frac{1+y_1^2}{y_2}\right]$

This is the expression for the co-ordinates of the centre of curvature in the Cartesian form.

Note. $(\bar{x}, \bar{y})$ can also be obtained in the form.

$$(\bar{x}, \bar{y}) = \left[x + \frac{1+x_1^2}{x_2},\ y - \frac{x_1}{x_2}(1+x_1^2)\right] \text{ when } x_1 = \frac{dx}{dy},\ x_2 = \frac{d^2x}{dy^2}$$

III. PROBLEMS BASED ON CENTRE OF CURVATURE

Example 4.3.29. Find the centre of curvature of $y = x^2$ at the origin.

[A.U. CBT J/F 2010] [A.U M/J 2014]

Solution : Formula : $\bar{x} = x - \frac{y_1}{y_2}(1+y_1^2)$; $\bar{y} = y + \frac{1+y_1^2}{y_2}$

$y = x^2$	at (0, 0)
$y_1 = 2x$	$y_1 = 0$
$y_2 = 2$	$y_2 = 2$

$$\therefore \bar{x} = 0 - \frac{0}{2}(1+0) = 0$$

$$\bar{y} = 0 + \frac{1+0}{2} = 0 + \frac{1}{2} = \frac{1}{2}$$

$\therefore$ Centre of curvature is $\left(0, \frac{1}{2}\right)$.

Example 4.3.30. Find the centre of curvature at any point $(am^2, 2am)$ on the parabola $y^2 = 4ax$.

Solution :

$y = 2\sqrt{a}\sqrt{x}$	at $(am^2, 2am)$
$y_1 = 2\sqrt{a}\frac{1}{2\sqrt{x}} = \frac{\sqrt{a}}{\sqrt{x}}$	$y_1 = \frac{\sqrt{a}}{\sqrt{am^2}} = \frac{1}{m}$
$y_2 = \frac{-\sqrt{a}}{2x\sqrt{x}}$	$y_2 = \frac{-\sqrt{a}}{2am^2\sqrt{am^2}} = \frac{-1}{2am^3}$

$$\therefore \bar{x} = x - \frac{y_1}{y_2}\left(1 + y_1^2\right)$$

$$= am^2 + \frac{\left(\frac{1}{m}\right)}{\left(\frac{1}{2am^3}\right)}\left(1 + \frac{1}{m^2}\right)$$

$$= am^2 + 2am^2\left(1 + \frac{1}{m^2}\right)$$

$$= am^2 + 2am^2 + 2a$$

$$= 3am^2 + 2a$$

$$\bar{y} = y + \frac{\left(1 + y_1^2\right)}{y_2} = 2am + \frac{1 + \left(\frac{1}{m}\right)^2}{\left(\frac{-1}{2am^3}\right)}$$

$$= 2am + \left(1 + \frac{1}{m^2}\right)(-2a\,m^3)$$

$$= 2am - 2am^3 - 2am = -2\,am^3$$

The centre of curvature is $(3am^2 + 2a, -2am^3)$.

Example 4.3.31. Show that the line joining any point θ on $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ to its centre of curvature is bisected by the line $y = 2a$.

Solution : $x = a(\theta + \sin\theta)$ $\qquad y = a(1 - \cos\theta)$

$$\frac{dx}{d\theta} = a(1 + \cos\theta) \qquad \frac{dy}{d\theta} = a[\sin\theta]$$

$$= a\,2\cos^2\frac{\theta}{2} \qquad = 2a\sin\frac{\theta}{2}.\cos\frac{\theta}{2}$$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{2a\sin\frac{\theta}{2}.\cos\frac{\theta}{2}}{2a\cos^2\frac{\theta}{2}} = \tan\frac{\theta}{2}$$

$$\frac{d^2y}{dx^2} = \left(\sec^2\frac{\theta}{2}\right)\frac{1}{2}\frac{d\theta}{dx} = \frac{1}{2}\sec^2\frac{\theta}{2}\frac{1}{2a\cos^2\frac{\theta}{2}}$$

$$y_2 = \frac{d^2y}{dx^2} = \frac{1}{4a}\sec^4\frac{\theta}{2}$$

Mid-point of (x, y) and $(\bar{x}, \bar{y})$ is $\left[\frac{x + \bar{x}}{2}, \frac{y + \bar{y}}{2}\right]$

$$\bar{y} = y + \frac{1 + y_1^2}{y_2}$$

$$\frac{y + \bar{y}}{2} = \frac{1}{2}\left[y + y + \frac{1 + y_1^2}{y_2}\right]$$

$$= \frac{1}{2}\left[2y + \frac{1 + y_1^2}{y_2}\right] = y + \frac{1 + y_1^2}{2y_2}$$

$$= a(1 - \cos\theta) + \frac{1 + \tan^2\frac{\theta}{2}}{2\left[\frac{1}{4a}\sec^4\frac{\theta}{2}\right]}$$

$$= a(1 - \cos\theta) + \frac{\sec^2\frac{\theta}{2}}{\frac{1}{2a}\sec^4\frac{\theta}{2}}$$

$$= a(1 - \cos\theta) + 2a\left[\frac{1}{\sec^2\frac{\theta}{2}}\right]$$

$$= a(1 - \cos\theta) + 2a\cos^2\frac{\theta}{2}$$

$$= a(1 - \cos\theta) + a[1 + \cos\theta]$$

$$= a[1 - \cos\theta + 1 + \cos\theta] = 2a$$

y co-ordinate of the mid-point $= 2a$

$\therefore$ It lies on the line $y - 2a = 0$.

EXERCISE 4.3

1. Find the radius of curvature at (x,y) for the curve $a^2y = x^3 - a^3$.

[Ans. $\rho = \dfrac{(a^4 + 9x^4)^{3/2}}{6a^4x}$]

2. Find the radius of curvature for the curve $y^2 = x^3 + 8$ at $(-2, 0)$.

[Ans. $\rho = \dfrac{(1+0)^{3/2}}{\left(\frac{1}{6}\right)} = 6$]

3. Show that at any point on the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$, the radius of curvature is twice the normal at that point.

4. Find the radius of curvature at any point θ on the curve $x = 3a\cos\theta - a\cos 3\theta$, $y = 3a\sin\theta - a\sin 3\theta$. **[A.U A/M 2011]**

[Ans. $\rho = 3a\sin\theta$]

5. Show that the radius of curvature at the origin of the curve $ax + by + lx^2 + 2mxy + ny^2 = 0$ is $\dfrac{\left(a^2 + b^2\right)^{3/2}}{2(b^2 l - 2abm + a^2 n)}$

6. Prove that the chord of curvature parallel to the y axis for the curve $y = a\log\sec\left(\frac{x}{a}\right)$ is of constant length.

7. Find the radius of curvature at any point on $x = e^t\cos t$, $y = e^t\sin t$

[Ans. $\rho = e^t\sqrt{2}$] **[A.U M/J 2014]**

8. Find the radius of curvature for the curve $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$. [Ans. $\rho = at$] **[A.U M/J 2013]**

9. Find the radius of curvature of $r = a\sin n\theta$ at the origin.

[Ans. $\frac{an}{2}$]

10. Find the position on $y^2 = 8x$ at which the curvature is $\frac{16}{125}$

11. For the curve $r^n = a^n\cos n\theta$, prove $\rho = \dfrac{a^n r^{1-n}}{n+1}$

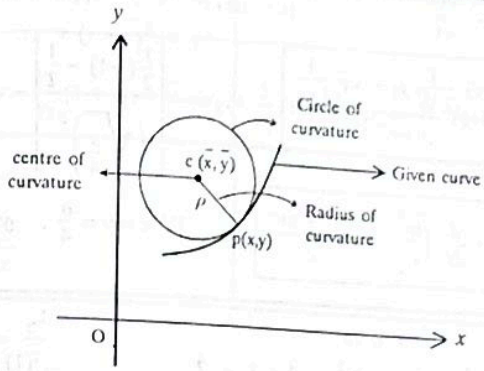
12. Find the curvature for $r = ae^{\theta\cot\alpha}$ [Ans. $r\operatorname{cosec}\alpha$]

13. Find ρ at $(-2, 0)$ on $y^2 = x^3 + 8$. [Ans. $\rho = 6$]

4.4 CIRCLE OF CURVATURE

◆ 4.4.(a) Circle of Curvature

The centre of curvature for a point $P(x, y)$ on a curve is the point $C(\bar{x}, \bar{y})$ which lies on the inward normal at P and is at a distance ρ from it.



The circle of curvature for a point $P(x, y)$ on a curve is the circle whose centre is at the point $C(\bar{x}, \bar{y})$ and whose radius is ρ. The equation of the circle of curvature is $(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$.

IV. PROBLEMS BASED ON CIRCLE OF CURVATURE

Example 4.4.1 Find the circle of curvature of the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point $\left(\frac{a}{4}, \frac{a}{4}\right)$.

[A.U. May, 2003] [A.U. CBT J/F 2009] [A.U. Trichy Jan. 2009] [A.U M/J 2010, N/D 2010, A/M 2011] [A.U. CBT Jan. 2011] [A.U CBT A/M 2011] [A.U. Jan. 2012, M/J 2012] [A.U A/M 2015 R-08] [A.U N/D 2014 R-13] [A.U D15/J16 R-13, R-08] [A.U M/J 2016 R-08, R-13]

Solution : The given curve is not a circle.

∴ The equation of the circle of curvature is

$$(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2,$$

where $\bar{x} = x - \dfrac{y_1}{y_2}(1 + y_1^2)$, $\bar{y} = y + \dfrac{1}{y_2}(1 + y_1^2)$, $\rho = \dfrac{1}{y_2}\left(1 + y_1^2\right)^{3/2}$

Given : $\sqrt{x} + \sqrt{y} = \sqrt{a}$

$\Rightarrow \sqrt{y} = \sqrt{a} - \sqrt{x}$

$\sqrt{y} = \sqrt{a} - \sqrt{x}$	at $\left(\frac{a}{4}, \frac{a}{4}\right)$
$\frac{1}{2\sqrt{y}} y_1 = 0 - \frac{1}{2\sqrt{x}}$ $\Rightarrow y_1 = -\frac{\sqrt{y}}{\sqrt{x}}$	$y_1 = -1 \quad [\because x = y = \frac{a}{4}]$
$y_2 = -\left[\frac{\sqrt{x}\frac{1}{2\sqrt{y}} y_1 - \sqrt{y}\frac{1}{2\sqrt{x}}}{(\sqrt{x})^2}\right]$ Formula : $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{vu' - uv'}{v^2}$	$y_2 = -\left[\frac{\frac{1}{2}(-1) - \frac{1}{2}}{\left(\sqrt{\frac{a}{4}}\right)^2}\right] = -\left[\frac{-1}{\frac{a}{4}}\right] = \frac{4}{a}$ $[\because x = y = \frac{a}{4}, \quad y_1 = -1]$

$$1 + y_1^2 = 1 + (-1)^2 = 1 + 1 = 2 \qquad \ldots (1)$$

$$\frac{1}{y_2} = \frac{1}{\left(\frac{4}{a}\right)} = \frac{a}{4} \qquad \ldots (2)$$

$$\frac{y_1}{y_2} = \frac{-1}{(4/a)} = -\frac{a}{4} \qquad \ldots (3)$$

$$\rho = \left(\frac{1}{y_2}\right)\left(1 + y_1^2\right)^{3/2} = \left(\frac{a}{4}\right)(2)^{3/2} = \left(\frac{a}{4}\right)(2\sqrt{2}) = \frac{a}{\sqrt{2}} \quad \text{by (1) \& (2)}$$

$$\bar{x} = x - \left(\frac{y_1}{y_2}\right)(1 + y_1^2) = \frac{a}{4} - \left(\frac{-a}{4}\right)(2) = \frac{a}{4} + \frac{2a}{4} = \frac{3a}{4} \quad \text{by (1) \& (3)}$$

$$\bar{y} = y + \left(\frac{1}{y_2}\right)(1 + y_1^2) = \frac{a}{4} + \left(\frac{a}{4}\right)(2) = \frac{a}{4} + \frac{2a}{4} = \frac{3a}{4} \quad \text{by (1) \& (2)}$$

Hence, the equation of the circle of curvature is

$$(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$$

$$\Rightarrow \left(x - \frac{3a}{4}\right)^2 + \left(y - \frac{3a}{4}\right)^2 = \left(\frac{a}{\sqrt{2}}\right)^2$$

Example 4.4.2. Find the equation of the circle of curvature at (c, c) on $xy = c^2$. **[A.U. A/M 2005] [A.U. N/D 2014 R-08]**

Solution : The given curve is not a circle.

∴ The equation of the circle of curvature is

$$(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2,$$

where $\bar{x} = x - \frac{y_1}{y_2}(1 + y_1^2)$, $\bar{y} = y + \frac{1}{y_2}(1 + y_1^2)$, $\rho = \frac{1}{y_2}\left(1 + y_1^2\right)^{3/2}$

Given : $xy = c^2$

$$\Rightarrow y = \frac{c^2}{x}$$

$y = \frac{c^2}{x}$	at (c, c)
$y_1 = -\frac{c^2}{x^2}$	$y_1 = \frac{-c^2}{c^2} = -1$
$y_2 = \frac{2c^2}{x^3}$	$y_2 = \frac{2c^2}{c^3} = \frac{2}{c}$

$$1 + y_1^2 = 1 + (-1)^2 = 1 + 1 = 2 \qquad \ldots (1)$$

$$\frac{1}{y_2} = \frac{1}{\left(\frac{2}{c}\right)} = \frac{c}{2} \qquad \ldots (2)$$

$$\frac{y_1}{y_2} = \frac{-1}{\left(\frac{2}{c}\right)} = -\frac{c}{2} \qquad \ldots (3)$$

$$\rho = \left(\frac{1}{y_2}\right)\left(1 + y_1^2\right)^{3/2} = \left(\frac{c}{2}\right)(2)^{3/2} = \frac{c}{2}(2\sqrt{2}) = \sqrt{2}\,c \quad \text{by (1) \& (2)}$$

$$\bar{x} = x - \left(\frac{y_1}{y_2}\right)(1 + y_1^2) = c - \left(\frac{-c}{2}\right)(2) = c + c = 2c \quad \text{by (1) \& (3)}$$

$$\bar{y} = y + \left(\frac{1}{y_2}\right)(1 + y_1^2) = c + \left(\frac{c}{2}\right)(2) = c + c = 2c \quad \text{by (1) \& (2)}$$

Hence, the required circle of curvature is

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$$

$$(x-2c)^2 + (y-2c)^2 = (c\sqrt{2})^2$$

Example 4.4.3 Find the equation of the circle of curvature of $\sqrt{x} + \sqrt{y} = 1$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$ **[AU, Nov. 2001]**

Solution :

∴ The equation of the circle of curvature is

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2 ,$$

where $\bar{x} = x - \frac{y_1}{y_2}(1+y_1^2)$, $\bar{y} = y + \frac{1}{y_2}(1+y_1^2)$, $\rho = \frac{1}{y_2}\left(1+y_1^2\right)^{3/2}$

Given : $\sqrt{x} + \sqrt{y} = 1$

$\Rightarrow \sqrt{y} = 1 - \sqrt{x}$

$\sqrt{y} = 1 - \sqrt{x}$	at $\left(\frac{1}{4}, \frac{1}{4}\right)$
$\frac{1}{2\sqrt{y}} y_1 = 0 - \frac{1}{2\sqrt{x}}$ $\Rightarrow y_1 = -\frac{\sqrt{y}}{\sqrt{x}}$	$y_1 = -1$ $\quad [\because x = y = \frac{1}{4}]$
$y_2 = -\left[\frac{\sqrt{x}\,\frac{1}{2\sqrt{y}} y_1 - \sqrt{y}\,\frac{1}{2\sqrt{x}}}{(\sqrt{x})^2}\right]$	$y_2 = -\left[\frac{\frac{1}{2}(-1) - \frac{1}{2}}{\left(\sqrt{\frac{1}{4}}\right)^2}\right] = -\left[\frac{-1}{(1/4)}\right] = 4$ $[\because x = y = \frac{1}{4}$, y_1 at $\left(\frac{1}{4}, \frac{1}{4}\right) = -1]$

$$1 + y_1^2 = 1 + (-1)^2 = 1 + 1 = 2 \quad \ldots (1)$$

$$\frac{1}{y_2} = \frac{1}{4} \quad \ldots (2)$$

$$\frac{y_1}{y_2} = -\frac{1}{4} \quad \ldots (3)$$

$$\rho = \left(\frac{1}{y_2}\right)\left(1+y_1^2\right)^{3/2} = \left(\frac{1}{4}\right)(2)^{3/2} = \left(\frac{1}{4}\right)(2\sqrt{2}) = \frac{1}{\sqrt{2}} \text{ by (1) \& (2)}$$

$$\bar{x} = x - \left(\frac{y_1}{y_2}\right)(1+y_1^2) = \frac{1}{4} - \left(\frac{-1}{4}\right)(2) = \frac{1}{4} + \frac{2}{4} = \frac{3}{4} \text{ by (1) \& (3)}$$

$$\bar{y} = y + \left(\frac{1}{y_2}\right)(1+y_1^2) = \frac{1}{4} + \left(\frac{1}{4}\right)(2) = \frac{1}{4} + \frac{2}{4} = \frac{3}{4} \text{ by (1) \& (2)}$$

Hence, the equation of the circle of curvature is

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$$

$$\left(x - \frac{3}{4}\right)^2 + \left(y - \frac{3}{4}\right)^2 = \left(\frac{1}{\sqrt{2}}\right)^2$$

Example 4.4.4. Find the equation of circle of curvature of the rectangular hyperbola xy = 12 at the point (3, 4) **[A.U Jan. 2010]**

Solution :

∴ The equation of the circle of curvature is

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2 ,$$

where $\bar{x} = x - \frac{y_1}{y_2}(1+y_1^2)$, $\bar{y} = y + \frac{1}{y_2}(1+y_1^2)$, $\rho = \frac{1}{y_2}\left(1+y_1^2\right)^{3/2}$

Given : $xy = 12$

$\Rightarrow y = \frac{12}{x}$

$y = \frac{12}{x}$	*at* (3, 4)
$y_1 = \frac{-12}{x^2}$	$y_1 = \frac{-12}{(3)^2} = \frac{-12}{9} = \frac{-4}{3}$
$y_2 = \frac{24}{x^3}$	$y_2 = \frac{24}{3^3} = \frac{24}{27} = \frac{8}{9}$

$$1 + y_1^2 = 1 + \left(\frac{-4}{3}\right)^2 = 1 + \frac{16}{9} = \frac{25}{9} \quad \ldots (1)$$

$$\frac{1}{y_2} = \frac{1}{\left(\frac{8}{9}\right)} = \frac{9}{8} \quad \ldots (2)$$

$$\frac{y_1}{y_2} = \frac{(-4/3)}{(8/9)} = \left(\frac{-4}{3}\right)\left(\frac{9}{8}\right) = -\frac{3}{2} \quad \ldots (3)$$

$$\rho = \left(\frac{1}{y_2}\right)(1+y_1^2)^{3/2} = \left(\frac{9}{8}\right)\left(\frac{25}{9}\right)^{3/2} = \left(\frac{9}{8}\right)\left(\frac{25\sqrt{25}}{9\sqrt{9}}\right)$$

$$= \left(\frac{9}{8}\right)\left[\frac{(25)(5)}{(9)(3)}\right] = \frac{125}{24} \quad \text{by (1) \& (2)}$$

$$\bar{x} = x - \left(\frac{y_1}{y_2}\right)(1+y_1^2) = 3 - \left(\frac{-3}{2}\right)\left(\frac{25}{9}\right) = 3 + \frac{25}{6} = \frac{43}{6} \quad \text{by (1) \& (3)}$$

$$\bar{y} = y + \left(\frac{1}{y_2}\right)(1+y_1^2) = 4 + \left(\frac{9}{8}\right)\left(\frac{25}{9}\right) = 4 + \frac{25}{8} = \frac{57}{8} \quad \text{by (1) \& (2)}$$

Hence, the equation of circle of curvature is

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$$

$$\Rightarrow \quad \left(x - \frac{43}{6}\right)^2 + \left(y - \frac{57}{8}\right)^2 = \left(\frac{125}{24}\right)^2$$

EXERCISE 4.4

1. Find the equation of the circle of curvature at the point (2, 3) on $\frac{x^2}{4} + \frac{y^2}{9} = 2$.

   [Ans. $\left(x + \frac{5}{4}\right)^2 + \left(y - \frac{5}{6}\right)^2 = \frac{13^2}{12^2}$**] [A.U M/J 2014]**

2. Find the equation of the circle of curvature of the parabola $y^2 = 12x$ at the point (3, 6). **[Ans.** $(x-15)^2 + (y+6)^2 = 288$**]**

   [A.U. Jan. 2009, Tvli. M/J 2011, Madurai Jan. 2011]

4.5 EVOLUTES

♦ 4.5 (a) Involutes and Evolutes [A.U M/J 2014] [A.U N/D 2014 R-13] [A.U D15 / J16 R-13] [A.U M/J 2016 R-08, R-13]

The locus of the centre of curvature of the given curve is called the evolute of the curve. The given curve is called the involute of its evolute.

♦ 4.5 (b) Working procedure to find the evolute of a curve

Step 1. We take the equation of the curve in the parametric form $x = x(t)$, $y = y(t)$ and compute y_1 and y_2 interms of t.

Step 2. We consider $(\bar{x}, \bar{y})$ in the Cartesian form and simplify the term $\alpha = \frac{1+y_1^2}{y_2}$ as it is common in both $\bar{x}, \bar{y}$. Thus, we obtain

$$\bar{x} = x - \frac{y_1}{y_2}(1+y_1^2) = x - y_1\alpha$$

$$\bar{y} = y + \frac{1}{y_2}(1+y_1^2) = y + \alpha$$

Step 3. We eliminate the parameter t from $\bar{x}$ and $\bar{y}$ to obtain the equation in the form $g(\bar{x}, \bar{y}) = c$.

Step 4. Taking the locus of $(\bar{x}, \bar{y})$, the required evolute is $g(x, y) = c$.

The following basic information regarding standard curves will be highly useful.

Curve	Cartesian equation	Parametric equations
Parabola	1. $y^2 = 4ax$ 2. $x^2 = 4ay$	1. $x = at^2, y = 2at$ 2. $x = 2at, y = at^2$
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$x = a\cos\theta, y = b\sin\theta$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$x = a\sec\theta, y = b\tan\theta$
Rectangular hyperbola	$xy = c^2$	$x = ct, y = \frac{c}{t}$
Astroid	$x^{2/3} + y^{2/3} = a^{2/3}$	$x = a\cos^3\theta, y = a\sin^3\theta$

♦ 4.5 (c) Evolute of a curve in two ways

Define evolute of a curve in two ways.

Evolute of a curve is the locus of the centre of curvature at any general point on the curve.

Also, evolute of a curve is the envelope of its normals.

V. PROBLEMS BASED ON EVOLUTES

Example 4.5.1. State two properties of the evolute of a curve.

Sol. (i) The normal at any point of a curve touches the evolute at the corresponding centre of curvature.

(ii) The length of an arc of the evolute is equal to the difference between the radii of curvature at the points on the original curve corresponding to the extremities of the arc.

Example 4.5.2. Find the equation of the evolute of the parabola $y^2 = 4ax$.

[A.U. Tvli. A/M 2009, A.U. Jan. 2010, A.U M/J 2010, CBT. Jan. 2011] [A.U Jan. 2011, Jan. 2012, M/J 2012, Jan. 2013] [A.U N/D 2014 R-08] [A.U D15 / J16 R-13, R-08] [A.U M/J 2016, R-13]

Solution : Given : $y^2 = 4ax$

The parametric equations are $x = at^2$, $y = 2at$

$x = at^2$	$y = 2at$
$\frac{dx}{dt} = 2at$	$\frac{dy}{dt} = 2a$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{2a}{2at} = \frac{1}{t}$$

$$y_2 = \frac{d^2y}{dx^2} = -\frac{1}{t^2}\frac{dt}{dx} = -\frac{1}{t^2}\frac{1}{\left(\frac{dx}{dt}\right)} = -\frac{1}{t^2}\left(\frac{1}{2at}\right) = -\frac{1}{2at^3}$$

$\frac{1}{y_2} = -2at^3$, $1 + y_1^2 = 1 + \frac{1}{t^2}$
$\frac{1}{y_2}(1 + y_1^2) = -2at^3\left(1 + \frac{1}{t^2}\right) = -2at^3 - 2at$ ... (1)
$\frac{y_1}{y_2}(1 + y_1^2) = y_1\left[\frac{1}{y_2}(1 + y_1^2)\right] = \frac{1}{t}[-2at^3 - 2at] = -2at^2 - 2a$ ... (2)

If (X, Y) is the centre of curvature at 't', then

$X = x - \frac{y_1}{y_2}(1 + y_1^2)$	$Y = y + \frac{1}{y_2}(1 + y_1^2)$
$X = at^2 + 2at^2 + 2a$ by (2) $= 3at^2 + 2a$ ... (3)	$Y = 2at - 2at^3 - 2at$ by (1) $= -2at^3$ ... (4)

Eliminate 't' between (3) & (4),

(3) $\Rightarrow 3at^2 = X - 2a$	(4) $\Rightarrow -2at^3 = Y$
$t^2 = \frac{X - 2a}{3a}$	$t^3 = -\frac{Y}{2a}$
$(t^2)^3 = \left(\frac{X - 2a}{3a}\right)^3$	$(t^3)^2 = \left(\frac{-Y}{2a}\right)^2$
$t^6 = \frac{(X - 2a)^3}{27a^3}$ ... (5)	$t^6 = \frac{Y^2}{4a^2}$ ... (6)

From (5) & (6), we get

$$\frac{(X - 2a)^3}{27a^3} = \frac{Y^2}{4a^2} \Rightarrow 4(X - 2a)^3 = 27aY^2$$

Locus of (X, Y) is $4(x - 2a)^3 = 27ay^2$

which is the equation of evolute of the given curve.

Example 4.5.3. Find the equation of the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

[A.U CBT J/F 2010, Tvli M/J 2011] [A.U June 2011] [A.U M/J 2014 R-08]

Solution :

Given : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The parametric equations are $x = a\cos\theta$, $y = b\sin\theta$

$x = a\cos\theta$	$y = b\sin\theta$
$\frac{dx}{d\theta} = -a\sin\theta$	$\frac{dy}{d\theta} = b\cos\theta$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{b\cos\theta}{-a\sin\theta} = -\frac{b\cos\theta}{a\sin\theta} = \frac{-b}{a}\cot\theta$$

$$y_2 = \frac{d^2y}{dx^2} = \frac{-b}{a}[-\operatorname{cosec}^2\theta]\frac{d\theta}{dx}$$

$$= \frac{b}{a}\frac{1}{\sin^2\theta}\frac{1}{\left(\frac{dx}{d\theta}\right)}$$

$$= \frac{b}{a}\frac{1}{\sin^2\theta}\left[\frac{1}{-a\sin\theta}\right] = -\frac{b}{a^2\sin^3\theta}$$

$\frac{1}{y_2} = -\frac{a^2}{b}\sin^3\theta$, $\quad 1+y_1^2 = 1+\frac{b^2\cos^2\theta}{a^2\sin^2\theta}$

$$\frac{1}{y_2}(1+y_1^2) = \left(\frac{-a^2}{b}\sin^3\theta\right)\left(1+\frac{b^2}{a^2}\frac{\cos^2\theta}{\sin^2\theta}\right) = -\frac{a^2}{b}\sin^3\theta - b\cos^2\theta\sin\theta \quad \ldots (1)$$

$$\frac{y_1}{y_2}(1+y_1^2) = y_1\left[\frac{1}{y_2}(1+y_1^2)\right] = \left(\frac{-b}{a}\frac{\cos\theta}{\sin\theta}\right)\left[\frac{-a^2}{b}\sin^3\theta - b\cos^2\theta\sin\theta\right]$$

$$= a\cos\theta\sin^2\theta + \frac{b^2}{a}\cos^3\theta \quad \ldots (2)$$

If (X, Y) is the centre of curvature of θ, then

$X = x - \frac{y_1}{y_2}(1+y_1^2)$	$Y = y + \frac{1}{y_2}(1+y_1^2)$
$X = a\cos\theta - a\cos\theta\sin^2\theta - \frac{b^2}{a}\cos^3\theta$ by(2)	$Y = b\sin\theta - \frac{a^2}{b}\sin^3\theta - b\cos^2\theta\sin\theta$ by(1)
$= a\cos\theta(1-\sin^2\theta) - \frac{b^2}{a}\cos^3\theta$	$= b\sin\theta(1-\cos^2\theta) - \frac{a^2}{b}\sin^3\theta$
$= a\cos\theta(\cos^2\theta) - \frac{b^2}{a}\cos^3\theta$	$= b\sin\theta(\sin^2\theta) - \frac{a^2}{b}\sin^3\theta$
$= a\cos^3\theta - \frac{b^2}{a}\cos^3\theta$	$= b\sin^3\theta - \frac{a^2}{b}\sin^3\theta$
$= \left(\frac{a^2-b^2}{a}\right)\cos^3\theta \quad \ldots (3)$	$= \left(\frac{b^2-a^2}{b}\right)\sin^3\theta \quad \ldots (4)$
Eliminate 'θ' between (3) & (4),	
(3) $\Rightarrow \cos^3\theta = \frac{aX}{a^2-b^2}$	(4) $\Rightarrow \sin^3\theta = \frac{bY}{b^2-a^2} = \frac{-bY}{a^2-b^2}$
$(\cos^3\theta)^{2/3} = \frac{(aX)^{2/3}}{(a^2-b^2)^{2/3}}$	$(\sin^3\theta)^{2/3} = \frac{(-bY)^{2/3}}{(a^2-b^2)^{2/3}}$
$\cos^2\theta = \frac{(aX)^{2/3}}{(a^2-b^2)^{2/3}} \ldots (5)$	$\sin^2\theta = \frac{(bY)^{2/3}}{(a^2-b^2)^{2/3}} \quad \ldots (6)$

From (5) & (6), we get

$$\cos^2\theta + \sin^2\theta = \frac{(aX)^{2/3}}{(a^2-b^2)^{2/3}} + \frac{(bY)^{2/3}}{(a^2-b^2)^{2/3}}$$

$$1 = \frac{(aX)^{2/3} + (bY)^{2/3}}{(a^2-b^2)^{2/3}}$$

i.e., $(aX)^{2/3} + (bY)^{2/3} = (a^2-b^2)^{2/3}$

Locus of (X, Y) is $(ax)^{2/3} + (by)^{2/3} = (a^2-b^2)^{2/3}$

which is the equation of evolute of the given curve.

Example 4.5.4. Find the evolute of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

[A.U. Tvli. Jan. 2011] [A.U. N/D 2010, June 2011] [A.U A/M 2015 R-08]

Solution : Given : $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **[A.U N/D 2015 R-13] [A.U M/J 2016, R-08]**

The parametric equations are $x = a\sec\theta$, $y = b\tan\theta$

$x = a\sec\theta$	$y = b\tan\theta$
$\frac{dx}{d\theta} = a\sec\theta\tan\theta$	$\frac{dy}{d\theta} = b\sec^2\theta$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{b\sec^2\theta}{a\sec\theta\tan\theta} = \frac{b\sec\theta}{a\tan\theta} = \frac{b}{a\sin\theta}$$

$$y_2 = \frac{d^2y}{dx^2} = \left[\frac{0 - b(a\cos\theta)}{a^2\sin^2\theta}\right]\left(\frac{d\theta}{dx}\right)$$

$$= \frac{-b\cos\theta}{a\sin^2\theta}\frac{1}{\left(\frac{dx}{d\theta}\right)}$$

$$= \frac{-b\cos\theta}{a\sin^2\theta}\left[\frac{1}{a\sec\theta\tan\theta}\right]$$

$$= -\frac{b}{a^2\tan^3\theta} = -\frac{b}{a^2}\frac{\cos^3\theta}{\sin^3\theta}$$

$$\frac{1}{y_2} = -\frac{a^2}{b}\tan^3\theta,\ 1 + y_1^2 = 1 + \frac{b^2\sec^2\theta}{a^2\tan^2\theta}$$

$$\frac{1}{y_2}(1+y_1^2) = \left(\frac{-a^2}{b}\tan^3\theta\right)\left(1 + \frac{b^2\sec^2\theta}{a^2\tan^2\theta}\right)$$

$$= -\frac{a^2}{b}\tan^3\theta - b\tan\theta\sec^2\theta \quad \ldots (1)$$

$$\frac{y_1}{y_2}(1+y_1^2) = y_1\left[\frac{1}{y_2}(1+y_1^2)\right] = \left(\frac{b\sec\theta}{a\tan\theta}\right)\left(-\frac{a^2}{b}\tan^3\theta - b\tan\theta\sec^2\theta\right)$$

$$= -a\sec\theta\tan^2\theta - \frac{b^2}{a}\sec^3\theta \quad \ldots (2)$$

If (X, Y) is the centre of curvature at 'θ', then

$$X = x - \frac{y_1}{y_2}(1+y_1^2) \qquad Y = y + \frac{1}{y_2}(1+y_1^2)$$

$$X = a\sec\theta + a\sec\theta\tan^2\theta + \frac{b^2}{a}\sec^3\theta \quad \text{by (2)}$$

$$= a\sec\theta(1+\tan^2\theta) + \frac{b^2}{a}\sec^3\theta$$

$$= a\sec\theta\sec^2\theta + \frac{b^2}{a}\sec^3\theta$$

$$= a\sec^3\theta + \frac{b^2}{a}\sec^3\theta$$

$$= \left(\frac{a^2+b^2}{a}\right)\sec^3\theta \quad \ldots (3)$$

$$Y = b\tan\theta - \frac{a^2}{b}\tan^3\theta - b\tan\theta\sec^2\theta \quad \text{by (1)}$$

$$= -b\tan\theta[\sec^2\theta - 1] - \frac{a^2}{b}\tan^3\theta$$

$$= -b\tan\theta\tan^2\theta - \frac{a^2}{b}\tan^3\theta$$

$$= -b\tan^3\theta - \frac{a^2}{b}\tan^3\theta$$

$$= \left(\frac{-b^2-a^2}{b}\right)\tan^3\theta$$

$$= -\left(\frac{a^2+b^2}{b}\right)\tan^3\theta \quad \ldots (4)$$

Eliminate 'θ' between (3) & (4),

$$(3) \Rightarrow \sec^3\theta = \frac{aX}{a^2+b^2}$$

$$(\sec^3\theta)^{2/3} = \frac{(aX)^{2/3}}{(a^2+b^2)^{2/3}}$$

$$\sec^2\theta = \frac{(aX)^{2/3}}{(a^2+b^2)^{2/3}} \quad \ldots (5)$$

$$(4) \Rightarrow \tan^3\theta = \frac{-bY}{a^2+b^2}$$

$$(\tan^3\theta)^{2/3} = \frac{(bY)^{2/3}}{(a^2+b^2)^{2/3}}$$

$$\tan^2\theta = \frac{(bY)^{2/3}}{(a^2+b^2)^{2/3}} \quad \ldots (6)$$

From (5) & (6), we get

$$\sec^2\theta - \tan^2\theta = \frac{(aX)^{2/3}}{(a^2+b^2)^{2/3}} - \frac{(bY)^{2/3}}{(a^2+b^2)^{2/3}}$$

$$1 = \frac{(aX)^{2/3} - (bY)^{2/3}}{(a^2+b^2)^{2/3}} \quad [\because \sec^2\theta - \tan^2\theta = 1]$$

$$\Rightarrow (aX)^{2/3} - (bY)^{2/3} = (a^2+b^2)^{2/3}$$

Locus of (X, Y) is $(ax)^{2/3} - (by)^{2/3} = (a^2+b^2)^{2/3}$

which is the equation of the evolute of the given curve.

Example 4.5.5. Find the evolute of the rectangular hyperbola $xy = c^2$.

[A.U Tvli. Jan. 2010], [A.U. CBT A/M 2011] [A.U. Madurai Jan. 2011]

Solution :

Given : $xy = c^2$

The parametric equations are $x = ct,\ y = \frac{c}{t}$

$$x = ct \qquad y = \frac{c}{t}$$

$$\frac{dx}{dt} = c \qquad \frac{dy}{dt} = -\frac{c}{t^2}$$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{\left(\frac{-c}{t^2}\right)}{c} = -\frac{1}{t^2}$$

$$y_2 = \frac{d^2y}{dx^2} = \frac{2}{t^3}\frac{dt}{dx} = \frac{2}{t^3}\frac{1}{\left(\frac{dx}{dt}\right)} = \frac{2}{t^3}\left(\frac{1}{c}\right) = \frac{2}{ct^3}$$

$$\frac{1}{y_2} = \frac{ct^3}{2}, \qquad 1 + y_1^2 = 1 + \frac{1}{t^4}$$

$$\frac{1}{y_2}(1+y_1^2) = \left(\frac{c}{2}t^3\right)\left(1 + \frac{1}{t^4}\right) = \frac{c}{2}t^3 + \frac{c}{2t} \quad \ldots (1)$$

$$\frac{y_1}{y_2}(1+y_1^2) = y_1\left[\frac{1}{y_2}(1+y_1^2)\right] = \left(\frac{-1}{t^2}\right)\left(\frac{c}{2}t^3 + \frac{c}{2t}\right) = \frac{-ct}{2} - \frac{c}{2t^3} \quad \ldots (2)$$

If (X, Y) is the centre of curvature at 't', then

$X = x - \frac{y_1}{y_2}(1 + y_1^2)$	$Y = y + \frac{1}{y_2}(1 + y_1^2)$
$X = ct + \frac{ct}{2} + \frac{c}{2t^3}$ by (2)	$Y = \frac{c}{t} + \frac{c}{2}t^3 + \frac{c}{2t}$ by (1)
$= \frac{3}{2}ct + \frac{c}{2t^3}$ ... (3)	$= \frac{3}{2}\frac{c}{t} + \frac{c}{2}t^3$ ... (4)

Eliminate 't' between (3) & (4), we get

$X + Y = \frac{3ct}{2} + \frac{c}{2t^3} + \frac{3c}{2t} + \frac{ct^3}{2}$	$X - Y = \frac{3ct}{2} + \frac{c}{2t^3} - \frac{3c}{2t} - \frac{ct^3}{2}$
$= \frac{c}{2}\left[3t + \frac{1}{t^3} + \frac{3}{t} + t^3\right]$	$= \frac{c}{2}\left[3t + \frac{1}{t^3} - \frac{3}{t} - t^3\right]$
$= \frac{c}{2}\left[t + \frac{1}{t}\right]^3$	$= -\frac{c}{2}\left[t^3 - 3t + \frac{3}{t} - \frac{1}{t^3}\right]$
$(X + Y)^{2/3} = \left(\frac{c}{2}\right)^{2/3}\left(t + \frac{1}{t}\right)^2$ ... (5)	$(X - Y) = \left(-\frac{c}{2}\right)\left(t - \frac{1}{t}\right)^3$
	$(X - Y)^{2/3} = \left(\frac{c}{2}\right)^{2/3}\left(t - \frac{1}{t}\right)^2$ ... (6)

From (5) & (6), we get

$$(X + Y)^{2/3} - (X - Y)^{2/3} = \left(\frac{c}{2}\right)^{2/3}\left[\left(t + \frac{1}{t}\right)^2 - \left(t - \frac{1}{t}\right)^2\right]$$

$$= \left(\frac{c}{2}\right)^{2/3}\left[\left(t^2 + \frac{1}{t^2} + 2\right) - \left(t^2 + \frac{1}{t^2} - 2\right)\right]$$

$$= \left(\frac{c}{2}\right)^{2/3}[4]$$

$$= (4c)^{2/3}$$

Locus of (X, Y) is $(x + y)^{2/3} - (x - y)^{2/3} = (4c)^{2/3}$

which is the equation of the evolute of the given curve.

Note : $\frac{4}{2^{2/3}} = \frac{4}{4^{1/3}} = 4^{1 - \frac{1}{3}} = 4^{2/3}$

Example 4.5.6. Find the equation of the evolute of the curve $x^{2/3} + y^{2/3} = a^{2/3}$.

Solution :

Given : $x^{2/3} + y^{2/3} = a^{2/3}$

The parametric equations are $x = a\cos^3\theta$, $y = a\sin^3\theta$

$x = a\cos^3\theta$	$y = a\sin^3\theta$
$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta$	$\frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta} = -\frac{\sin\theta}{\cos\theta} = -\tan\theta$$

$$y_2 = \frac{d^2y}{dx^2} = -\sec^2\theta\frac{d\theta}{dx} = -\sec^2\theta\frac{1}{\left(\frac{dx}{d\theta}\right)} = -\sec^2\theta\left(\frac{1}{-3a\cos^2\theta\sin\theta}\right)$$

$$= \frac{1}{3a\sin\theta\cos^4\theta}$$

$\frac{1}{y_2} = 3a\sin\theta\cos^4\theta$, $1 + y_1^2 = 1 + \tan^2\theta = \sec^2\theta = \frac{1}{\cos^2\theta}$
$\frac{1}{y_2}(1 + y_1^2) = (3a\sin\theta\cos^4\theta)\left(\frac{1}{\cos^2\theta}\right) = 3a\sin\theta\cos^2\theta$ ... (1)
$\frac{y_1}{y_2}(1 + y_1^2) = y_1\left[\frac{1}{y_2}(1 + y_1^2)\right] = \left(\frac{-\sin\theta}{\cos\theta}\right)(3a\sin\theta\cos^2\theta)$
$= -3a\sin^2\theta\cos\theta$ ... (2)

If (X, Y) is the centre of curvature at 'θ', then

$X = x - \frac{y_1}{y_2}(1 + y_1^2)$	$Y = y + \frac{1}{y_2}(1 + y_1^2)$
$X = a\cos^3\theta + 3a\sin^2\theta\cos\theta$ by (2)	$Y = a\sin^3\theta + 3a\sin\theta\cos^2\theta$ by (1)

Eliminate 'θ' between X and Y, we get

$X + Y = a[\cos^3\theta + 3\sin^2\theta\cos\theta + 3\sin\theta\cos^2\theta + \sin^3\theta]$	$X - Y = a[\cos^3\theta + 3\sin^2\theta\cos\theta - 3\sin\theta\cos^2\theta - \sin^3\theta]$
$= a(\cos\theta + \sin\theta)^3$	$= a(\cos\theta - \sin\theta)^3$
$(X+Y)^{2/3} = a^{2/3}(\cos\theta + \sin\theta)^2$	$(X-Y)^{2/3} = a^{2/3}(\cos\theta - \sin\theta)^2$
$= a^{2/3}[\cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta]$	$= a^{2/3}[\cos^2\theta + \sin^2\theta - 2\sin\theta\cos\theta]$
$= a^{2/3}[1 + \sin 2\theta]$ ... (3)	$= a^{2/3}[1 - \sin 2\theta]$ ... (4)

From (3) & (4), we get

$$(X+Y)^{2/3} + (X-Y)^{2/3} = a^{2/3}\ [2]$$

Locus of (X, Y) is $(x+y)^{2/3} + (x-y)^{2/3} = 2a^{2/3}$

which is the equation of the evolute of the given curve.

Example 4.5.7. Show that the evolute of the cycloid $x = a(\theta - \sin\theta)$ $y = a(1 - \cos\theta)$ is another equal cycloid.

[A.U. Madurai June 2011] [A.U. A/M 2011] [A.U N/D 2014 R-13]

Solution : **[A.U A/M 2015 R-13]**

Given :

$x = a(\theta - \sin\theta)$	$y = a(1 - \cos\theta)$
$\frac{dx}{d\theta} = a(1 - \cos\theta)$	$\frac{dy}{d\theta} = a(\sin\theta)$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a\sin\theta}{a(1-\cos\theta)} = \frac{\sin\theta}{1-\cos\theta}$$

$$y_2 = \frac{d^2y}{dx^2} = \frac{(1-\cos\theta)(\cos\theta) - \sin\theta(\sin\theta)}{(1-\cos\theta)^2}\left(\frac{d\theta}{dx}\right)$$

$$= \frac{\cos\theta - \cos^2\theta - \sin^2\theta}{(1-\cos\theta)^2}\frac{1}{\left(\frac{dx}{d\theta}\right)}$$

$$= \frac{\cos\theta - (\cos^2\theta + \sin^2\theta)}{(1-\cos\theta)^2}\frac{1}{\left(\frac{dx}{}\right)}$$

$$= \frac{\cos\theta - 1}{(1-\cos\theta)^2}\frac{1}{a(1-\cos\theta)} = -\frac{1}{a(1-\cos\theta)^2}$$

$\frac{1}{y_2} = -a(1-\cos\theta)^2,\ (1+y_1^2) = 1 + \frac{\sin^2\theta}{(1-\cos\theta)^2}$	
$\frac{1}{y_2}(1+y_1^2)$	$= [-a(1-\cos\theta)^2]\left[1 + \frac{\sin^2\theta}{(1-\cos\theta)^2}\right]$
	$= -a(1-\cos\theta)^2 - a\sin^2\theta$
	$= -a[1 + \cos^2\theta - 2\cos\theta + \sin^2\theta]$
	$= -a[2 - 2\cos\theta] = -2a[1-\cos\theta]$ ... (1)
$\frac{y_1}{y_2}(1+y_1^2)$	$= y_1\left[\frac{1}{y_2}(1+y_1^2)\right] = \left(\frac{\sin\theta}{1-\cos\theta}\right)[-2a(1-\cos\theta)]$
	$= -2a\sin\theta$ ... (2)

If (X, Y) is the centre of curvature at 'θ', then

$X = x - \frac{y_1}{y_2}(1+y_1^2)$	$Y = y + \frac{1}{y_2}(1+y_1^2)$
$X = a\theta - a\sin\theta + 2a\sin\theta$ by (2)	$Y = a(1-\cos\theta) - 2a(1-\cos\theta)$ by (1)
$= a\theta + a\sin\theta$	$= -a(1-\cos\theta)$ ... (4)
$= a(\theta + \sin\theta)$ ... (3)	

Elimination of θ from (3) & (4) is very difficult.

Locus of (X, Y) is given by the parametric equations.

$x = a(\theta + \sin\theta)$ and $y = -a(1 - \cos\theta)$

$\therefore$ Clearly, this represents the equation of another cycloid.

Example 4.5.8. Find the evolute of the tractrix $x = a\left(\cos t + \log\tan\frac{t}{2}\right)$, $y = a\sin t$.

Solution : Given : $x = a\left(\cos t + \log\tan\frac{t}{2}\right)$

$$\frac{dx}{dt} = a\left[-\sin t + \frac{1}{\tan\frac{t}{2}}\sec^2\frac{t}{2}\left(\frac{1}{2}\right)\right]$$

$$= a\left[-\sin t + \frac{1}{2\sin\frac{t}{2}\cos\frac{t}{2}}\right] = a\left[-\sin t + \frac{1}{\sin t}\right]$$

$$= a\left[\frac{1-\sin^2 t}{\sin t}\right] = a\left[\frac{\cos^2 t}{\sin t}\right]$$

Given : $y = a\sin t \Rightarrow \frac{dy}{dt} = a\cos t$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{a\cos t}{\left(\frac{a\cos^2 t}{\sin t}\right)} = \frac{\sin t}{\cos t} = \tan t$$

$$y_2 = \frac{d^2y}{dx^2} = sec^2 t\left(\frac{dt}{dx}\right) = sec^2 t\,\frac{1}{\left(\frac{dx}{dt}\right)}$$

$$= \frac{1}{\cos^2 t}\,\frac{1}{\left(\frac{a\cos^2 t}{\sin t}\right)} = \frac{\sin t}{a\cos^4 t}$$

$\frac{1}{y_2} = \frac{a\cos^4 t}{\sin t}$, $1 + y_1^2 = 1 + \tan^2 t = sec^2 t = \frac{1}{\cos^2 t}$

$$\frac{1}{y_2}(1+y_1^2) = \left(\frac{a\cos^4 t}{\sin t}\right)\left(\frac{1}{\cos^2 t}\right) = \frac{a\cos^2 t}{\sin t} \quad \ldots (1)$$

$$\frac{y_1}{y_2}(1+y_1^2) = y_1\left[\frac{1}{y_2}(1+y_1^2)\right] = \left(\frac{\sin t}{\cos t}\right)\left(\frac{a\cos^2 t}{\sin t}\right) = a\cos t \quad \ldots (2)$$

If (X, Y) is the centre of curvature at 't', then

$$X = x - \frac{y_1}{y_2}(1+y_1^2), \quad Y = y + \frac{1}{y_2}(1+y_1^2)$$

$$X = a\cos t + a\log\tan\frac{t}{2} - a\cos t \text{ by (2)}$$

$$= a\log\tan\frac{t}{2}$$

$$\frac{X}{a} = \log\tan\frac{t}{2}$$

$$\Rightarrow \quad e^{\frac{X}{a}} = \tan\frac{t}{2} \Rightarrow e^{\frac{-X}{a}} = \frac{1}{\tan\frac{t}{2}},$$

Formula: $\frac{e^{\frac{X}{a}} + e^{\frac{-X}{a}}}{2} = \cosh\frac{X}{a}$

$$\cosh\frac{X}{a} = \frac{\tan\frac{t}{2} + \frac{1}{\tan\frac{t}{2}}}{2} = \frac{\tan^2\frac{t}{2}+1}{2\tan\frac{t}{2}}$$

$$= \frac{1}{\sin t} \quad \ldots (3)$$

$\because \sin t = \frac{2\tan t/2}{1+\tan^2\frac{t}{2}}$

$$Y = a\sin t + a\frac{\cos^2 t}{\sin t} \text{ by (1)}$$

$$= a\left[\frac{\sin^2 t + \cos^2 t}{\sin t}\right] = \frac{a}{\sin t}$$

$$\frac{Y}{a} = \frac{1}{\sin t} \quad \ldots (4)$$

From (3) & (4), we get

$$\frac{Y}{a} = \cosh\frac{X}{a}$$

$$Y = a\cosh\frac{X}{a} \quad \text{[Catenary]}$$

Locus of (X, Y) is $y = a\cosh\frac{x}{a}$

which is the equation of the evolute of the given curve.

Example 4.5.9. Show that the evolute of the curve $x = a(\cos\theta + \theta\sin\theta)$, $y = a(\sin\theta - \theta\cos\theta)$ is a circle. **[A.U. M/J 2009] [A.U Jan. 2014, R2008]**

Solution :

Given :

$x = a(\cos\theta + \theta\sin\theta)$	$y = a(\sin\theta - \theta\cos\theta)$
$\frac{dx}{d\theta} = a(-\sin\theta + \theta\cos\theta + \sin\theta)$	$\frac{dy}{d\theta} = a[\cos\theta - (-\theta\sin\theta + \cos\theta)]$
$= a\theta\cos\theta$	$= a[\cos\theta + \theta\sin\theta - \cos\theta]$
	$= a\theta\sin\theta$

$$y_1 = \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a\theta\sin\theta}{a\theta\cos\theta} = \tan\theta$$

$$y_2 = \frac{d^2y}{dx^2} = sec^2\theta\frac{d\theta}{dx} = sec^2\theta\frac{1}{\left(\frac{dx}{d\theta}\right)}$$

$$= sec^2\theta\left(\frac{1}{a\theta\cos\theta}\right)$$

$$= \frac{1}{a\theta\cos^3\theta}$$

$\frac{1}{y_2} = a\theta\cos^3\theta$,	$1 + y_1^2 = 1 + \tan^2\theta = sec^2\theta = \frac{1}{\cos^2\theta}$
$\frac{1}{y_2}(1+y_1^2) = (a\theta\cos^3\theta)\left(\frac{1}{\cos^2\theta}\right) = a\theta\cos\theta$	... (1)
$\frac{y_1}{y_2}(1+y_1^2) = y_1\left[\frac{1}{y_2}(1+y_1^2)\right] = \left(\frac{\sin\theta}{\cos\theta}\right)(a\theta\cos\theta) = a\theta\sin\theta$	... (2)

If (X, Y) is the centre of curvature at 'θ', then

$X = x - \frac{y_1}{y_2}(1+y_1^2)$	$Y = y + \frac{1}{y_2}(1+y_1^2)$
$X = a\cos\theta + a\theta\sin\theta - a\theta\sin\theta$ by (2)	$Y = a\sin\theta - a\theta\cos\theta + a\theta\cos\theta$ by (1)
$X = a\cos\theta$	$Y = a\sin\theta$
$X^2 = a^2\cos^2\theta$ ... (3)	$Y^2 = a^2\sin^2\theta$ ... (4)

Eliminate 'θ' between (3) & (4), we get

$$X^2 + Y^2 = a^2[\cos^2\theta + \sin^2\theta] = a^2$$

Locus of (X, Y) is $x^2 + y^2 = a^2$ [circle]

which is the equation of the evolute of the given curve.

EXERCISE 4.5

1. Find the evolute of the parabola $x^2 = 4ay$. [Ans. $27ax^2 = 4(y - 2a)^3$]] **[A.U Trichy Jan 2010, M/J 2013]**
2. Find the equation in parametric form of the evolute of the curve $x = 2\cos t + \cos 2t$, $y = 2\sin t + \sin 2t$.
   [Ans. $x = \frac{2}{3}\cos t - \frac{1}{3}\cos 2t$, $y = \frac{2}{3}\sin t - \frac{1}{3}\sin 2t$**]**
3. Find the evolute of $\sqrt{x} + \sqrt{y} = \sqrt{a}$
   [Ans. Parametric equation to the evolute are
   $x = a\cos^4\theta + 2a\sin^2\theta\cos^4\theta + 2a\sin^6\theta$
   $y = a\sin^4\theta + 2a\sin^4\theta\cos^2\theta + 2a\cos^6\theta$
   (or) $108a(xy - 2a^2) = -9a(x + y - 3a)^2 - 8(x + y - 3a)^3 + 108a^2(x + y - 3a)$ **[A.U M/J 2014]**

4.6 ENVELOPES

♦ 4.6 (a) Envelopes

[A.U. N/D 2015 R-13]

A curve which touches each member of a family of curve is called the envelope of that family curves.

♦ Method of obtaining Envelope

1. The envelope of the family of curves $f(x, y, \alpha) = 0$ which is rewritten as a quadratic equation $A\alpha^2 + B\alpha + C = 0$ where A, B, C are functions of x and y, is given by $B^2 - 4AC = 0$
   Note : $B^2 - 4AC = 0$ is the magic formula to find envelope!
2. Envelope is obtained generally by eliminating the parameter α between the equation of the given family curves
   $f(x, y, \alpha) = 0$ ... (1) and $\frac{\partial f}{\partial \alpha} = 0$ ... (2)
3. For a given two parameter family of curves $f(x, y, \alpha, \beta) = 0$ ... (1) with a given relation $g(\alpha, \beta) = 0$ ... (2) between the parameters α, β (1) can be reduced to a one parameter family by elimination of one of the parameters say β interms of α by using the relation $g(\alpha, \beta)$. Then proceed asusual.
4. In case α cannot be eliminated between $f(x, y, \alpha) = 0$ ... (1) and $\frac{\partial f}{\partial \alpha} = 0$ ... (2) then solve (1) and (2) for x and y interms of α.
   Then, the envelope is given in the parametric form by the equations $x = x(\alpha)$ and $y = y(\alpha)$.
5. Envelope of the family of normals to a given curve C is the evolute of the curve C.

VI. PROBLEMS BASED ON ENVELOPES

Example 4.6.1. Define envelope of a family of curves.

Sol. The locus of the ultimate points of intersection of consecutive members of a family of curves is called the envelope of the family of curves.

Example 4.6.2. Give the analytical method of finding the equation to the envelope of the family of curves, whose equation is $f(x, y, c) = 0$ where c is a parameter.

Sol. The envelope is got by eliminating c between the equations $f(x, y, c) = 0$ and $\frac{\partial}{\partial c} f(x, y, c) = 0$.

VI. (a) Problems under $B^2 - 4AC = 0$ which is the equation of the envelope of the family $A\alpha^2 + B\alpha + C = 0$ where α is the parameter.

Example 4.6.3. Find the envelope of the family of straight lines $y = mx + am^2$, m being the parameter.

Solution : Given $y = mx + am^2$

$$am^2 + mx - y = 0$$

which is a quadratic equation in 'm'

So, the envelope is $B^2 - 4AC = 0$ ... (1) [Here, $A = a$, $B = x$, $C = -y$]

(1) $\Rightarrow$ $x^2 - 4a(-y) = 0$

$$x^2 + 4ay = 0$$

Example 4.6.4. Find the envelope of $y = mx + \frac{1}{m}$ where m is the parameter.

[A.U Tvli. Jan. 2010] [A.U Jan. 2010] [A.U. Madurai June 2011] [A.U Jan. 2014] [A.U M/J 2016, R-13]

Solution : Given $y = mx + \frac{1}{m}$

$$my = m^2x + 1$$

$m^2x - my + 1 = 0$, which is a quadratic equation in the parameter 'm'.

So, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = x$, $B = -y$, $C = 1$]

(1) $\Rightarrow$ $(-y)^2 - 4x = 0$

i.e., $y^2 = 4x$.

Example 4.6.5. Find the envelope of the family $x^2(x-a)+(x+a)(y-m)^2=0$ where a is a constant and m is a parameter.

Solution : The given equation is

$$x^2(x-a) + (x+a)(y^2 - 2my + m^2) = 0$$

$$m^2(x+a) - 2my(x+a) + [y^2(x+a) + x^2(x-a)] = 0,$$

which is a quadratic equation in the parameter 'm'.

So, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = x + a$, $B = -2y(x+a)$, $C = y^2(x+a) + x^2(x-a)$]

(1) $\Rightarrow$ $[-2y(x+a)]^2 - 4(x+a)[y^2(x+a) + x^2(x-a)] = 0$

$$4y^2(x+a)^2 - 4y^2(x+a)^2 - 4x^2(x^2 - a^2) = 0$$

$$-4x^2(x^2 - a^2) = 0$$

i.e., $x^2(x^2 - a^2) = 0$

Example 4.6.6. Find the envelope of the family of lines $y = mx + \frac{a}{m}$ where a is a constant. **[A.U M/J 2009] [A.U Jan. 2010], A.U CBT J/F 2010]**

Solution : Given : $y = mx + \frac{a}{m}$ **[A.U. CBT A/M 2011] [A.U M/J 2014 R-08] [A.U A/M 2015 R-08]**

$$\Rightarrow ym = m^2x + a$$

$m^2x - ym + a = 0$, which is a quadratic equation in the parameter 'm'.

So, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = x$, $B = -y$, $C = a$]

(1) $\Rightarrow$ $y^2 - 4xa = 0$

i.e., $y^2 = 4ax$

Example 4.6.7. Find the envelope of the family of straight lines $x\cos\alpha + y\sin\alpha = a\sec\alpha$, α being the parameter.

Solution : Given $x\cos\alpha + y\sin\alpha = a\sec\alpha$

$\div \cos\alpha \Rightarrow$ $x + y\tan\alpha = a\sec^2\alpha$

$$x + y\tan\alpha = a(1 + \tan^2\alpha)$$

$$a\tan^2\alpha - y\tan\alpha + a - x = 0,$$

which is a quadratic equation in $\tan\alpha$.

$\therefore$ The envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = a$, $B = -y$, $C = a - x$]

(1) $\Rightarrow$ $(-y)^2 - 4a(a - x) = 0$

i.e., $y^2 = 4a(a - x)$.

Example 4.6.8. Find the envelope of $y = mx + \sqrt{a^2 m^2 + b^2}$ where m is a parameter. [A.U. Jan. 2009] [A.U. D15 / J16 R-13]

Solution : $y - mx = \sqrt{a^2 m^2 + b^2}$ [A.U Madurai Jan. 2011]

$(y - mx)^2 = a^2 m^2 + b^2$

$y^2 + m^2 x^2 - 2mxy = a^2 m^2 + b^2$

$m^2 (x^2 - a^2) - 2m xy + y^2 - b^2 = 0$,

which is a quadratic equation in the parameter 'm'.

Hence, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = (x^2 - a^2)$, $B = -2xy$, $C = y^2 - b^2$]

(1) $\Rightarrow$ $4x^2 y^2 - 4(x^2 - a^2)(y^2 - b^2) = 0$

$x^2 y^2 - (x^2 - a^2)(y^2 - b^2) = 0$

$x^2 y^2 - [x^2 y^2 - b^2 x^2 - a^2 y^2 + a^2 b^2] = 0$

$\Rightarrow$ $b^2 x^2 + a^2 y^2 = a^2 b^2$

$\div a^2 b^2$ $\Rightarrow$ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Example 4.6.9. Find the envelope of $(x - \alpha)^2 + y^2 = 4\alpha$, α being the parameter.

Solution : Given : $(x - \alpha)^2 + y^2 = 4\alpha$ [A.U N/D 2014 R-08]

$x^2 + \alpha^2 - 2x\alpha + y^2 - 4\alpha = 0$

$\alpha^2 - \alpha(2x + 4) + x^2 + y^2 = 0$

which is a quadratic equation in the parameter 'α'.

So, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = 1$, $B = -(2x + 4)$, $C = x^2 + y^2$]

(1) $\Rightarrow (2x + 4)^2 - 4(1)(x^2 + y^2) = 0$

$4x^2 + 16 + 16x - 4x^2 - 4y^2 = 0$

$16 + 16x - 4y^2 = 0$

$4y^2 = 16 + 16x$

$y^2 = 4 + 4x$

$y^2 = 4(1 + x)$

Example 4.6.10. Find the envelope of $y = mx + \sqrt{1 + m^2}$ where m is a parameter. [A.U Madurai Jan. 2011]

Solution : Given : $y = mx + \sqrt{1 + m^2}$

$y - mx = \sqrt{1 + m^2}$

Squaring on both sides

$(y - mx)^2 = 1 + m^2$

$y^2 + m^2 x^2 - 2mxy = 1 + m^2$

$m^2 (x^2 - 1) - 2mxy + y^2 - 1 = 0$

which is a quadratic equation in the parameter 'm'.

So, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = x^2 - 1$, $B = -2xy$, $C = y^2 - 1$]

(1) $\Rightarrow$ $(-2xy)^2 - 4(x^2 - 1)(y^2 - 1) = 0$

$4x^2 y^2 - 4(x^2 - 1)(y^2 - 1) = 0$

$x^2 y^2 - [x^2 y^2 - x^2 - y^2 + 1] = 0$

$x^2 y^2 - x^2 y^2 + x^2 + y^2 - 1 = 0$

$x^2 + y^2 = 1$ which is a circle.

Example 4.6.11 Find the envelope of the family of curves $y = x \tan\alpha - \frac{gx^2}{2u^2 \cos^2\alpha}$, α being the parameter.

Solution : Given : $y = x \tan x - \frac{gx^2}{2u^2} \sec^2\alpha$

$y = x \tan\alpha - \frac{gx^2}{2u^2}(1 + \tan^2\alpha)$

$$\frac{gx^2}{2u^2}(1+\tan^2\alpha) = x\tan\alpha - y$$

$$gx^2 + gx^2\tan^2\alpha = 2u^2 x\tan\alpha - 2u^2 y$$

$$gx^2\tan^2\alpha - 2u^2 x\tan\alpha + gx^2 + 2u^2 y = 0,$$

which is a quadratic equation in $\tan\alpha$

So, the envelope is $B^2 - 4AC = 0$ ... (1)

[Here, $A = gx^2$, $B = -2u^2 x$, $C = gx^2 + 2u^2 y$]

$$(1) \Rightarrow (-2u^2 x)^2 - 4[gx^2][gx^2 + 2u^2 y] = 0$$

$$4u^4 x^2 - 4g^2x^4 - 8gx^2u^2 y = 0$$

$$\div 4x^2 \Rightarrow \quad u^4 - g^2x^2 - 2gu^2 y = 0$$

$$u^4 - 2gu^2 y = g^2x^2$$

$$2gu^2 y = u^4 - g^2x^2$$

$$y = \frac{u^4}{2gu^2} - \frac{g^2x^2}{2gu^2}$$

$$y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}$$

$$y + \frac{gx^2}{2u^2} = \frac{u^2}{2g}$$

Example 4.6.12. Find the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ where the parameters a, b are related by $ab = c^2$, where c is known.

(OR)

Find the envelope of straight lines which move such that the product of the intercepts on the two axes is constant.

Solution : Given : $b = \frac{c^2}{a}$.

Hence, the straight line becomes

$$\frac{x}{a} + \frac{ay}{c^2} = 1, \text{ eliminating } b$$

$$c^2x + a^2y = ac^2$$

$$a^2y - ac^2 + c^2x = 0 \text{ is a quadratic in } a.$$

Hence, the envelope is $B^2 - 4AC = 0$ [Here, $A = y$, $B = -c^2$, $C = c^2x$]

$$(-c^2)^2 - 4y(c^2x) = 0$$

$$c^4 - 4yc^2x = 0$$

i.e., $$c^2 - 4yx = 0$$

$$4xy = c^2$$

Example 4.6.13. Find the envelope of $x\sec^2\theta + y\,\text{cosec}^2\theta = a$ where θ is the parameter. **[A.U. N/D 2015 R-13]**

Solution : Given : $x\sec^2\theta + y\,\text{cosec}^2\theta = a$ ... (1)

i.e., $$x[1+\tan^2\theta] + y[1+\cot^2\theta] = a$$

$$x + x\tan^2\theta + y + y\cot^2\theta = a$$

$$x + x\tan^2\theta + y + y\frac{1}{\tan^2\theta} = a$$

$$(x+y-a)\tan^2\theta + x\tan^4\theta + y = 0$$

Let $m = \tan^2\theta$, we get

$$xm^2 + (x+y-a)m + y = 0$$

which is a quadratic equation in $m = \tan^2\theta$

∴ So, the envelope is

$$B^2 - 4AC = 0 \quad \text{... (1)}$$

[Here, $A = x$, $B = x + y - a$, $C = y$]

$$(1) \Rightarrow (x+y-a)^2 - 4(x)(y) = 0$$

i.e., $$x^2 + y^2 + a^2 + 2xy - 2xa - 2ya - 4xy = 0$$

$$x^2 + y^2 - 2xy - 2ax - 2ay + a^2 = 0$$

$$(x-y)^2 - 2ax - 2ay + a^2 = 0$$

Example 4.6.14. Show that the envelope of the family of circles described on the double ordinates of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ as diameters, is the ellipse $\frac{x^2}{a^2+b^2} + \frac{y^2}{b^2} = 1$ **[A.U. Jan. 2011, Jan. 2013]**

Solution : The double ordinates on the envelope $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are $(a\cos t, b\sin t)$ and $(a\cos t, -b\sin t)$ with 't' as the parameter.

$(1)^2 + (2)^2 \Rightarrow$

$$\left[\frac{x\cos\alpha}{a} + \frac{y\sin\alpha}{b}\right]^2 + \left[\frac{-x\sin\alpha}{a} + \frac{y\cos\alpha}{b}\right]^2 = 1^2 + 0^2$$

$$\frac{x^2\cos^2\alpha}{a^2} + \frac{y^2\sin^2\alpha}{b^2} + \frac{2xy\sin\alpha\cos\alpha}{ab} + \frac{x^2\sin^2\alpha}{a^2} + \frac{y^2\cos^2\alpha}{b^2} - \frac{2xy\sin\alpha\cos\alpha}{ab} = 1$$

$$\frac{x^2}{a^2}[\cos^2\alpha + \sin^2\alpha] + \frac{y^2}{b^2}[\cos^2\alpha + \sin^2\alpha] = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

which represents an ellipse.

Thus, the envelope of the given family of curves is an ellipse.

Example 4.6.18. Find the envelope of the family of circles $x^2 + y^2 - 2ax\cos\theta - 2ay\sin\theta = c^2$ where θ is the parameter.

Solution : Given $x^2 + y^2 - 2ax\cos\theta - 2ay\sin\theta = c^2$

$$-2ax\cos\theta - 2ay\sin\theta = -x^2 - y^2 + c^2$$

$$2ax\cos\theta + 2ay\sin\theta = x^2 + y^2 - c^2 \quad \ldots (1)$$

differentiate p.w.r.to θ, we get

$$-2ax\sin\theta + 2ay\cos\theta = 0 \quad \ldots (2)$$

$(1)^2 + (2)^2 \Rightarrow$

$$(2ax\cos\theta + 2ay\sin\theta)^2 + (-2ax\sin\theta + 2ay\cos\theta)^2 = (x^2 + y^2 - c^2)^2$$

$$4a^2x^2\cos^2\theta + 4a^2y^2\sin^2\theta + 8a^2xy\sin\theta\cos\theta$$
$$4a^2x^2\sin^2\theta + 4a^2y^2\cos^2\theta - 8a^2xy\sin\theta\cos\theta = (x^2 + y^2 - c^2)^2$$

$$4a^2x^2[\sin^2\theta + \cos^2\theta] + 4a^2y^2[\sin^2\theta + \cos^2\theta] = [x^2 + y^2 - c^2]^2$$

$$4a^2x^2 + 4a^2y^2 = (x^2 + y^2 - c^2)^2$$

$$4a^2(x^2 + y^2) = (x^2 + y^2 - c^2)^2$$

Example 4.6.19. Find the envelope of the family of straight lines $y\cos\alpha - x\sin\alpha = a\cos 2\alpha$ where α is a parameter.

Solution : $y\cos\alpha - x\sin\alpha = a\cos 2\alpha \quad \ldots (1)$ **[A.U. Madurai June 2011]**

Differentiate w.r.t. α, we get

$$-y\sin\alpha - x\cos\alpha = -2a\sin 2\alpha \quad \ldots (2)$$

Eliminate y between (1) & (2), we get

$(1)\times\sin\alpha \Rightarrow y\sin\alpha\cos\alpha - x\sin^2\alpha = a\sin\alpha\cos 2\alpha \quad \ldots (3)$

$(2)\times\cos\alpha \Rightarrow -y\sin\alpha\cos\alpha - x\cos^2\alpha = -2a\sin 2\alpha\cos\alpha \quad \ldots (4)$

$(3) + (4) \Rightarrow -x[\sin^2\alpha + \cos^2\alpha] = a[\cos 2\alpha \cdot \sin\alpha - 2\sin 2\alpha\cos\alpha]$

$$-x = a[(\cos^2\alpha - \sin^2\alpha)\sin\alpha - 4\sin\alpha\cos^2\alpha]$$

$$= a[-3\sin\alpha\cos^2\alpha - \sin^3\alpha]$$

$$\boxed{x = a[\sin^3\alpha + 3\sin\alpha\cos^2\alpha]}$$

Eliminate x between (1) & (2), we get

$(1) \times \cos\alpha \Rightarrow y\cos^2\alpha - x\sin\alpha\cos\alpha = a\cos 2\alpha\cos\alpha \quad \ldots (5)$

$(2) \times \sin\alpha \Rightarrow -y\sin^2\alpha - x\sin\alpha\cos\alpha = -2a\sin 2\alpha\sin\alpha \quad \ldots (6)$

$(5) - (6) \Rightarrow y[\sin^2\alpha + \cos^2\alpha] = a[\cos 2\alpha\cos\alpha + 2\sin 2\alpha\sin\alpha]$

$$y = a[(\cos^2\alpha - \sin^2\alpha)\cos\alpha + 4\sin^2\alpha\cos\alpha]$$

$$\boxed{y = a[\cos^3\alpha + 3\sin^2\alpha\cos\alpha]}$$

$$x + y = a[\sin^3\alpha + 3\sin^2\alpha\cos\alpha + 3\sin\alpha\cos^2\alpha + \cos^3\alpha]$$

$$= a[\sin\alpha + \cos\alpha]^3$$

$$x - y = a[\sin^3\alpha - 3\sin^2\alpha\cos\alpha + 3\sin\alpha\cos^2\alpha - \cos^3\alpha]$$

$$= a[\sin\alpha - \cos\alpha]^3$$

$$(x+y)^{2/3} + (x-y)^{2/3} = a^{2/3}[\sin\alpha + \cos\alpha]^2 + a^{2/3}[\sin\alpha - \cos\alpha]^2$$

$$= a^{2/3}[\sin^2\alpha + \cos^2\alpha + 2\sin\alpha\cos\alpha] + a^{2/3}[\sin^2\alpha + \cos^2\alpha - 2\sin\alpha\cos\alpha]$$

$$= a^{2/3}[1 + 2\sin\alpha\cos\alpha] + a^{2/3}[1 - 2\sin\alpha\cos\alpha]$$

$$= a^{2/3}[1 + 2\sin\alpha\cos\alpha + 1 - 2\sin\alpha\cos\alpha]$$

$= a^{2/3}[2] = 2a^{2/3}$ is the envelope of the given system.

Example 4.6.20. Find the envelope of the family of straight lines $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$, θ being the parameter. **[A.U Trichy Jan. 2010]**

Solution : **[A.U D15/J16 R-8]**

$$\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$$

$$ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$$

differentiate p.w.r.t. θ, we get

$$ax[\sec\theta \tan\theta] - by[-\operatorname{cosec}\theta \cot\theta] = 0$$

$$\frac{ax \sin\theta}{\cos^2\theta} + \frac{by\cos\theta}{\sin^2\theta} = 0$$

$$\frac{ax\sin\theta}{\cos^2\theta} = \frac{-by\cos\theta}{\sin^2\theta}$$

$$\frac{ax}{\cos^3\theta} = -\frac{by}{\sin^3\theta}$$

$$\frac{\left(\frac{ax}{\cos\theta}\right)}{\cos^2\theta} = -\frac{\left(\frac{by}{\sin\theta}\right)}{\sin^2\theta} = \frac{\frac{ax}{\cos\theta} - \frac{by}{\sin\theta}}{\cos^2\theta + \sin^2\theta} = \frac{a^2 - b^2}{1}$$

$$\therefore \frac{ax}{\cos^3\theta} = -\frac{by}{\sin^3\theta} = a^2 - b^2$$

$$ax = (a^2 - b^2)\cos^3\theta, \quad by = -(a^2 - b^2)\sin^3\theta$$

$$(ax)^{2/3} = (a^2 - b^2)^{2/3}\cos^2\theta, \; (by)^{2/3} = (a^2 - b^2)^{2/3}\sin^2\theta$$

$$(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}\cos^2\theta + (a^2 - b^2)^{2/3}\sin^2\theta$$

$$= (a^2 - b^2)^{2/3}[\cos^2\theta + \sin^2\theta]$$

$$= (a^2 - b^2)^{2/3}$$

is the envelope.

Example 4.6.21. Find the envelope of the family of straight lines $y = mx - 2am - am^3$, where m is the parameter. **[A.U Jan. 2014]** **[A.U M/J 2016, R-13]**

Solution : Given : $y = mx - 2am - am^3$ ... (1)

Differentiate partially w.r.to 'm', we get

$$0 = x - 2a - 3am^2$$

$$3am^2 = x - 2a$$

$$m^2 = \frac{x - 2a}{3a} \quad \ldots (2)$$

Eliminate 'm' from (1) & (2), we get

$$(1) \Rightarrow y = m[x - 2a - am^2]$$

$$= m\left[x - 2a - a\left(\frac{x - 2a}{3a}\right)\right] \quad \text{by (2)}$$

$$= m\left[(x - 2a) - \left(\frac{x - 2a}{3}\right)\right]$$

$$y = m\left[\frac{2}{3}(x - 2a)\right]$$

Squaring on both sides, we get

$$y^2 = m^2\frac{4}{9}(x - 2a)^2$$

$$= \frac{4}{9}m^2(x - 2a)^2$$

$$\Rightarrow y^2 = \frac{4}{9}\left(\frac{x - 2a}{3a}\right)(x - 2a)^2 \quad \text{by (2)}$$

$$= \frac{4}{27a}(x - 2a)^3$$

$$27ay^2 = 4(x - 2a)^3$$

Example 4.6.22 Find the envelope of $y = mx + m^3$, m being the parameter.

Solution :

Given : $y = mx + m^3$

Differentiate partially w.r.t 'm', we get

$$0 = x + 3m^2$$

i.e., $m^2 = -\frac{x}{3}$ ... (1)

$$y = mx + m^3 \text{ (Given)}$$

$$y = mx + m^3 = m(x + m^2)$$

Squaring on both sides, we get

$$y^2 = m^2 (x + m^2)^2$$

$$= \left(\frac{-x}{3}\right)\left(x - \frac{x}{3}\right)^2$$

$$= \left(\frac{-x}{3}\right)\left(\frac{2}{3}x\right)^2$$

$$= \left(\frac{-x}{3}\right)\frac{4}{9}x^2$$

$$= \frac{-4}{27}x^3$$

$27y^2 + 4x^3 = 0$ is the required envelope.

Example 4.6.23. Find the envelope of the family of lines $y = mx + am^p$, m being a parameter.

Solution : Given : $y = mx + am^p$ ... (1)

Differentiate p.w.r.to m, we get

$$0 = x + p\,am^{p-1}$$

$$\Rightarrow apm^{p-1} = -x \quad \Rightarrow am^{p-1} = \frac{-x}{p} \quad \text{... (2)}$$

$$\Rightarrow m^{p-1} = \frac{-x}{ap} \quad \text{... (3)}$$

$$(1) \Rightarrow y = m\,(x + a\,m^{p-1})$$

$$y = m\left[x + \left(\frac{-x}{p}\right)\right]$$

$$y = mx\left[\frac{p-1}{p}\right]$$

$$yp = mx\,[p-1]$$

$$y^{p-1}\,p^{p-1} = m^{p-1}x^{p-1}\,[p-1]^{p-1}$$

$$y^{p-1}\,p^{p-1} = \frac{-x}{ap}x^{p-1}[p-1]^{p-1} \quad \text{by (3)}$$

$$ay^{p-1}p^p = -x^p\,[p-1]^{p-1}$$

$$\Rightarrow ap^p y^{p-1} + (p-1)^{p-1}x^p = 0$$

Example 4.6.24. Find the envelope at $x^2 + y^2 - ax\cos\theta - by\sin\theta = 0$ where θ is the parameter.

Solution : Given : $x^2 + y^2 - ax\cos\theta - by\sin\theta = 0$

$$ax\cos\theta + by\sin\theta = x^2 + y^2 \quad \text{... (1)}$$

differentiate p.w.r. to θ, we get

$$-ax\sin\theta + by\cos\theta = 0$$

i.e., $$ax\sin\theta - by\cos\theta = 0 \quad \text{... (2)}$$

$$(1)^2 + (2)^2 \Rightarrow (ax\sin\theta + by\cos\theta)^2 + (ax\cos\theta - by\sin\theta)^2 = (x^2 + y^2)^2 + 0$$

$$a^2x^2\sin^2\theta + b^2y^2\cos^2\theta + 2ab\,xy\sin\theta\cos\theta$$
$$+ a^2x^2\cos^2\theta + b^2y^2\sin^2\theta - 2ab\,xy\sin\theta\cos\theta = (x^2 + y^2)^2$$

$$a^2x^2[\sin^2\theta + \cos^2\theta] + b^2y^2[\sin^2\theta + \cos^2\theta] = (x^2 + y^2)^2$$

$$a^2x^2 + b^2y^2 = (x^2 + y^2)^2$$

which is the required envelope.

Example 4.6.25. Prove that every member of the family of parabolas given by $y^2 = 4b\,(x - b)$ with b as the parameter is touched by one or the other of the straight lines $y = \pm x$.

Solution : Given $y^2 = 4b\,(x - b)$

i.e. $$y^2 = 4bx - 4b^2 \quad \text{... (1)}$$

Differentiate p.w.r to 'b', we get

$$0 = 4x - 8b \quad \text{... (2)}$$

i.e., $$b = \frac{x}{2}$$

$$\therefore (1) \Rightarrow y^2 = 4\frac{x}{2}x - 4\left(\frac{x}{2}\right)^2$$

$$= 2x^2 - x^2$$

$$= x^2$$

The required envelopes are the straight lines $y = \pm x$ which touch every parabola of the given family.

Example 4.6.26. Find the envelope of the system of straight lines $2y - 3tx + at^3 = 0$ where 't' is the parameter. Find the coordinates of the point of contact of the envelope and the particular member of the system when t = 2.

Solution : Given : $2y - 3tx + at^3 = 0$ ... (1)

Differentiate p.w.r to 't', we get

$$-3x + 3at^2 = 0$$

$$x = at^2 \quad \text{... (2)}$$

$$\frac{x}{a} = t^2$$

$$\left(\frac{x}{a}\right)^3 = t^6 \quad \text{... (3)}$$

(1) $\Rightarrow$ $2y - 3t(at^2) + at^3 = 0$ by (2)

$$2y - 3at^3 + at^3 = 0$$

$$2y - 2at^3 = 0$$

$$y = at^3 \quad \text{... (4)}$$

$$\frac{y}{a} = t^3$$

$$\left(\frac{y}{a}\right)^2 = t^6 \quad \text{... (5)}$$

From (3) & (5), we get

$$\left(\frac{x}{a}\right)^3 = \left(\frac{y}{a}\right)^2$$

$$\frac{x^3}{a^3} = \frac{y^2}{a^2}$$

$$x^3 = ay^2$$

Which is the required envelope.

when $t = 2$, (2) $\Rightarrow x = 4a$

when $t = 2$, (4) $\Rightarrow y = 8a$

Thus, the point of contact of the envelope with the given family when $t = 2$ is $(x = 4a, y = 8a)$

VI. (c) Envelope of two parameter family of curves

Example 4.6.27. Find the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ subject to $a + b = c$ where c is a known constant. [A.U M/J 2009] [A.U Trichy Jan. 2009,Tvli. Jan. 2010, CBT A/M 2011, Madurai Jan 2011]

Solution :

$\frac{x}{a} + \frac{y}{b} = 1$ ... (1)	$a + b = c$ ... (2)
Taking differentials, we get	Taking differentials, we get
$-\frac{x}{a^2}da - \frac{y}{b^2}db = 0$	$da + db = 0$
$\frac{x}{a^2}da = -\frac{y}{b^2}db$ ... (3)	$da = -db$ ... (4)

$$\frac{(3)}{(4)} \Rightarrow \left(\frac{x}{a^2} = \frac{y}{b^2}\right) = \left(\frac{\frac{x}{a}}{a} = \frac{\frac{y}{b}}{b}\right) = \frac{\frac{x}{a} + \frac{y}{b}}{a + b} = \frac{1}{c} \text{ by (1) \& (2)}$$

i.e., $\frac{x}{a^2} = \frac{1}{c}$	$\frac{y}{b^2} = \frac{1}{c}$
$\Rightarrow a^2 = xc$	$\Rightarrow b^2 = yc$
$\Rightarrow a = \sqrt{x}\sqrt{c}$	$\Rightarrow b = \sqrt{y}\sqrt{c}$

(2) $\Rightarrow$ $a + b = c$

$$\sqrt{x}\sqrt{c} + \sqrt{y}\sqrt{c} = c$$

$$\sqrt{x} + \sqrt{y} = \sqrt{c}$$

Example 4.6.28 Find the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ subject to $a^n + b^n = c^n$ given c is a known constant.

Solution : **[A.U M/J 2014]**

$\frac{x}{a} + \frac{y}{b} = 1$ ... (1)	$a^n + b^n = c^n$ ... (2)
Taking differentials, we get	Taking differentials, we get

$-\frac{x}{a^2}da - \frac{y}{b^2}db = 0$	$n a^{n-1} da + n b^{n-1} db = 0$
$\frac{x}{a^2}da = -\frac{y}{b^2}db$ ... (3)	$n a^{n-1} da = -nb^{n-1} db$ ... (4)

$$\frac{(3)}{(4)} \Rightarrow \left[\frac{\left(\frac{x}{a^2}\right)}{na^{n-1}} = \frac{\left(\frac{y}{b^2}\right)}{nb^{n-1}}\right] = \left[\frac{\left(\frac{x}{a}\right)}{a^n} = \frac{\left(\frac{y}{b}\right)}{b^n}\right] = \frac{\frac{x}{a}+\frac{y}{b}}{a^n+b^n} = \frac{1}{c^n} \text{ by (1) \& (2)}$$

i.e., $\frac{x}{a^{n+1}} = \frac{y}{b^{n+1}} = \frac{1}{c^n}$

i.e., $\frac{x}{a^{n+1}} = \frac{1}{c^n}$	i.e., $\frac{y}{b^{n+1}} = \frac{1}{c^n}$
$a^{n+1} = x c^n$,	$b^{n+1} = y c^n$
$\left[a^{n+1}\right]^{\frac{n}{n+1}} = x^{\frac{n}{n+1}}\left[c^n\right]^{\frac{n}{n+1}}$	$\left[b^{n+1}\right]^{\frac{n}{n+1}} = y^{\frac{n}{n+1}}\left[c^n\right]^{\frac{n}{n+1}}$
$a^n = x^{\frac{n}{n+1}}\left[c^n\right]^{\frac{n}{n+1}}$,	$b^n = y^{\frac{n}{n+1}}\left[c^n\right]^{\frac{n}{n+1}}$

(2) $\Rightarrow$ $a^n + b^n = c^n$

$$x^{\frac{n}{n+1}}\left[c^n\right]^{\frac{n}{n+1}} + y^{\frac{n}{n+1}}\left[c^n\right]^{\frac{n}{n+1}} = c^n$$

$$x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} = \left[c^n\right]\left[c^n\right]^{\frac{-n}{n+1}} = \left[c^n\right]^{1-\frac{n}{n+1}}$$

$$x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} = \left[c^n\right]^{\frac{1}{n+1}}$$

$$\Rightarrow x^{\frac{n}{n+1}} + y^{\frac{n}{n+1}} = c^{\frac{n}{n+1}}$$

Note (1) If $n = 1$, the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ subject to

$a + b = c$ is $x^{1/2} + y^{1/2} = c^{1/2}$

Note (2) If $n = 2$, the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ subject to $a^2 + b^2 = c^2$ is $x^{2/3} + y^{2/3} = c^{2/3}$

Example 4.6.29 Find the envelope of $\frac{x}{a} + \frac{y}{b} = 1$ where a & b are connected by $a^2 + b^2 = c^2$, c being a constant. [A.U Tvli. Jan. 2009] [A.U N/D 2002] [A.U N/D 2010] [A.U Jan. 2013] [A.U M/J 2014 R-08] [A.U N/D 2014 R-13] [A.U A/M 2015 R-13]

Solution :

$\frac{x}{a} + \frac{y}{b} = 1$ ... (1)	$a^2 + b^2 = c^2$ ... (2)
Taking differentials, we get	Taking differentials, we get
$-\frac{x}{a^2}da - \frac{y}{b^2}db = 0$	$2a\,da + 2b\,db = 0$
$\frac{x}{a^2}da = -\frac{y}{b^2}db$ ... (3)	$a\,da = -b\,db$ ... (4)

$$\frac{(3)}{(4)} \Rightarrow \left[\frac{\left(\frac{x}{a^2}\right)}{a} = \frac{\left(\frac{y}{b^2}\right)}{b}\right] \Rightarrow \left[\frac{\left(\frac{x}{a}\right)}{a^2} = \frac{\left(\frac{y}{b}\right)}{b^2}\right] = \frac{\frac{x}{a}+\frac{y}{b}}{a^2+b^2} = \frac{1}{c^2} \text{ by (1) \& (2)}$$

i.e., $\frac{x}{a^3} = \frac{y}{b^3} = \frac{1}{c^2}$

$\frac{x}{a^3} = \frac{1}{c^2}$	$\frac{y}{b^3} = \frac{1}{c^2}$
$a^3 = x c^2$	$b^3 = y c^2$
$(a^3)^{2/3} = x^{2/3}(c^2)^{2/3}$	$(b^3)^{2/3} = y^{2/3}(c^2)^{2/3}$
$a^2 = x^{2/3}(c^2)^{2/3}$	$b^2 = y^{2/3}(c^2)^{2/3}$

(2) $\Rightarrow$ $a^2 + b^2 = c^2$

$$x^{2/3}\left(c^2\right)^{2/3} + y^{2/3}\left(c^2\right)^{2/3} = c^2$$

$$x^{2/3} + y^{2/3} = \left(c^2\right)\left(c^2\right)^{-2/3} = \left(c^2\right)^{1-\frac{2}{3}} = \left(c^2\right)^{1/3}$$

$$\Rightarrow x^{2/3} + y^{2/3} = c^{2/3}$$

Example 4.6.30. Find the envelope of the family of ellipses

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$ **where the two parameters are connected by the relation** $a+b=c$ **where c is a constant.**

[A.U. Trichy J/J 2010]

Solution :

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ... (1)	$a+b=c$ ... (2)
Taking differentials, we get	Taking differentials, we get
$\frac{-2x^2}{a^3}da-\frac{2y^2}{b^3}db=0$	$da+db=0$
$\frac{x^2}{a^3}da=-\frac{y^2}{b^3}db$ ... (3)	$da=-db$ ... (4)

$$\frac{(3)}{(4)} \Rightarrow \left(\frac{x^2}{a^3}=\frac{y^2}{b^3}\right) \Rightarrow \left[\frac{\left(\frac{x^2}{a^2}\right)}{a}=\frac{\left(\frac{y^2}{b^2}\right)}{b}\right]=\frac{\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)}{a+b}=\frac{1}{c} \text{ by (1) \& (2)}$$

i.e.,

$\frac{x^2}{a^3}=\frac{1}{c}$	$\frac{y^2}{b^3}=\frac{1}{c}$
$a^3=x^2c$	$b^3=y^2c$
$a=(x^2c)^{1/3}$	$b=(y^2c)^{1/3}$
$a=x^{2/3}c^{1/3}$	$b=y^{2/3}c^{1/3}$

(2) $\Rightarrow$ $a+b=c$

$$x^{2/3}c^{1/3}+y^{2/3}c^{1/3}=c$$

$$x^{2/3}+y^{2/3}=cc^{-1/3}=c^{1-\frac{1}{3}}=c^{2/3}$$

$$\Rightarrow \quad x^{2/3}+y^{2/3}=c^{2/3}$$

Example 4.6.31. Find the envelope of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ **given** $a^n+b^n=c^n$ **where c is a known constant.** **[A.U Jan. 2014, R-2008]**

Solution :

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ... (1)	$a^n+b^n=c^n$ ... (2)
Taking differentials, we get	Taking differentials, we get
$\frac{-2x^2}{a^3}da-\frac{2y^2}{b^3}db=0$	$na^{n-1}da+nb^{n-1}db=0$
$\frac{x^2}{a^3}da=-\frac{y^2}{b^3}db$ ... (3)	$na^{n-1}da=-nb^{n-1}db$ ... (4)

$$\frac{(3)}{(4)} \Rightarrow \left[\frac{\left(\frac{x^2}{a^3}\right)}{na^{n-1}}=\frac{\frac{y^2}{b^3}}{nb^{n-1}}\right] \Rightarrow \left[\frac{\left(\frac{x^2}{a^2}\right)}{a^n}=\frac{\left(\frac{y^2}{b^2}\right)}{b^n}\right]=\frac{\frac{x^2}{a^2}+\frac{y^2}{b^2}}{a^n+b^n}=\frac{1}{c^n} \text{ by (1)\&(2)}$$

i.e.,

$\frac{x^2}{a^{n+2}}=\frac{1}{c^n}$	$\frac{y^2}{b^{n+2}}=\frac{1}{c^n}$
$a^{n+2}=x^2c^n$	$b^{n+2}=y^2c^n$
$\left[a^{n+2}\right]^{\frac{n}{n+2}}=\left[x^2c^n\right]^{\frac{n}{n+2}}$	$\left[b^{n+2}\right]^{\frac{n}{n+2}}=\left[y^2c^n\right]^{\frac{n}{n+2}}$
$a^n=x^{\frac{2n}{n+2}}\left[c^n\right]^{\frac{n}{n+2}}$	$b^n=y^{\frac{2n}{n+2}}\left[c^n\right]^{\frac{n}{n+2}}$

(2) $\Rightarrow$ $a^n+b^n=c^n$

$$x^{\frac{2n}{n+2}}\left[c^n\right]^{\frac{n}{n+2}}+y^{\frac{2n}{n+2}}\left[c^n\right]^{\frac{n}{n+2}}=c^n$$

$$x^{\frac{2n}{n+2}}+y^{\frac{2n}{n+2}}=\left[c^n\right]\left[c^n\right]^{\frac{-n}{n+2}}=\left[c^n\right]^{1-\frac{n}{n+2}}=\left[c^n\right]^{\frac{2}{n+2}}$$

$$\text{i.e., } x^{\frac{2n}{n+2}}+y^{\frac{2n}{n+2}}=c^{\frac{2n}{n+2}}$$

Example 4.6.32. Find the envelope of the straight line $\frac{x}{a}+\frac{y}{b}=1$, **where a and b are connected by the relation** $ab=c^2$, **c is a constant.**

[A.U. Jan 2010, M/J 2010, Tvli M/J 2011]

Solution :

$\frac{x}{a}+\frac{y}{b}=1$ ... (1)	$ab=c^2$ ... (2)
Taking differentials, we get	Taking differentials, we get
$-\frac{x}{a^2}da-\frac{y}{b^2}db=0$	$a\,db+b\,da=0$
$\frac{x}{a^2}da=-\frac{y}{b^2}db$ ... (3)	$b\,da=-a\,db$ ... (4)

$$\frac{(3)}{(4)} \Rightarrow \left[\frac{\left(\frac{x}{a^2}\right)}{b}=\frac{\left(\frac{y}{b^2}\right)}{a}\right] \Rightarrow \left[\frac{\frac{x}{a}}{ab}=\frac{\frac{y}{b}}{ab}\right]$$

$$= \left[\frac{\left(\frac{x}{a}\right)}{1}=\frac{\left(\frac{y}{b}\right)}{1}\right] = \frac{\frac{x}{a}+\frac{y}{b}}{1+1} = \frac{1}{2} \text{ by (1)}$$

i.e.,

$\frac{x}{a}=\frac{1}{2}$	$\frac{y}{b}=\frac{1}{2}$
$a=2x$	$b=2y$

(2) $\Rightarrow$ $ab=c^2$

$(2x)(2y)=c^2$

$\Rightarrow$ $4xy=c^2$

♦ VI. (d) In case α cannot be eliminated between $f(x,y,\alpha)=0$ and $\frac{\partial f}{\partial \alpha}=0$,

Then the envelope is given in the parametric form by the equations $x=x(\alpha)$ and $y=y(\alpha)$

Example 4.6.33. Determine the envelope of $x\sin t - y\cos t = at$ where 't' is the parameter.

Solution : Given $x\sin t - y\cos t = at$ ... (1)

Differentiating (1) p.w.r.to 't', we get

$x\cos t + y\sin t = a$ ... (2)

As 't' the parameter cannot be eliminated between (1) and (2)

Solve (1) and (2), for x and y interms of 't'.

(1) $\times \sin t$ $\Rightarrow$ $x\sin^2 t - y\sin t\cos t = at\sin t$ ... (3)

(2) $\times \cos t$ $\Rightarrow$ $x\cos^2 t + y\sin t\cos t = a\cos t$ ... (4)

(3) + (4) $\Rightarrow$ $x[\sin^2 t + \cos^2 t] = a[t\sin t + \cos t]$

$x = a(t\sin t + \cos t)$

i.e., $x(t) = a(t\sin t + \cos t)$

(1) $\times \cos t$ $\Rightarrow$ $x\cos t\sin t - y\cos^2 t = at\cos t$ ... (5)

(2) $\times \sin t$ $\Rightarrow$ $x\cos t\sin t + y\sin^2 t = a\sin t$ ... (6)

(6) − (5) $\Rightarrow$ $y = a[\sin t - t\cos t]$

i.e., $y(t) = a[\sin t - t\cos t]$

VI(e). No envelope exists problems

Example 4.6.34. Show that the family of straight lines $2y - 4x + \alpha = 0$ has no envelope, where α being the parameter. [A.U A/M 2008]

Solution : Given : $2y - 4x + \alpha = 0$ ... (1)

Differentiating (1) p.w.t to 'α', we get

$1 = 0$ [absurd]

Note that the given family of straight lines $y = 2x - \frac{\alpha}{2}$ are all parallel with common slope $m = 2$. Therefore no curve (envelope) exists which touches each member of this parallel straight lines.

EXERCISE 4.6

1. Find the envelope of the family of straight lines $y = mx + \frac{1}{m}$. [Ans. $y^2 = 4x$]

2. Find the equation of the envelope of the lines $x\sec\alpha + y\,\text{cosec}\,\alpha = a$, where α is the parameter and a is a constant. [Ans. $x^{2/3} + y^{2/3} = a^{2/3}$]

3. Find the envelope of the family of lines $x\cos^3\alpha + y\sin^3\alpha = a$ the parameter being α. [Ans. $x^2y^2 = a^2(x^2 + y^2)$]

4. Find the envelope of the system of lines $\frac{x}{l} + \frac{y}{m} = 1$, where l and m are connected by the relation $\frac{l}{a} + \frac{m}{b} = 1$ (l and m are the parameters). [Ans. $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$.] [A.U. Jan. 2012]

5. Find the envelope of a system of concentric ellipses with their axes along the co-ordinate axes and of constant area.

[Ans. $2\pi xy = c$.]

6. Find the envelope of the family of straight lines $\frac{x}{a} + \frac{y}{b} = 1$ where $a^2 + b^2 = k^2$, k being a constant. [Ans. $x^{2/3} + y^{2/3} = k^{2/3}$]

7. Find the envelopes of the following family of straight lines

(a) $y = x \tan\alpha + 2 \sec\alpha$ [Ans. $x^2 + y^2 = 4$]

(b) $x \cos\theta + y \sin\theta = 1$ [Ans. $x^2 + y^2 = 1$]

(c) $x \cos\alpha + y \sin\alpha = c \sin\alpha \cos\alpha$ [A.U A/M 2011] [Ans. $x^{2/3} + y^{2/3} = a^{2/3}$]

(d) $x \operatorname{cosec}\alpha - y \cot\alpha = a$ [A.U Jan. 2013] [Ans. $x^2 - y^2 = a^2$]

(e) $y = mx + am^p$; m being a parameter.

[Ans. $(p-1)^{p-1} x^p + p^p a y^{p-1} = 0$]

(f) $x \cos^3\theta + y \sin^3\theta = a$, θ being a parameter.

[Ans. $x^2 y^2 = a^2 (x^2 + y^2)$]

(g) $\frac{x}{a} \cos\theta + \frac{y}{b} \sin\theta = 1$ [Ans. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$]

8. Find the envelope of the family of

(a) $y = mx + a\sqrt{1 + m^2}$ [Ans. $x^2 + y^2 = a^2$]

(b) $y \operatorname{cosec}^2\alpha + x \sec^2\alpha = a$ where α is the parameter.

[Ans. $(x + y - a)^2 = 4xy$]

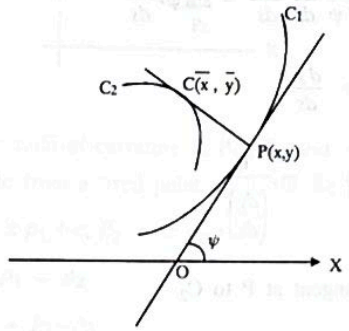
(c) $(x - a)^2 + y^2 = 4a$ where a is the parameter. [Ans. $y^2 = 4(x + 1)$]

(d) $\frac{x^2}{\alpha} + \frac{y^2}{1 - \alpha} = 1$. [Ans. $(y^2 - x^2 - 1)^2 = 4x^2$]

4.7 PROPERTIES OF ENVELOPES AND EVOLUTES

PROPERTY 1. The normal at any point of a curve is a tangent to its evolute touching at the corresponding centre of curvature.

Proof :



Let $P(x, y)$ be any point on the given curve and $C(\bar{x}, \bar{y})$ be the corresponding centre of curvature.

Now, CP is normal to the curve at P and CP $= \rho$.

Let the tangent at P make an angle ψ with OX.

C_1 is the given curve and C_2 be its evolute.

We have, $\bar{x} = x - \rho \sin\psi$, $\bar{y} = y + \rho \cos\psi$

Now, $\frac{d\bar{y}}{ds} = \frac{dy}{ds} + \rho(-\sin\psi)\frac{d\psi}{ds} + \cos\psi \frac{d\rho}{ds}$

$= \frac{dy}{ds} - \rho \sin\psi \frac{d\psi}{ds} + \cos\psi \frac{d\rho}{ds}$

$= \frac{dy}{ds} - \frac{ds}{d\psi}\frac{dy}{ds}\frac{d\psi}{ds} + \cos\psi \frac{d\rho}{ds}$

$\left[\because \rho = \frac{ds}{d\psi},\ \sin\psi = \frac{dy}{ds},\ \cos\psi = \frac{dx}{ds}\right]$

$= \frac{dy}{ds} - \frac{dy}{ds} + \cos\psi \frac{d\rho}{ds}$

$= \cos\psi \frac{d\rho}{ds}$ ... (1)

$$\frac{d\bar{x}}{ds} = \frac{dx}{ds} - \rho\cos\psi\frac{d\psi}{ds} - \sin\psi\frac{d\rho}{ds}$$

$$= \frac{dx}{ds} - \frac{ds}{d\psi}\frac{dx}{ds}\frac{d\psi}{ds} - \sin\psi\frac{d\rho}{ds}$$

$$= -\sin\psi\frac{d\rho}{ds} \quad \ldots (2)$$

$$\frac{(1)}{(2)} \Rightarrow \frac{d\bar{y}}{dx} = -\cot\psi = -\frac{1}{\left(\frac{dy}{dx}\right)}$$

(i.e.,) Slope of the tangent at P to C_2

= Slope of the normal at P to C_1

$\therefore$ CP touches C_2 at C.

PROPERTY 2. The difference between the radii of curvatures at two points of a curve is equal to the length of the arc of the evolute between the two corresponding points.

Proof : If $(\bar{x}, \bar{y})$ is the centre of curvature corresponding a point (x, y) of the given curve, we know that, from property (1)

$$d\bar{x} = -\sin\psi\, d\rho, \qquad d\bar{y} = \cos\psi\, d\rho$$

Let β be the length of the arc of the evolute measured from source fixed point A on it. Then, the elementary length $d\beta$ is given by

$$(d\beta)^2 = (d\bar{x})^2 + (d\bar{y})^2 = \sin^2\psi\,(d\rho)^2 + \cos^2\psi\,(d\rho)^2$$

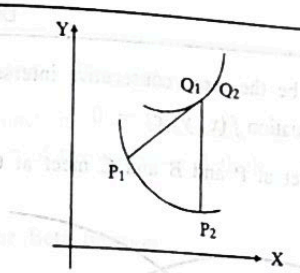
$$= (d\rho)^2[\sin^2\psi + \cos^2\psi]$$

$$= (d\rho)^2$$

$$\therefore d\beta = \pm d\rho$$

Integrating, we get $\beta = \pm\rho + c$ ... (1)

where c is a constant. Let Q_1 and Q_2 be two points on the evolute corresponding to the points P_1 and P_2 on the given curve.



If ρ_1, ρ_2 be the radii of curvature at P_1, P_2 and β_1, β_2 be the lengths of arcs of the evolute from a fixed point on it, till the points Q_1, Q_2, then

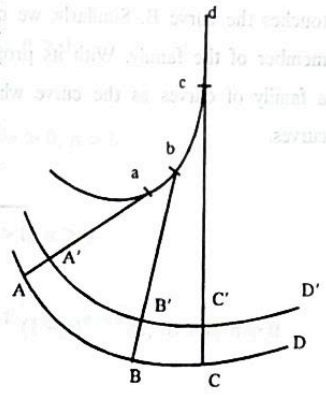
$$\beta_1 = \pm\rho_1 + c,\ \beta_2 = \pm\rho_2 + c$$

$$\beta_1 - \beta_2 = \rho_1 \sim \rho_2$$

(i.e.,) arc $Q_1Q_2 = \sim \rho_1 \sim \rho_2$

PROPERTY 3. There is one evolute, but an infinite number of involutes.

Let ABCD ... be the original curve and *abcd* ... be its evolute. Take *abcd* ... as the original curve, ABCD... is an involute of *abcd*.... we suppose any equal lengths AA', BB',CC', DD'.... to be taken along each normal, as indicated in the adjoining figure, then a new curve A' B' C' D'... is formed which may be called a *parallel* to the original curve, having the same normals as the original curve and therefore having the same evolute. It is therefore clear that if any curve is given, it can have one evolute, but an infinite number of curves may have the same evolute and hence *any curve may possess an infinite number of involutes.* The involutes of a given curve form a family of parallel curves.

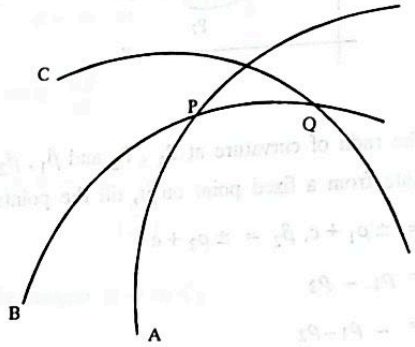


Remarks : If the evolute *abcd* ... be regarded as a rigid curve and a string be unwound from it, keeping the string tight, the points of the unwinding string trace a family of parallel curves, each of which is an involute of the curve *abcd*...

PROPERTY 4. The envelope of a family of curves touches at each of its points the corresponding member of that family.

Proof : Let A, B, C be the three consecutive intersecting members of the family given by the equation $f(x, y, c) = 0$

Assume, A and B meet at P and B and C meet at Q.



Now by definition, P and Q are points on the envelope. Also P and Q being two continuous points common to B and the envelope, ultimately the curve B and the envelope have a common tangent and therefore the envelope touches the curve B. Similarly, we can show that the envelope touches every member of the family. With its property we can also define the envelope of a family of curves as the curve which touches every member of the family curves.

4.8 Beta Functions - Properties :

The Gamma function is a function of single real variable and the Beta function is a function of two variables. Both the functions are special type of improper integrals.

Definition : The Beta function :

The Beta function is defined by

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1}\, dx, \ (m > 0, n > 0)$$

is convergent for $m > 0$, $n > 0$.

This function is also known as Euler's integral of the first kind.

Note : Different forms for $\beta(m, n)$

1. $\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\, dx, \ m > 0, n > 0$

   $\beta(n, m) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\, dx, \ m > 0, n > 0$

2. $\beta(m, n) = \frac{1}{2} \int_0^\infty \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx, \ m > 0, n > 0$

3. $\beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}}\, dx, \ m > 0, n > 0$

4. $\beta(m, m) = \frac{1}{2^{m+n-1}} \int_{-1}^1 (1+x)^{m-1} (1-x)^{n-1}\, dx, \ m > 0, n > 0$

5. $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} x \cos^{2n-1} x\, dx, \ m > 0 \ n > 0$

6. $\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2 \int_0^{\pi/2} \sin^p x \ \cos^q x\, dx, \ p > -1, q > -1$

7. $\beta(m, n) = 2a^m b^n \int_0^{\pi/2} \frac{\sin^{2m-1} x \cos^{2n-1} x}{(a \sin^2 x + b \cos^2 x)^{m+n}}\, dx$

♦ Standard Results

Example 4.8.1. Symmetry

$\beta(m, n) = \beta(n, m)$.

Proof : $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

put $x = 1-y$. Hence $1-x = y$.

Also $dx = -dy$

$\therefore$ when $x = 0$, $y = 1$ and

when $x = 1$, $y = 0$

$$\therefore \beta(m,n) = \int_1^0 (1-y)^{m-1} y^{n-1}(-dy)$$

$$= \int_0^1 y^{n-1}(1-y)^{m-1}\,dy = \beta(n,m)$$

Example 4.8.2. Beta function interms of trigonometric functions

$$\beta(m, n) = 2\int_0^{\pi/2} (\sin x)^{2m-1} (\cos x)^{2n-1}\,dx$$

Proof : We know that $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

put $x = \sin^2 t$. Hence $dx = 2\sin t \cos t\,dt$

when $x = 0$, $t = 0$ and

when $x = 1$, $t = \frac{\pi}{2}$

$$\therefore \beta(m,n) = \int_0^{\pi/2} (\sin^2 t)^{m-1} (\cos^2 t)^{n-1} 2\sin t \cos t\,dt$$

$$= 2\int_0^{\pi/2} (\sin x)^{2m-1} (\cos x)^{2n-1}\,dx$$

Example 4.8.3. Beta function expressed as an improper integral

$$\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx$$

Proof : $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

Put $x = \frac{y}{1+y}$. Hence $1-x = 1-\frac{y}{1+y} = \frac{1}{1+y}$

when $x = 0$, $y = 0$; when $x \to 1$, $y \to \infty$

Also $dx = \frac{dy}{(1+y)^2}$

$$\therefore \beta(m,n) = \int_0^\infty \frac{y^{m-1}}{(1+y)^{m-1}(1+y)^{n-1}(1+y)^2}\,dy$$

$$= \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}}\,dy$$

$$= \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx \quad [\because y \text{ is a dummy variable}]$$

Example 4.8.4. $\beta(m, n) = \beta(m+1, n) + \beta(m, n+1)$.

Proof : $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

$$= \int_0^1 x^{m-1}(1-x)^{n-1}(x+1-x)\,dx$$

$$= \int_0^1 x^m (1-x)^{n-1}\,dx + \int_0^1 x^{m-1}(1-x)^n\,dx$$

$$= \beta(m+1, n) + \beta(m, n+1)$$

Example 4.8.5. $\beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2\int_0^{\pi/2} \sin^m x \cos^n x\,dx$

Solution : We know that

$$\beta(p, q) = 2\int_0^{\pi/2} (\sin x)^{2p-1} (\cos x)^{2q-1}\,dx$$

put $p = \frac{m+1}{2}$, $q = \frac{n+1}{2}$

(i.e.,) $2p - 1 = m$, $2q - 1 = n$

$$\therefore \beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right) = 2\int_0^{\pi/2} (\sin x)^m (\cos x)^n\,dx$$

$$= 2\int_0^{\pi/2} \sin^m x \cos^n x\,dx$$

Example 4.8.6. Show that $\beta(m,n) = \int_0^1 \frac{x^{n-1}+x^{m-1}}{(1+x)^{m+n}}\,dx$.

Solution : We know that

$$\beta(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy$$

$$\beta(m,n) = \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}}\,dy + \int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy \quad \ldots (1)$$

In the second integral,

Put $y = \frac{1}{u}$ when $y = 1$, $u = 1$

$dy = -\frac{1}{u^2}\,du$ when $y = \infty$, $u = 0$

$$\therefore \int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy = \int_1^0 \frac{\left(\frac{1}{u}\right)^{n-1}}{\left(1+\frac{1}{u}\right)^{m+n}} \cdot \left(-\frac{1}{u^2}\right) du$$

$$= -\int_1^0 \frac{1}{u^{n-1} \cdot \frac{(u+1)^{m+n}}{u^{m+n}}} \cdot \frac{1}{u^2}\,du$$

$$= \int_0^1 \frac{du}{(1+u)^{m+n} \cdot u^{-m-n+n-1+2}}$$

$$= \int_0^1 \frac{u^{m-1}\,du}{(1+u)^{m+n}}$$

$$\therefore \int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy = \int_0^1 \frac{y^{m-1}}{(1+y)^{m+n}}\,dy \quad \ldots (2)$$

[u can be replaced by 'y' since the given integral is definite integral]
Substituting (2) in (1), we get,

$$\beta(m,n) = \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}}\,dy + \int_0^1 \frac{y^{m-1}}{(1+y)^{m+n}}\,dy$$

$$= \int_0^1 \frac{x^{n-1}+x^{m-1}}{(1+x)^{m+n}}\,dx \qquad [\text{change } y \text{ into } x']$$

Example 4.8.7. $\beta(m,n) = \frac{1}{2}\int_0^\infty \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}\,dx$

Solution : We know that $\beta(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx$ ... (1)

$$\beta(n,m) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx \quad \ldots (2)$$

By result 1 : $\beta(m,n) = \beta(n,m)$

$$\therefore (1) + (2) \Rightarrow \beta(m,n) + \beta(n,m) = \int_0^\infty \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}\,dx$$

$$2\beta(m,n) = \int_0^\infty \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}\,dx$$

$$\therefore \beta(m,n) = \frac{1}{2}\int_0^\infty \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}\,dx$$

Example 4.8.8. $\int_0^1 x^p (1-x)^r\,dx = \beta(p+1, r+1)$

Solution : We know that

$$\int_0^1 x^p (1-x^q)\,dx = \frac{1}{q}\,\beta\left(\frac{p+1}{q}, r+1\right)$$

Put $q = 1$ we get

$$\int_0^1 x^p (1-x)^r\,dx = \beta(p+1, r+1)$$

Example 4.8.9. Show that $\int_0^1 \frac{x^{m-1}(1-x)^{n-1}}{(a+x)^{m+n}}\,dx = \frac{\beta(m,n)}{a^n(1+a)^m}$

Solution : $I = \int_0^1 \frac{x^{m-1}(1-x)^{n-1}}{(a+x)^{m+n}}\,dx$. Put $\frac{x}{a+x} = \frac{t}{a+1}$

$$\therefore x(a+1) = t(a+x) \text{ or } x.(a+1-t) = at$$

or $$x = \frac{at}{a+1-t}$$

$$\therefore dx = \frac{(a+1-t)(a) - at(-1)}{(a+1-t)^2}.dt = \frac{a(a+1)\,dt}{(a+1-t)^2}$$

Also $1 - x = 1 - \frac{at}{a+1-t} = \frac{a+1-t-at}{a+1-t}$

$= \frac{a+1-t(a+1)}{a+1-t} = \frac{(a+1)(1-t)}{a+1-t}$

$a + x = a + \frac{at}{a+1-t} = \frac{a(a+1)-at+at}{a+1-t} = \frac{a(a+1)}{(a+1-t)}$

When $x = 0$, $0 = \frac{at}{a+1-t} \Rightarrow t = 0$, when $x = 1$, $\frac{1}{a+1} = \frac{t}{a+1} \Rightarrow t = 1$

x	0	1
t	0	1

$$\therefore I = \int_0^1 \frac{a^{m-1} t^{m-1} (a+1)^{n-1} (1-t)^{n-1} a(a+1)\,dt}{(a+1-t)^{m-1} (a+1-t)^{n-1} (a+1-t)^2} \cdot \frac{(a+1-t)^{m+n}}{a^{m+n} (a+1)^{m+n}}$$

$$= \frac{1}{a^n (1+a)^m} \int_0^1 t^{m-1} (1-t)^{n-1}\,dt = \frac{\beta(m,n)}{a^n (a+1)^m}$$

Example 4.8.10. Prove that $\int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx = \frac{1}{a^n b^m} \beta(m,n)$

Solution : $I = \int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx \qquad bx = at;\ dx = \frac{a}{b} dt$

$$= \int_0^\infty \frac{a^{m-1} t^{m-1}}{b^{m-1}} \cdot \frac{a}{b} \cdot dt \frac{1}{(a+at)^{m+n}}$$

x	0	∞
t	0	∞

$$= \int_0^\infty \frac{a^m t^{m-1} dt}{b^m a^{m+n} (1+t)^{m+n}} = \frac{1}{a^n b^m} \int_0^\infty \frac{t^{m-1} dt}{(1+t)^{m+n}} = \frac{1}{a^n b^m} B(m,n)$$

EXERCISE

1. Prove the following :

(a) $\int_0^\infty \frac{x^4 (1+x^5)}{2(1+x)^{15}} dx = \beta(5,10)$ (b) $\int_0^\infty \frac{x^8 (1-x^6)}{(1+x)^{24}} dx = 0$

(c) $\int_0^2 \frac{x^2}{\sqrt{1-x/2}} dx = 8\beta\left(3, \frac{1}{2}\right)$ (d) $\int_0^1 \frac{5x\,dx}{\sqrt{1-x^5}} = \beta\left(\frac{2}{5}, \frac{1}{2}\right)$

2. Show that $\int_0^1 (1-x^n)^{1/n} dx = \frac{1}{2n} \beta\left(\frac{1}{n}, \frac{1}{n}\right)$

3. Show that $\int_0^1 \frac{x^{n-1}}{(x+a)^{m+n}} = \frac{1}{a^n} \beta(m,n)$

4. Show that $\int_a^b \frac{dx}{(b-x)^{1-n} (x-a)^n} = \beta(1-n, n)$

5. Show that $\int_0^a \frac{dx}{(a^n - x^n)^{1/n}} = \frac{1}{n} \beta\left(\frac{1}{n}, 1 - \frac{1}{n}\right)$

Gamma Functions - Properties :

Definition : Gamma function

Gamma function denoted by $\Gamma(n)$ is defined by the improper integral which is dependent on the parameter n.

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx \qquad (x > 0)$$

Note : Gamma function is also known as Euler's integral of the second kind.

♦ Standard Results

Example 4.8.11. Recurrence formula for $\Gamma(n)$ (or) Reduction formula.

$$\Gamma_{n+1} = n \Gamma_n$$

$$\Gamma_n = \int_0^\infty x^{n-1} e^{-x} dx$$

$$\Gamma_{n+1} = \int_0^\infty x^n e^{-x} dx = \int_0^\infty x^n d\left[-e^{-x}\right]$$

$$= x^n (-e^{-x})\Big]_0^\infty - \int_0^\infty (-e^{-x}) n x^{n-1} dx$$

$\because$ Integration by parts $\int u\,dv = uv - \int v\,du$

$$= -x^n e^{-x}\Big]_0^\infty + n \int_0^\infty e^{-x} x^{n-1} dx$$

$$= (0-0) + n \Gamma_n \qquad [\because e^{-\infty} = 0]$$

$$= n \Gamma_n$$

Note : $\Gamma_{n+1} = n!$ of n is a positive integer.

Note : Integrals which reduce to Gamma functions.

(i) $\int_0^\infty x^m e^{-nx}\,dx$ put $nx = t$

(ii) $\int_0^\infty x^m e^{-nx^2}\,dx$ put $nx^2 = t$

(iii) $\int_0^\infty x^m e^{-nx^3}\,dx$ put $nx^3 = t$

(iv) $\int_0^1 x^m (\log x)^n\,dx$ put $\log x = -t$

(v) $\int_0^1 x^m \left(\log \frac{1}{x}\right)^n dx$ put $\log x = -t$

Note 2. $\Gamma_{(n)}$ in different forms.

$$(1)\ \Gamma(n) = \frac{1}{n}\int_0^\infty e^{-x^{1/n}}\,dx$$

$$(2)\ \Gamma(n) = 2a^n \int_0^\infty x^{2n-1} e^{-ax^2}\,dx$$

$$(3)\ \Gamma(n) = \int_0^1 \left(\log \frac{1}{x}\right)^{n-1} dx$$

Example 4.8.12 Prove that $\Gamma(n) = \frac{1}{n}\int_0^\infty e^{-x^{1/n}}\,dx$

Put $x^{1/n} = t$ (or) $x = t^n$ $\therefore$

$dx = n t^{n-1}\,dt$

when $x = 0$, $t = 0$
when $x = \infty$, $t = \infty$

$$\therefore \text{R.H.S} = \frac{1}{n}\int_0^\infty e^{-t}(n t^{n-1})\,dt = \int_0^\infty e^{-t} t^{n-1}\,dt$$

$$= \Gamma_{(n)} = \text{L.H.S.}$$

Example 4.8.13 Prove that $\Gamma(n) = 2a^n \int_0^\infty x^{2n-1} e^{-ax^2}\,dx$

Solution : Put $ax^2 = t$

$$x = \frac{t^{1/2}}{a^{1/2}}$$

$$dx = \frac{1}{a^{1/2}}\frac{1}{2\sqrt{t}}\,dt = \frac{1}{2a^{1/2}} t^{-1/2}\,dt$$

when $x = 0$, $t = 0$
when $x = \infty$, $t = \infty$

$$\therefore \text{R.H.S.} = 2a^n \int_0^\infty \left[\frac{t^{1/2}}{a^{1/2}}\right]^{2n-1} e^{-t}\,\frac{1}{2a^{1/2}}\,t^{-1/2}\,dt$$

$$= 2a^n \int_0^\infty \frac{\left[t^{1/2}\right]^{2n-1}}{\left[a^{1/2}\right]^{2n-1}} e^{-t}\,\frac{1}{2a^{1/2}}\left[t^{-1/2}\right] dt$$

$$= a^n \int_0^\infty \frac{t^{n-1/2}}{a^{n-1/2}} e^{-t}\,\frac{1}{a^{1/2}}\,t^{-1/2}\,dt$$

$$= \int_0^\infty t^{n-1} e^{-t}\,dt = \Gamma(n) = \text{L.H.S.}$$

Example 4.8.14 Prove that $\Gamma(n) = \int_0^1 \left[\log \frac{1}{x}\right]^{n-1} dx$

Solution : Put $-\log x = t$

$$x = e^{-t}$$

$$dx = -e^{-t}\,dt$$

when $x = 0$, $t = -\log 0 = \infty$
when $x = 1$, $t = -\log 1 = 0$

$$\therefore \text{R.H.S} = \int_0^1 [\log 1 - \log x]^{n-1}\,dx$$

$$= \int_0^1 [-\log x]^{n-1}\,dx \qquad \text{since } \log 1 = 0$$

$$= \int_0^\infty t^{n-1} \left(-e^{-t}\right) dt$$

$$= \int_0^\infty e^{-t} t^{n-1} dt = \Gamma(n) = \text{L.H.S.}$$

Example 4.8.15. Prove that $\Gamma(1) = 1$.

Proof : $\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx$

$$\Gamma(1) = \int_0^\infty e^{-x} dx$$

$$= \left[-e^{-x}\right]_0^\infty$$

$$= [0 - (-1)] = 1$$

Example 4.8.16. Prove that $\Gamma(n+1) = n!$ where n is a positive integer.

Proof : We have $\Gamma(n+1) = n\,\Gamma(n)$

$$= n(n-1)\,\Gamma(n-2)$$

$$= n(n-1)(n-2) \ldots \Gamma(1) = n!$$

Example 4.8.17. Prove that $\Gamma(n) = 2\int_0^\infty e^{-y^2} y^{2n-1} dy$.

Proof : We have $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

put $x = y^2$.

Hence $dx = 2y\,dy$

$$\therefore \Gamma(n) = \int_0^\infty e^{-y^2} (y^2)^{n-1} 2y\,dy = 2\int_0^\infty e^{-y^2} y^{2n-1} dy$$

Example 4.8.18. Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Proof : $\Gamma(n) = 2\int_0^\infty e^{-x^2} x^{2n-1} dx$

$$\Gamma\left(\frac{1}{2}\right) = 2\int_0^\infty e^{-x^2} x^{2\left(\frac{1}{2}\right)-1} dx = 2\int_0^\infty e^{-x^2} dx = 2\left(\frac{\sqrt{\pi}}{2}\right) = \sqrt{\pi}.$$

Example 4.8.19. Prove that $\Gamma(n) = \int_0^\infty \left[\log\left(\frac{1}{x}\right)\right]^{n-1} dx$

Proof : We have $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

put $x = \log\left(\frac{1}{y}\right)$

Hence $dx = -\frac{1}{y} dy$

when $x = 0$, $y = 1$ and when $x = \infty$, $y = 0$

$$\Gamma(n) = \int_1^0 y \left[\log\left[\frac{1}{y}\right]\right]^{n-1} \left[\frac{-1}{y}\right] dy = \int_0^1 \log\left[\frac{1}{x}\right]^{n-1} dx$$

Example 4.8.20. Evalute the following : (i) $\frac{\Gamma(7)}{\Gamma(5)}$ (ii) $\frac{\Gamma(5/2)}{\Gamma(3/2)}$ (iii) $\frac{\Gamma(8/3)}{\Gamma(2/3)}$ (iv) $\Gamma\left(\frac{9}{2}\right)$

Solution : (i) $\Gamma(7) = \Gamma(6+1) = 6!$ since $\Gamma_{n+1} = n!$

$$\Gamma(5) = \Gamma(4+1) = 4!$$

$$\therefore \frac{\Gamma(7)}{\Gamma(5)} = \frac{6!}{4!} = \frac{1.2.3.4.5.6}{1.2.3.4} = 30$$

(ii) $\Gamma(5/2) = \Gamma(3/2+1) = (3/2)\,\Gamma(3/2)$

$$\therefore \frac{\Gamma(5/2)}{\Gamma(3/2)} = \frac{(3/1)\,\Gamma(3/2)}{\Gamma(3/2)} = \frac{3}{2}$$

(iii) $\Gamma(8/3) = \Gamma(5/3+1) = \frac{5}{3}\frac{2}{3}\Gamma(2/3)$

$$\therefore \frac{\Gamma(8/3)}{\Gamma(2/3)} = \frac{\left(\frac{5}{3}\right)\left(\frac{2}{3}\right)\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{2}{3}\right)} = \frac{10}{9}$$

(iv) $\Gamma 9/2 = \Gamma\left(\frac{7}{2}+1\right) = \frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\Gamma\frac{1}{2}$

$$= \frac{105}{16}\sqrt{\pi}$$

Note : Gamma function for the negative value of n

$$\Gamma(n) = \frac{\Gamma(n+1)}{n}, \; n < 0$$

This is called analytic continuation.

For example

(i) $\Gamma_{(-1/2)} = \dfrac{\Gamma(-1/2+1)}{-1/2} = \dfrac{\Gamma\frac{1}{2}}{\frac{-1}{2}} = -2\Gamma\frac{1}{2} = -2\sqrt{\pi}$

(ii) $\Gamma(-3/2) = \dfrac{\Gamma(-3/2+1)}{-3/2} = -\frac{2}{3}\Gamma-\frac{1}{2} = -\frac{2}{3}(-2\sqrt{\pi}) = \frac{4}{3}\sqrt{\pi}$

(iii) $\Gamma(-5/2) = \dfrac{\Gamma(-5/2+1)}{-5/2} = -\frac{2}{5}\Gamma(-3/2) = -\frac{2}{5}\frac{4}{3}\sqrt{\pi} = \frac{-8}{15}\sqrt{\pi}$

Example 4.8.21. Evaluate : (i) $\int_0^\infty x^4 e^{-x}\,dx$ (ii) $\int_0^\infty e^{-3x}x^6\,dx$

(iii) $\int_0^\infty x^2 e^{-2x^2}\,dx$ (iv) $\int_0^1 \frac{dx}{\sqrt{(-\log x)}}$

Solution : (i) $\int_0^\infty x^4 e^{-x}\,dx = \int_0^\infty x^{5-1}e^{-x}\,dx$

$= \Gamma(5) = 4! = 24$

(ii) $\int_0^\infty e^{-3x}x^6\,dx$

Put $3x = t$ when $x = 0,\ t = 0$

$3dx = dt$ when $x = \infty,\ t = \infty$

$$\therefore \int_0^\infty e^{-3x}x^6\,dx = \int_0^\infty e^{-t}\left(\frac{t}{3}\right)^6\frac{dt}{3} = \frac{1}{3^7}\int_0^\infty e^{-t}\,t^6\,dt$$

$$= \frac{1}{3^7}\Gamma(7) = \frac{1}{37}(6!) = \frac{80}{243}$$

(iii) $\int_0^\infty x^2 e^{-2x^2}\,dx$

Put $2x^2 = t$ when $x = 0,\ t = 0$

$4x\,dx = dt$ when $x = \infty,\ t = \infty$

$$dx = \frac{dt}{4x} = \frac{dt}{4\sqrt{t}/2}$$

$$\therefore \int_0^\infty x^2 e^{-2x^2}\,dx = \int_0^\infty \left(\frac{t}{2}\right)e^{-t}\frac{dt}{4\left(\frac{\sqrt{t}}{\sqrt{2}}\right)}$$

$$= \frac{\sqrt{2}}{2\times 4}\int_0^\infty t^{1/2}e^{-t}\,dt = \frac{\sqrt{2}}{8}\int_0^\infty t^{\frac{3}{2}-1}e^{-t}\,dt$$

$$= \frac{\sqrt{2}}{8}\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{2}}{8}\Gamma\frac{1}{2}+1 = \frac{\sqrt{2}}{8}\frac{1}{2}\Gamma\frac{1}{2}$$

$$= \frac{\sqrt{2}}{8}\frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{8\sqrt{2}}$$

(iv) $\int_0^1 \frac{dx}{\sqrt{(-\log x)}}$

Put $\log x = -t$ when $x = 0,\ t = \infty$

$x = e^{-t}$ when $x = 1,\ t = 0$

$dx = -e^{-t}\,dt$

$$\therefore \int_0^1 \frac{dx}{\sqrt{(-\log x)}} = \int_\infty^0 \frac{1}{\sqrt{t}}(-e^{-t})\,dt$$

$$= \int_0^\infty t^{-1/2}e^{-t}\,dt = \Gamma\frac{1}{2} = \sqrt{\pi}$$

EXERCISE

1. Evaluate : (i) $\dfrac{\Gamma(7)}{2\Gamma(4)\Gamma(3)}$ (ii) $\dfrac{\Gamma(3)\Gamma(3/2)}{\Gamma(9/2)}$ (iii) $\Gamma(1/2)\Gamma(3/2)\Gamma(5/2)$

(iv) $\dfrac{\Gamma(7/3)}{\Gamma(4/3)}$ (v) $\dfrac{\Gamma(3)\Gamma(5/2)}{\Gamma(11/2)}$

2. Evaluate : (i) $\int_0^\infty x^5 e^{-x}\,dx$ (ii) $\int_0^\infty \sqrt{x}\,e^{-x}\,dx$ (iii) $\int_0^\infty x^{3/2}e^{-x}\,dx$

(iv) $\int_0^\infty x^3 e^{-2x}\,dx$ (v) $\int_0^\infty x^6 e^{-2x}\,dx$ (vi) $\int_0^\infty x^5 e^{-x^2}\,dx$

(vii) $\int_0^\infty e^{-x^3}\,dx$ (viii) $\int_0^1 (\log x)^3\,dx$ (ix) $\int_0^1 (x\log x)^4\,dx$

(x) $\int_0^1 \left(\log\frac{1}{x}\right)^{3/2}$ (xi) $\int_0^\infty 2^{-3x^2}\,dx$ (xii) $\int_0^1 x^2\left(\log\frac{1}{x}\right)^3\,dx$

3. Evaluate :
   (i) $\Gamma(-5/2)$ (ii) $\Gamma(-7/2)$ (iii) $\Gamma(-9/2)$ (iv) $\Gamma(-1/3)$

4. Prove that $\Gamma(n) = \int_0^1 \left(\log \frac{1}{x}\right)^{n-1} dx,\ n > 0$

5. Show that $\int_0^\infty \frac{e^{-st}}{\sqrt{t}} dt = \sqrt{(\pi/s)},\ s > 0$

6. Prove that $\Gamma(n) = 2a^n \int_0^\infty x^{2n-1} e^{-ax^2} dx$

7. Prove that $\Gamma(n) = a^n \int_0^\infty e^{-ax} x^{n-1} dx$

8. Show that $\int_0^\infty e^{(2ax - x^2)} dx = \frac{\sqrt{\pi}}{2} e^{a^2}$

9. Prove that :
   (i) $\int_0^\infty x^{n-1} \cos ax\, dx = \frac{1}{a^n} \Gamma(n) \cos\left(\frac{n\pi}{2}\right)$
   (ii) $\int_0^\infty x^{n-1} \sin ax\, dx = \frac{1}{a^n} \Gamma(n) \sin\left(\frac{n\pi}{2}\right)$

10. Prove that :
   (i) $\int_0^\infty \cos(ax^{1/n})\, dx = \frac{1}{a^n} \Gamma(n+1) \cos\left(\frac{n\pi}{2}\right)$
   (ii) $\int_0^\infty \sin(ax^{1/n})\, dx = \frac{1}{a^n} \Gamma(n+1) \sin\left(\frac{n\pi}{2}\right)$

11. Show that $\frac{d^k}{dx^k} \Gamma(x) = (\log t)^k \int_0^\infty t^{x-1} e^{-x} dx$

ANSWERS

1. (i) 30 (ii) 16/105 (iii) $3\pi\sqrt{\pi}/8$ (iv) 4/3 (v) 16/315
2. (i) 120 (ii) $\sqrt{\pi}/2$ (iii) $3\sqrt{\pi}/4$ (iv) 3/8 (v) 45/8
   (vi) $105\sqrt{\pi}/8$ (vii) $(1/3)\Gamma(1/3)$ (viii) –6 (ix) 94/625
   (x) $-2\sqrt{\pi}$ (xi) $\frac{\sqrt{\pi}}{2\sqrt{3}\log 2}$ (xiii) 2/27
3. (i) $(-8\sqrt{\pi})/15$ (ii) $16\sqrt{\pi}/105$ (iii) $(-32\sqrt{\pi})/945$ (iv) $-3\Gamma(2/3)$

Beta and Gamma functions

♦ Standard Results

Relation between Beta and Gamma functions

The Beta and Gamma functions are related by

$$B(m,n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$$

Example 4.8.22. Evaluate : (i) $\beta(3, 5)$ (ii) $\beta(3/2, 2)$ (iii) $\beta(1/3, 2/3)$

Solution : (i) Using $\beta(m,n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$

we have $\beta(3, 5) = \frac{\Gamma(3)\,\Gamma(5)}{\Gamma(3+5)} = \frac{2!\,4!}{7!} = \frac{1}{105}$

(ii) $\beta\left(\frac{3}{2}, 2\right) = \frac{\Gamma(3/2)\,\Gamma(2)}{\Gamma(3/2+2)} = \frac{\Gamma(3/2)\,\Gamma(2)}{\Gamma(7/2)} = \frac{\frac{1}{2}\Gamma(1/2)\,(1!)}{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\Gamma(1/2)} = \frac{4}{15}$

(iii) $\beta\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{\Gamma(1/3)\,\Gamma(2/3)}{\Gamma\left(\frac{1}{3}+\frac{2}{3}\right)} = \Gamma(1/3)\,\Gamma\left(1-\frac{1}{3}\right) = \frac{\pi}{\sin\frac{\pi}{3}} = \frac{2\pi}{\sqrt{3}}$

Example 4.8.23. Evaluate the following:

(i) $\int_0^1 x^4 (1-x)^3\, dx$ (ii) $\int_0^2 \frac{x^2}{\sqrt{2-x}} dx$ (iii) $\int_0^a y^4 \sqrt{a^2 - y^2}\, dy$

Solution :

(i) $\int_0^1 x^4 (1-x)^3\, dx = \beta(5, 4) = \frac{\Gamma(5)\,\Gamma(4)}{\Gamma(5+4)} = \frac{4!\,3!}{8!} = \frac{1}{280}$

(ii) $\int_0^2 \frac{x^2}{\sqrt{2-x}} dx$

Substitute $x = 2t$ then $dx = 2\,dt$. Therefore

$$\int_0^2 \frac{x^2}{\sqrt{2-x}} dx = \int_0^1 \frac{4t^2}{\sqrt{2-2t}} 2\,dt$$

$$= 4\sqrt{2} \int_0^1 t^2 (1-t)^{-1/2} dt$$

$$= 4\sqrt{2}\, \beta\left(3, \frac{1}{2}\right) = 4\sqrt{2}\, \frac{\Gamma(3)\,\Gamma(1/2)}{\Gamma(7/2)} = \frac{64\sqrt{2}}{15}$$

(iii) $\int_0^a y^4 \sqrt{a^2 - y^2}\, dy$

Substitute $y^2 = a^2 t$ or $y = a\sqrt{t}$. Then $dy = \frac{a}{2\sqrt{t}} dt$

Given integral becomes

$$\int_0^1 (a\sqrt{t})^4 \sqrt{a^2 - a^2 t}\, \frac{a}{2}\frac{dt}{\sqrt{t}} = \frac{a^6}{2}\int_0^1 t^{3/2}(1-t)^{1/2}\, dt$$

$$= \frac{a^6}{2}\beta\left(\frac{5}{2}, \frac{3}{2}\right) = \frac{a^6}{2}\cdot\frac{\Gamma(5/2)\,\Gamma(3/2)}{\Gamma(5/2+3/2)} = \frac{\pi a^6}{32}$$

Example 4.8.24. Evaluate :

(i) $\int_0^{\pi/2} \sin^6\theta\, d\theta$ (ii) $\int_0^{\pi} \cos^4\theta\, d\theta$ (iii) $\int_0^{\pi/2} \sin^4\theta \cos^5\theta\, d\theta$

Solution : (i) $\int_0^{\pi/2} \sin^6\theta\, d\theta$

Note : Property : $\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2\int_0^{\pi/2} \sin^p\cos^q\theta\, d\theta$

$$= 2\int_0^{\pi/2} \sin^q\theta\, \cos^p\theta\, d\theta \quad \ldots (1)$$

(i) Taking $p = 6$, $q = 0$ in (1), we get

$$\int_0^{\pi/2} \sin^6\theta\, d\theta = \frac{1}{2}\beta\left(\frac{6+1}{2}, \frac{0+1}{2}\right) = \frac{1}{2}\beta\left(\frac{7}{2}, \frac{1}{2}\right)$$

$$= \frac{1}{2}\cdot\frac{\Gamma(7/2)\,\Gamma(1/2)}{\Gamma(4)} = \frac{5\pi}{32}$$

(ii) $\int_0^{\pi} \cos^4\theta\, d\theta$

From (1), $\int_0^{\pi} \cos^4\theta\, d\theta = 2\cdot\int_0^{\pi/2} \cos^4\theta\, d\theta = 2\cdot\frac{1}{2}\cdot\beta\left(\frac{0+1}{2}, \frac{4+1}{4}\right)$

$$= \beta\left(\frac{1}{2}, \frac{5}{2}\right) = \frac{\Gamma(1/2)\,\Gamma(5/2)}{\Gamma(3)} = \frac{3\pi}{8}$$

(iii) $\int_0^{\pi/2} \sin^4\theta \cos^5\theta\, d\theta$

Here $p = 4$, $q = 3$ hence from (1), we get

$$\int_0^{\pi/2} \sin^4\theta \cos^5\theta\, d\theta = \frac{1}{2}\beta\left(\frac{4+1}{2}, \frac{5+1}{2}\right) = \frac{1}{2}\beta\left(\frac{5}{2}, 3\right)$$

$$= \frac{1}{2}\cdot\frac{\Gamma(5/2)\,\Gamma(3)}{\Gamma(11/2)} = \frac{8}{315}$$

Example 4.8.25. Evaluate : (i) $\int_0^{\infty} \frac{x\, dx}{1+x^6}$ (ii) $\int_0^{\infty} \frac{y^2\, dy}{1+y^4}$

Solution : (i) Letting $x^6 = t$ or $x = t^{1/6}$ we have $dx = \frac{1}{6}t^{-5/6}\, dt$ the given integral becomes,

$$\int_0^{\infty} \frac{t^{1/6}(1/6)\,t^{-5/6}\, dt}{1+t} = \frac{1}{6}\int_0^t \frac{t^{-2/3}}{1+t}\, dt = \frac{1}{6}\int_0^{\infty} \frac{t^{1/3-1}}{(1+t)^{2/3+1/3}}\, dt$$

$$= \frac{1}{6}\beta\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{1}{6}\frac{\Gamma(1/3)\,\Gamma(2/3)}{\Gamma(1)}$$

$$= \frac{1}{6}\Gamma(1/3)\,\Gamma(1-1/3) = \frac{1}{6}\cdot\frac{\pi}{\sin(\pi/3)} = \frac{\pi}{3\sqrt{3}}$$

(ii) Substitute $y^4 = t$ or $y = t^{1/4}$. Then $dy = \frac{1}{4}t^{-3/4}\, dt$ so that

$$\int_0^{\infty} \frac{y^2\, dy}{1+y^4} = \int_0^{\infty} \frac{[t^{1/4}]^2\,(1/4)\,t^{-3/4}\, dt}{1+t} = \frac{1}{4}\int_0^{\infty} \frac{t^{-1/4}}{1+t}\, dt$$

$$= \frac{1}{4}\int_0^{\infty} \frac{t^{3/4-1}}{(1+t)^{3/4+1/4}}\, dt = \frac{1}{4}\beta\left(\frac{3}{4}, \frac{1}{4}\right)$$

$$= \frac{1}{4}\cdot\frac{\Gamma(3/4)\,\Gamma(1/4)}{\Gamma(1)} = \frac{1}{4}\Gamma\left(1-\frac{1}{4}\right)\Gamma(1/4)$$

$$= \frac{1}{4}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{2\sqrt{2}}$$

EXERCISE

1. Prove the following:

(i) $\beta(4, 5) = \frac{1}{280}$ (ii) $\beta\left(\frac{3}{2}, 2\right) = \frac{4}{15}$

(iii) $\beta\left(\frac{5}{2}, \frac{7}{2}\right) = \frac{3\pi}{256}$ (iv) $\beta\left(\frac{7}{2}, -\frac{1}{2}\right) = -\frac{15}{8}\pi$

2. Evaluate the following :

(i) $\int_0^1 x^5 (1-x^3)^3 \, dx$ (ii) $\int_0^4 x^{3/2} (4-x)^{5/2} \, dx$

(iii) $\int_0^1 \frac{x^3 \, dx}{\sqrt{1-x}}$ (iv) $\int_0^2 (4-x^2)^{3/2} \, dx$

(v) $\int_0^2 (8-x^3)^{-1/3} \, dx$ (vi) $\int_1^3 \frac{dx}{\sqrt{(x-1)(3-x)}}$

(vii) $\int_0^\infty \frac{x^2}{1+x^4} dx$ (viii) $\int_0^1 \sqrt{(1-x)/x} \, dx$

(ix) $\int_{-1}^{+1} \left(\frac{1+x}{1-x}\right)^{1/2} dx$ (x) $\int_0^1 \sqrt{(1-x^4)} \, dx$

(xi) $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$ (xiii) $\int_0^{\pi/2} \tan^p \theta \, d\theta, \; 0 < p < 1$

3. Show that

(i) $(m+n)\beta(m, n+1) = n\beta(m, n)$

(ii) $\beta(m, n)\beta(m+n, p) = \beta(n, p)\beta(n+p, m)$

(iii) $\frac{\beta(m+2, n-2)}{m(m+1)} = \frac{\beta(m, n)}{(n-1)(n-2)}$

(iv) $\beta(n, n)\beta(n+1/2, n+1/2) = 2^{1-4n}\left(\frac{\pi}{n}\right)$

4. Prove that $\int_0^a \frac{dx}{\sqrt{a^4-x^4}} = \frac{\Gamma(1/4)^2}{4a\sqrt{2\pi}}$

5. Prove that $\Gamma(n)\Gamma\left(\frac{1-n}{2}\right) = 2^{n-1}\sqrt{\pi}\,\Gamma\left(\frac{n}{2}\right) sec\left(\frac{n\pi}{2}\right)$

ANSWERS

2. (i) 1/60, (ii) 12π, (iii) 32/35, (iv) 3π

(v) $2\pi/3\sqrt{3}$, (vi) π, (vii) $\pi/2\sqrt{2}$, (viii) $\frac{\pi}{2}$

(ix) π, (x) $\frac{\sqrt{\pi}}{6}\frac{\Gamma(1/4)}{\Gamma(3/4)}$ (xi) $\frac{\sqrt{\pi}}{4}\frac{\Gamma(1/4)}{\Gamma(3/4)}$, (xii) $\frac{\pi}{2} sec\left(\frac{p\pi}{2}\right)$

Unit-4. DIFFERENTIAL CALCULUS

MULTIPLE-CHOICE QUESTION

1. The radius of the curvature of the curve $y = 4\sin x$ at $x = \frac{\pi}{2}$ is

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $-\frac{1}{4}$

Ans. (c)

2. The curvature of the straight line is

(a) 1 (b) 2 (c) −1 (d) 0

Ans. (d)

3. The envelope of the family of curves $A\alpha^2 + B\alpha + C = 0$ (α is parameter) is

(a) $B^2 + 4AC = 0$ (b) $B^2 - AC = 0$

(c) $B^2 + AC = 0$ (d) $B^2 - 4AC = 0$

Ans. (d)

4. The locus of center of curvature is called

(a) involute (b) evolute

(c) radius of curvature (d) envelope

Ans. (b)

5. If the radius of curvature and curvature of a curve at any point are ρ and κ respectively.

(a) $\rho = -\frac{1}{\kappa}$ (b) $\rho = \kappa$

(c) $\rho = -\kappa$ (d) $\rho = \frac{1}{\kappa}$

Ans. (d)

6. The radius of curvature of $y = e^x$ at $x = 0$ is

(a) $2\sqrt{2}$ (b) $\frac{2}{\sqrt{2}}$ (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

Ans. (a)

7. Find the envelope of the curve $y = mx + \frac{a}{m}$ where m is a parameter

(a) $y^2 - 4ax = 0$ (b) $y^2 + 4ax = 0$
(c) $x^2 + y^2 = 1$ (d) $xy = c^2$

Ans. (a)

8. What is the curvature of a circle of radius 3 ?

(a) 3 (b) -3 (c) $\frac{1}{3}$ (d) $-\frac{1}{3}$

Ans. (c)

9. What is the radius of curvature at (4, 3) on the curve $x^2 + y^2 = 25$

(a) 5 (b) -5 (c) 25 (d) -25

Ans. (a)

10. The curvature at any point of the circle is equal to ..... of its radius.

(a) square (b) same (c) reciprocal (d) constant

Ans. (c)

11. The envelope of $ty - x = at^2$, t is the parameter is

(a) $x^2 = 4ay$ (b) $y^2 = 4ax$
(c) $x^2 + y^2 = 1$ (d) $x^2 - y^2 = 1$

Ans. (b)

12. The radius of curvature in polar coordinates is

(a) $\rho = \dfrac{\left(r^2 + (r')^2\right)^{3/2}}{r^2 - rr' + 2(r')^2}$ (b) $\rho = \dfrac{\left(r^2 - (r')^2\right)^{3/2}}{r^2 - rr' + 2(r')^2}$

(c) $\rho = \dfrac{\left(r^2 - (r'')^2\right)^{3/2}}{r^2 - rr' + 2(r')^2}$ (d) $\rho = \dfrac{\left(r^2 + (r')^2\right)^{3/2}}{r^2 - rr'' + 2(r')^2}$

Ans. (d)

13. The radius of curvature in Cartesian coordinates is

(a) $\rho = \dfrac{(1 + y_1^2)^{3/2}}{y_2}$ (b) $\rho = \dfrac{(1 - y_1^2)^{3/2}}{y_2}$

(c) $\rho = \dfrac{(1 + y_1^2)^{2/3}}{y_2}$ (d) $\rho = \dfrac{(1 + y_2^2)^{3/2}}{y_1}$

Ans. (a)

14. The envelope of family of lines $y = mx + am^2$ (where m is the parameter) is

(a) $x^2 + 2ay = 0$ (b) $x^2 + 4ay = 0$
(c) $y^2 + 2ax = 0$ (d) $y^2 + 4ax = 0$

Ans. (b)

15. A curve which touches each number of a family of the curves is called ..... of that family.

(a) Evolute (b) Envelope
(c) Circle of curvature (d) Radius of curvature

Ans. (b)

16. If radius of curvature of the curve $r^n = a^n \cos n\theta$ is $\frac{a^n r^{-n+1}}{n+1}$, then the radius of the curvature of the curve $r^2 = a^2 \cos 2\theta$ is

(a) $\frac{a^3}{2r}$ (b) $\frac{a^3}{3r}$ (c) $\frac{a^2}{3r}$ (d) $\frac{a^2}{2r}$

Ans. (c)

17. The equation of circle of curvature at any point (x, y) with center of curvature $\bar{x}, \bar{y}$ and with radius of curvature ρ is

(a) $(x + \bar{x})^2 + (y + \bar{y})^2 = \rho^2$ (b) $(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$
(c) $(x - \bar{x})^2 - (y + \bar{y})^2 = \rho^2$ (d) $(x + \bar{x})^2 + (y + \bar{y})^2 = \rho$

Ans. (b)

18. The radius of curvature in parametric coordinates is

(a) $\rho = \dfrac{\left((x')^2 + (y')^2\right)^{3/2}}{x'y'' - y'x''}$ (b) $\rho = \dfrac{\left((x')^2 + (y')^2\right)^{3/2}}{x'y'' + y'x''}$

(c) $\rho = \dfrac{\left((x')^2 - (y')^2\right)^{3/2}}{x'y'' - y'x''}$ (d) $\rho = \dfrac{\left((x')^2 - (y')^2\right)^{3/2}}{x'y'' + y'x''}$

Ans. (a)

19. The evolute of the cycloid $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$ is

(a) astroid (b) parabola
(c) cycloid (d) circle

Ans. (c)

20. Evolute of a curve is the envelope of ..... of that curve.

(a) tangent (b) normal
(c) parallel (d) locus

Ans. (b)

21. The radius of curvature of the curve $x = t^2$, $y = t$ at $t = 1$ is

(a) $\frac{5\sqrt{5}}{2}$ (b) $\frac{\sqrt{5}}{2}$ (c) $\frac{5}{2}$ (d) $\sqrt{5}$

Ans. (a)

22. In an ellipse the radius of curvature at the end of which axis is equal to the semi-latus rectum of the ellipse.

(a) minor (b) major
(c) vertical (d) horizonal

Ans. (b)

23. The radius of curvature of the curve $y = \log \sec x$ at any point of it is

(a) $\sec x$ (b) $\tan x$
(c) $\cot x$ (d) $\operatorname{cosec} x$

Ans. (a)

24. If a curve β is the evolute of a curve α, then α is said to be an ..... of β.

(a) normal (b) tangent
(c) envelope (d) involute

Ans. (d)

25. The parametric coordinates of any point on the ellipse are

(a) $x = \cos\theta, y = \sin\theta$ (b) $x = a\cos^2\theta, y = b\sin^2\theta$
(c) $x = a\cos\theta, y = b\sin\theta$ (d) $x = a\cos^3\theta, y = b\sin^3\theta$

Ans. (c)

26. The parametric coordinates of any point on the hyperbola are

(a) $x = \sec\theta, y = \tan\theta$ (b) $x = a\sec\theta, y = b\tan\theta$
(c) $x = a\sec^2\theta, y = b\tan^2\theta$ (d) $x = a\sec^3\theta, y = b\tan^3\theta$

Ans. (b)

27. The parametric coordinates of any point on the parabola are

(a) $x = at^2, y = 2at$ (b) $x = at^3, y = at$
(c) $x = at^2, y = at$ (d) $x = t^3, y = 2t$

Ans. (a)

28. For a given curve, the number of evolute is

(a) 3 (b) 2 (c) 0 (d) 1

Ans. (d)

29. The evolute of $y^2 = 4ax$ is $27ay^2 = 4(x - 2a)^3$. The X-axis will ..... the evolute.

(a) cut (b) not cut
(c) touch (d) touch at $(at^2, 2at)$

Ans. (c)

30. The radius of curvature of $xy = 1$ at (1, 1) is

(a) 2 (b) $\frac{1}{2}$ (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

Ans. (c)

31. The parametric coordinates of any point on the curve $x^{2/3} + y^{2/3} = a^{2/3}$ are

(a) $x = \cos\theta, y = \sin\theta$ (b) $x = a\cos^2\theta, y = a\sin^2\theta$
(c) $x = a\cos\theta, y = a\sin\theta$ (d) $x = a\cos^3\theta, y = a\sin^3\theta$

Ans. (d)

32. The parametric coordinates of any point on the rectangular hyperbola are

(a) $x = ct^2, y = \frac{c}{t^2}$ (b) $x = ct, y = \frac{c}{t^2}$
(c) $x = ct^2, y = \frac{c}{t}$ (d) $x = ct, y = \frac{c}{t}$

Ans. (d)

33. The parametric coordinates of any point on the ellipse are

(a) $x = \cos\theta, y = \sin\theta$ (b) $x = a\cos^2\theta, y = b\sin^2\theta$
(c) $x = a\cos\theta, y = b\sin\theta$ (d) $x = a\cos^3\theta, y = b\sin^3\theta$

Ans. (c)

34. The envelope of $x\cos\theta + y\sin\theta = 1$ is

(a) $x^2 + y^2 = 1$ (b) $x^2 + y^2 = a^2$
(c) $x^2 - y^2 = 1$ (d) $x^2 + y^2 = 0$

Ans. (a)

35. The radius of curvature at any point on the curve $r = e^{\theta}$ is

(a) $\frac{\sqrt{2}}{r}$ (b) $\frac{r}{\sqrt{2}}$
(c) r (d) $\sqrt{2}\,r$

Ans. (d)

36. The envelope of the family of lines $\frac{x}{t} + yt = 2c$, t being the parameter is

(a) $x^2 + y^2 = c^2$ (b) $xy = c^2$
(c) $x^2y^2 = c^2$ (d) $x^2 - y^2 = c^2$

Ans. (b)

37. The radius of curvature at the point (s, χ) on the curve $s = \log\sec\chi$ is

(a) $c\sec\chi$ (b) $c\cot\chi$
(c) $c\sec\chi\tan\chi$ (d) $c\tan\chi$

Ans. (d)

38. The radius of curvature for the circle $y = e^x$ at $(0, 1)$ is

(a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2\sqrt{2}}$

Ans. (b)

39. The radius of curvature of the circle $x^2 + y^2 = 25$ at $(3, 4)$ is

(a) 25 (b) -5 (c) 5 (d) -25

Ans. (c)

40. The envelope of the family of lines $\frac{x}{t} + yt = 2c$, t being the parameter is

(a) $xy = c^2$ (b) $xy = c$
(c) $\frac{x}{y} = c^2$ (d) $\frac{x}{y} = c$

Ans. (a)

41. The envelope of the family of curves given by $y = \sin(x + c)$ where c is a parameter is

(a) $y = 1$ (b) $y^2 = 1$
(c) $y^2 = 4$ (d) $y = 4$

Ans. (b)

42. The envelope of the family of straight lines $y = mx + \frac{a}{m}$ is

(a) $y^2 = 2ax$ (b) $y^2 = 4ax$
(c) $y^2 = ax$ (d) $y = 4ax^2$

Ans. (b)

43. The envelope of a one-parameter family of straight lines $x\cos\alpha + y\sin\alpha = a$, where α is a parameter is

(a) $xy = a^2$ (b) $y^2 = 4a^2$
(c) $x^2 - y^2 = a^2$ (d) $x^2 + y^2 = a^2$

Ans. (d)

44. The centre of curvature of $y^2 = 4x$ at $(1, 2)$ is

(a) $(5, -2)$ (b) $(-5, 2)$
(c) $(-5, -2)$ (d) $(5, 2)$

Ans. (a)

45. The curvature of the curve $y = e^x$ at $x = 0$ is

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$
(c) $\frac{1}{2\sqrt{2}}$ (d) $\frac{1}{4\sqrt{2}}$

Ans. (c)

46. The curvature of the curve $2x^2 + 2y^2 + 5x - 2y + 1 = 0$ is

(a) $\frac{4}{\sqrt{21}}$ (b) $\frac{2}{\sqrt{21}}$
(c) $\frac{2}{21}$ (d) $\frac{4}{21}$

Ans. (a)

47. The radius of curvature of the curve $y = c \cosh\left(\frac{x}{c}\right)$ is

(a) $\cosh\frac{x}{c}$ (b) $\cosh\frac{x}{c}$

(c) c (d) $\frac{1}{c}\cosh\frac{x}{c}$

Ans. (b)

48. The radius of curvature of the curve $y = c \cosh\left(\frac{x}{c}\right)$ is

(a) $\cosh\frac{x}{c}$ (b) $c\cosh\frac{x}{c}$

(c) c (d) $\frac{1}{c}\cosh\frac{x}{c}$

Ans. (b)

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UNIT-V

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